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**GUIDING THE CHAIN OF THOUGHT OF
MIXTURE-OF-EXPERTS LLMS VIA ROUTING
INTERVENTION FOR EFFICIENT REASONING**

Master Thesis in
Big Data Analytics and Text Mining

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Abstract

Sparse mixture-of-experts language models offer a distinctive opportunity to study and influence reasoning through the experts activated for each token. This thesis develops an auditable framework for investigating that opportunity without updating model weights, connecting automatic annotation of seven reasoning activities, representation probing, routing analysis, and controlled interventions.

Experiments cover Qwen3.5-35B-A3B, Gemma-4-26B-A4B, and GPT-OSS-20B across six mathematical benchmarks, with 4,647 attempts from 1,547 problems per model. A key finding is that hidden representations contain information about upcoming reasoning activities: probes achieve macro-AUROC scores of 0.852, 0.788, and 0.780, respectively, outperforming a prefix-position baseline on held-out SAT responses. This supports their value as research instruments, while broader transfer and online detection remain unvalidated.

The intervention results reveal an important distinction between identifying informative patterns and using them to improve reasoning. Routing–correctness associations depend on the model, benchmark, and reasoning activity. In comparisons of 1,395 matched attempts per intervention, stronger fixed expert promotion reduces Qwen and GPT-OSS accuracy by 2.80 and 3.30 percentage points. Gemma changes by only -0.07 points, although non-inferiority is not established. No completed intervention establishes a statistically supported accuracy gain.

Cache simulations identify a separate opportunity: Qwen requires fewer expert loads per token under the tested conditions. However, total savings remain uncertain, and longer outputs can offset local benefits; actual SSD latency is not measured.

Together, these findings provide tools for studying reasoning, evidence about the limits of correlation-based steering, and a practical basis for evaluating future routing policies against both accuracy and computational cost.

Introduction

An answer to a mathematical problem provides a clear outcome, but it gives only a partial account of how a language model produced that outcome. A generated solution may alternate between interpreting the problem, planning, calculating, exploring alternatives, and checking conclusions. These activities can accompany both successful and unsuccessful solutions. Understanding their role requires measurements of the generated reasoning process as well as its final correctness.

Sparse mixture-of-experts (MoE) models offer a computational perspective on this question. Their routers select a subset of expert networks to process each token. Expert selection and routing weights can therefore vary as a solution develops. This creates an opportunity to examine whether particular routing patterns accompany particular reasoning activities, and whether changing those patterns affects subsequent generation. It also presents an interpretation problem: an expert frequently used in correct solutions may reflect the problems being solved, the position in the response, or its length, rather than a mechanism whose promotion improves correctness.

The objective of this thesis is to investigate guidance of MoE reasoning through interventions on routing, with the model weights held fixed. The central question is whether signals obtained from reasoning annotations and internal measurements can support useful intervention policies. Addressing this question requires separating three kinds of evidence. A probe can establish that a representation contains information predictive of a label. A correlation can establish that a routing statistic varies with an outcome in a given population. An intervention comparison measures what happens when a specified routing policy is applied during generation. Success at one of these levels does not

establish success at the next.

Research questions. The investigation is organised around three questions:

1. Can functional reasoning episodes be annotated automatically, and is information about an upcoming episode accessible from the hidden representation of the preceding response prefix?
2. How do expert selection, routing confidence, and representation measurements relate to reasoning episodes and answer correctness across models, benchmarks, and positions in a response?
3. Do policies calibrated from these observational measurements improve held-out mathematical accuracy, change token use, or reduce simulated expert-loading demand when applied to routing during generation?

Research approach. Our experimental framework connects four stages. First, an LLM-based annotator assigns sentences to seven adapted Schoenfeld episode classes: Read, Analyze, Plan, Implement, Explore, Verify, and Monitor. Prompt optimisation and validation assess the annotator as a measurement instrument; its predictions on generated solutions are not treated as human ground truth. Second, prospective probes test the accessibility of episode information before the next labelled unit and identify locations for internal measurement. These labels describe the function of visible text, without assuming that the text exhaustively represents the model’s internal computation.

Third, saved benchmark solutions are replayed through frozen models to extract routing and representation features. Analyses distinguish the whole continuation, reasoning tokens, labelled episode classes, and positional windows. Repeated attempts are grouped by source problem, and the reported bootstrap intervals preserve that dependence. Generation and replay configurations are recorded separately because corresponding model versions under different quantisation settings are not assumed to produce identical internal activations. Fourth, calibrated policies promote or suppress selected expert identities, or modify the selection-boundary margin during free generation. Evaluation compares each policy with its own matched baseline on problems excluded

from calibration. A complementary cache experiment records ordered routing decisions and compares replacement policies, separating changes in expert locality from changes in answer length and from any eventual effect on offloaded-inference speed.

The observational study covers Qwen3.5-35B-A3B, Gemma-4-26B-A4B, and GPT-OSS-20B on six mathematical benchmarks. Gemma and GPT-OSS share a stratified annotation procedure but retain model-specific selected cohorts; Qwen uses a separate annotation protocol. Completed intervention evidence includes Qwen and GPT-OSS identity and margin guiding, Gemma fixed identity guiding, and ordered-cache simulations on 100 matched problems for each of the three models. The interrupted Qwen paper-style run supplies a separate descriptive analysis of completed attempts.

Research contributions. The thesis contributes an integrated workflow linking functional descriptions of reasoning to measurements of sparse expert computation and to explicit routing interventions. Saved prompts, trace identities, sentence offsets, model configurations, and records of data provenance make the connection between these stages inspectable. The empirical contribution is a comparison of routing–correctness associations across models and reasoning classes, together with matched tests of whether selected associations support effective steering. The tested contribution is the connection between observational routing signals and generation-time intervention outcomes. Episode labels organise the measurements and probes inform measurement locations, but the evaluated controllers do not predict episodes online or condition routing on those predictions. Episode-aware control remains an extension requiring separate detector and policy validation.

The grouped results show prospective episode information on held-out responses within the probe corpus, beyond a simple prefix-position baseline. They do not establish transfer to other domains or the encoders’ own generations. Routing associations depend on the model and benchmark, including changes of sign within a reasoning class. Under the primary attempt-weighted objective, stronger fixed promotion reduces accuracy for Qwen and GPT-OSS, and paper-style promotion severely degrades GPT-OSS accuracy. After consistent offline

rescoring, all margin intervals contain zero and no complete intervention has a positive interval excluding zero under either reported weighting. These findings motivate a qualified view of routing-guided reasoning: observational structure provides testable hypotheses, while a reliable improvement requires additional evidence from controlled interventions. Annotation uncertainty, problem difficulty, response length, termination, and deployment differences remain important limits on interpretation.

Thesis organisation. Chapter 1 introduces the theoretical framework, and Chapter 2 reviews related research. Chapter 3 develops the procedure, while Chapter 4 specifies the data, model configurations, and evaluation protocols. Chapter 5 presents the findings, and Chapter 6 discusses their implications and limitations.

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Chapter 1

Theoretical Framework

The objective of this thesis is to investigate how the reasoning behaviour of a mixture-of-experts (MoE) language model can be guided through interventions on its routing mechanism. This objective connects a behavioural question, concerning the organisation of a generated solution, with a computational question, concerning which experts process its tokens. Establishing that connection requires a vocabulary for describing reasoning, measurements of internal model activity, and a distinction between observing an association and changing the mechanism that produces it.

The project develops four connected stages toward this objective. The first constructs an automatic annotator of reasoning episodes. The second tests whether episode information is accessible from prefix hidden states and selects measurement locations. The third measures associations between routing, representations, reasoning behaviour, and answer correctness. The fourth tests expert-identity and margin policies during free generation. Chapter 3 specifies their integration. This chapter defines the concepts needed to distinguish measurement, association, and the effect of a routing intervention.

1.1 Reasoning as a measurable process

Let q denote a problem and let $x_{1:T}$ be the model’s generated response, including intermediate reasoning and the final answer. An autoregressive

language model with parameters θ defines

$$P_{\theta}(x_{1:T} \mid q) = \prod_{t=1}^T P_{\theta}(x_t \mid q, x_{<t}). \quad (1.1)$$

Each generated token becomes part of the context for later decisions. Intermediate text can therefore affect subsequent computation, even when evaluation ultimately assigns a single correctness label to the final answer. Chain-of-thought prompting makes such intermediate text explicit and has demonstrated benefits on several reasoning benchmarks [1].

For analysis, the response is segmented into units $s_1, \dots, s_J$, usually sentences or short coherent spans. A unit can restate the problem, introduce a strategy, execute a calculation, or check a previous result. These functions provide a richer description than response length or final accuracy alone. However, a verbal explanation is an observable output of the model, rather than a complete record of its internal computation. Experiments on chain-of-thought faithfulness show that a response can omit factors that influenced its answer [2]. Accordingly, this thesis uses the term *reasoning episode* for a function assigned to a text unit, without assuming that the label identifies a unique internal cognitive state.

Three levels of observation will recur throughout the thesis. At the token level, the model produces hidden representations and expert-routing decisions. At the unit level, a segment receives a functional episode label. At the attempt level, the complete response has a final answer and possibly binary indicators of particular reasoning events. Repeated attempts add a fourth level: the source problem. Keeping these levels explicit is necessary both for temporal alignment and for statistical analysis.

1.2 Sparse mixture-of-experts computation

1.2.1 Conditional computation in sparse MoE layers

A sparse MoE layer contains several parameterised expert networks and a router that selects a subset for each token. The sparsely gated formulation of Shazeer et al. made it possible to increase model capacity while activating

only part of that capacity for an input [3]. In language-model backbones, an MoE module commonly replaces a dense feed-forward sublayer. The expert is therefore a neural transformation within a larger network; it is not an independent model that solves the whole problem.

Let $u_t^{(\ell)} \in \mathbb{R}^d$ be the input to the router at layer ℓ , and let E_ℓ be the number of routed experts there. For a softmax-based router, scores $z_t^{(\ell)}$ define

$$z_t^{(\ell)} = W_r^{(\ell)} u_t^{(\ell)} + b_r^{(\ell)}, \quad (1.2)$$

$$p_{t,e}^{(\ell)} = \frac{\exp(z_{t,e}^{(\ell)})}{\sum_{j=1}^{E_\ell} \exp(z_{t,j}^{(\ell)})}, \quad (1.3)$$

$$S_t^{(\ell)} \in \arg \max_{S \subseteq \{1, \dots, E_\ell\}, |S|=k_\ell} \sum_{e \in S} p_{t,e}^{(\ell)}. \quad (1.4)$$

The bias can be absent, and other routing architectures use different score transformations. These equations specify the softmax case used to define the measurements below, rather than a universal architectural definition.

Writing $f_e^{(\ell)}$ for expert e , a routed output has the form

$$m_t^{(\ell)} = \sum_{e \in S_t^{(\ell)}} g_{t,e}^{(\ell)} f_e^{(\ell)}(u_t^{(\ell)}) + m_{t,\text{shared}}^{(\ell)}. \quad (1.5)$$

Here $g_{t,e}^{(\ell)}$ denotes the effective mixing weight, which may be renormalised over the selected experts. The shared contribution is present only in architectures that include shared experts. This distinction matters: the full router distribution p , the selected set S , and the effective weights g are different objects. An intervention on one need not have the same effect as an intervention on another.

Switch Transformer uses one routed expert per token, whereas Mixtral uses two of eight experts [4, 5]. DeepSeekMoE combines finer expert partitioning with shared experts [6]. These architectural choices affect how routing statistics should be interpreted. They also illustrate why total parameter count and active parameter count must be distinguished: sparse activation reduces per-token expert computation, but does not eliminate the cost of storing expert weights or dispatching tokens.

The project’s Qwen3.5-35B-A3B subject provides a concrete instance. Its official model card describes a hybrid backbone with 40 layers, combining

Gated DeltaNet and gated attention, with 256 routed experts, eight selected experts per token, and a shared expert [7]. This makes the subject a hybrid sequence model. The general routing notation above applies independently of the particular sequence-mixing sublayer.

1.2.2 Interpreting expert identity

Expert identities are local to a layer. The correct identifier is the pair (ℓ, e) : expert 7 in one layer has different parameters from expert 7 in another layer. Likewise, selecting an expert frequently during a labelled episode does not establish that the expert implements that episode. Token identity, syntax, position, and problem content may all influence selection. Mixtral’s routing analysis, for example, reports syntactic regularities and locality between consecutive tokens [5].

It is useful to distinguish *statistical specialisation*, meaning that expert use differs across observed contexts, from *functional involvement*, meaning that changing the expert’s contribution changes a behaviour of interest. The first can nominate targets for study. The second requires an intervention and an appropriate comparison. Neither implies a one-to-one mapping between experts and semantic reasoning functions.

1.3 A vocabulary for reasoning episodes

1.3.1 Adapting episode theory to model-generated text

Schoenfeld’s account of mathematical problem solving emphasises the organisation and control of problem-solving activity in addition to the availability of mathematical knowledge [8]. Li et al. adapt this perspective to reasoning-model traces and release a corpus with sentence and paragraph annotations [9]. The present project uses seven sentence-level categories. Their operational meaning in the annotation experiment is summarised in Table 1.1.

These labels are nominal categories. They are not ordered from weak to strong reasoning, and a solution need not pass through them in a fixed sequence.

Table 1.1: Operational episode categories used by the project, based on the adapted framework of Li et al. [9]. The descriptions concern a unit’s function, rather than its mathematical validity.

Episode	Function of the text unit
Read	Restates information or the goal supplied by the problem.
Analyze	Interprets information, introduces structure, or derives a relationship.
Plan	Commits to a concrete operation or strategy before executing it.
Implement	Executes a procedure, calculation, substitution, or enumeration.
Explore	Tentatively proposes a hypothesis, option, or alternative path.
Verify	Checks an existing candidate, result, or consistency claim.
Monitor	Expresses content-light self-regulation, a transition, or a structural boundary.

A Verify unit can contain an invalid check; an Implement unit can contain an incorrect calculation. Similarly, repeated monitoring does not necessarily improve a solution. Episode analysis and correctness evaluation therefore measure different aspects of model behaviour.

Segmentation is part of this measurement. One sentence may both announce and execute a plan, while a short structural marker may have no mathematical content. Assigning one label requires a documented precedence rule and access to enough context to distinguish, for example, a given fact from a newly derived relation. These ambiguities limit the meaning of exact agreement with a reference annotation. Per-class errors and inspection of ambiguous boundaries are consequently useful alongside aggregate classification scores.

1.3.2 Automatic episode annotation

Let $a_j \in \mathcal{A}$ denote the reference episode for unit s_j , where $\mathcal{A}$ contains the seven categories. A prompted annotator with instruction ϕ returns

$$\hat{a}_j = A_\phi(q, s_{j-1}, s_j, s_{j+1}). \quad (1.6)$$

This is the information pattern used for automatic episode annotation: the problem and neighbouring units help identify the current unit’s function. Be-

cause the current and next units are visible, it is an offline annotation tool. It cannot be inserted unchanged into a controller that must act before s_j is generated.

GEPA searches over instructions using feedback and natural-language reflection, while keeping the underlying model weights fixed [10]. In this project, a same-model judge can provide structured optimisation feedback, but final prompt selection is assessed against reference labels. Such feedback is not independent ground truth. Splitting complete responses before their units are assigned to training, validation, and test sets prevents neighbouring passages from crossing evaluation boundaries.

Accuracy measures exact label agreement. Balanced accuracy averages recall across categories, and macro-F1 averages the category-specific F1 scores; both expose failures concealed by a dominant class. Cohen’s κ measures chance-adjusted nominal agreement. Rank-based agreement measures require an independently justified ordering and are not primary measures for these episode labels. The purpose of annotation optimisation is to establish a usable measurement instrument for later experiments, rather than to optimise the subject model’s routing directly.

1.4 Prospective representation probes

For a unit beginning at token index b_j , define a boundary representation $v_j^{(\ell)} = h_{b_j-1}^{(\ell)}$. Under causal processing, it depends on the problem and the preceding response, but not on the target unit. A binary linear probe for episode a has the form

$$\hat{P}(a_j = a \mid v_j^{(\ell)}) = \sigma \left(w_{a,\ell}^\top v_j^{(\ell)} + c_{a,\ell} \right), \quad (1.7)$$

where σ is the logistic function. Above-chance prediction indicates that information useful to this classifier is accessible at that boundary. The interpretation depends on probe capacity, train/test separation, and possible shortcuts [11].

The prospective probing experiment trains a separate one-versus-rest logistic regression for each label and hidden-state index. It teacher-forces released DeepSeek-R1 responses through frozen Qwen, GPT-OSS, and Gemma encoders.

Consequently, its subject is each encoder’s representation of supplied prefixes, rather than its own generation of the annotated solutions. The original replication uses a sentence-level split: units from a response can appear in both partitions, and test-based layer selection makes the selected maximum optimistic. A separate response-grouped evaluation selects layers on inner validation responses and measures predictions on outer test responses (Section 4.5.1).

Temporal availability and causal influence must also be distinguished. A pre-unit representation is prospective, but predicting a future label does not establish that the probed direction causes that label. A probe can exploit problem type, position, or formatting. Independently trained, class-balanced binary scores also need not form calibrated multiclass probabilities, even when their largest value is used to propose a label for inspection.

Finally, hidden-state indices must be tied to actual locations in the architecture. Let $h_t^{(\ell)}$ denote a block-input residual state and $u_t^{(\ell)}$ the router input from Section 1.2. An intervening sequence-mixing sublayer and normalisation can make these vectors different. The current correlation extraction retains block-input hidden states alongside the router distribution from the corresponding block; it does not directly capture the exact normalised router-input vector. Geometric relationships between the two measurements must be interpreted with this distinction in mind.

1.5 Statistical characterisation of routing

1.5.1 Routing descriptors

For a fixed layer and token, suppress the indices on p , and write $p_{(1)} \geq \dots \geq p_{(E)}$ for the sorted router probabilities. For $E > 1$, Shannon entropy and its normalised concentration transform are

$$H(p) = - \sum_{e=1}^E p_e \log p_e, \quad C(p) = 1 - \frac{H(p)}{\log E}. \quad (1.8)$$

$C = 0$ for a uniform distribution and approaches one as mass concentrates on one expert. Although C is termed routing confidence, it is not a calibrated probability that the answer is correct. Routing probabilities describe expert

allocation; vocabulary probabilities describe the next token. Their entropies concern different distributions.

For top- k routing with $k < E$, additional measures are

$$M_k(p) = \sum_{i=1}^k p_{(i)}, \quad (1.9)$$

$$B_k(p) = p_{(k)} - p_{(k+1)}, \quad (1.10)$$

$$G_k(p) = \frac{M_k(p)}{k} - \frac{1 - M_k(p)}{E - k}. \quad (1.11)$$

Selected mass M_k measures how much probability is assigned to the active set. The boundary margin B_k describes the separation between the last selected and first unselected expert. G_k compares mean selected and unselected probability. The top-1/top-2 margin alone does not describe the selection boundary when more than one expert is active.

Some of these quantities contain redundant information. For fixed E and k , G_k is an affine function of M_k ; their associations cannot be treated as independent evidence. Similarly, C is an affine sign reversal of H . Presenting a transformed measure can aid interpretation, but does not create an additional empirical finding.

For a trace of $T > 1$ tokens, the analysis summarises temporal routing with top-1 switching and top- k set overlap:

$$W^{(\ell)} = \frac{1}{T-1} \sum_{t=2}^T \mathbf{1} \left[\arg \max_e p_{t,e}^{(\ell)} \neq \arg \max_e p_{t-1,e}^{(\ell)} \right], \quad (1.12)$$

$$O^{(\ell)} = \frac{1}{T-1} \sum_{t=2}^T \frac{|S_{t-1}^{(\ell)} \cap S_t^{(\ell)}|}{k_\ell}. \quad (1.13)$$

The overlap denominator is the number of selected experts, not the union size used by Jaccard similarity. High overlap indicates persistent selection, but its relation to useful reasoning is an empirical question.

Identity features can instead measure the proportion of tokens selecting a particular (ℓ, e) , or the frequency of a particular expert pair. An ordered top-1/top-2 pair describes co-selection at one token; it does not describe a transition between tokens or every combination in a top- k set. Layer identity must be preserved before such features can nominate individual experts for intervention.

Hidden-state summaries provide complementary measurements, including vector norm and consecutive-token cosine distance,

$$D^{(\ell)} = \frac{1}{T-1} \sum_{t=2}^T \left(1 - \frac{\langle h_t^{(\ell)}, h_{t-1}^{(\ell)} \rangle}{\|h_t^{(\ell)}\| \|h_{t-1}^{(\ell)}\|} \right). \quad (1.14)$$

The analysis also constructs a geometry feature by correlating pairwise cosine similarities among sampled hidden states with the corresponding similarities among router distributions. That internal construction uses Pearson correlation. It is a scalar descriptive feature, distinct from the coefficient subsequently used to associate it with correctness; the many token pairs are not treated as independent experimental observations.

1.5.2 Measures of solution correctness

Let $Y_{ia} \in \{0, 1\}$ indicate whether attempt a on problem i produces a correct final answer, and let X_{ia} be a routing or representation feature. Their attempt-level association is measured by point-biserial correlation,

$$r_{\text{pb}} = \text{Corr}(X, Y). \quad (1.15)$$

This is exactly Pearson correlation when correctness is encoded as zero or one. A positive coefficient means that the feature tends to be larger among correct attempts. Its magnitude can depend on class prevalence and feature distributions; it is neither a causal effect nor a measure of probability calibration. The same coefficient applies to a continuous feature paired with a binary reasoning-event indicator. Liu et al. provide direct language model precedent for this continuous-versus-binary setting [12].

With N attempts per problem, define

$$\hat{A}_i^{(N)} = \frac{1}{N} \sum_{a=1}^N Y_{ia}, \quad \bar{X}_i = \frac{1}{N} \sum_{a=1}^N X_{ia}. \quad (1.16)$$

The first quantity is the empirical per-problem avg@ N . For $N = 32$, it takes values in $\{0, 1/32, \dots, 1\}$. It differs from pass@ N , which concerns at least one successful attempt among N samples. The primary repeated-attempt analysis uses one pair $(\bar{X}_i, \hat{A}_i^{(N)})$ per problem and computes Spearman correlation,

$$\rho_s = \text{Corr}(\text{rank}(\bar{X}_i), \text{rank}(\hat{A}_i^{(N)})), \quad (1.17)$$

using average ranks for ties. The scientific question is whether larger feature values tend to accompany larger success fractions, without requiring a linear relation between their numerical scales. Boundedness alone does not make Pearson invalid; Spearman is selected because monotone association is the target. Small problem counts and many tied success fractions still limit precision. Section 2.6 discusses the supporting literature.

Complete groups are necessary to make $\text{avg}@N$ comparable across problems. The analysis checks declared sample identifiers, outcome availability, and feature coverage, and checks for exact duplicate generations wherever response records permit this verification. A known gold answer with no parseable generated answer is counted as incorrect; a task without a scoring target remains unscored. Truncation and answer-parsing failures must be reported because they can alter both trace features and the measured success fraction.

1.5.3 Statistical inference under dependence

Attempts on the same problem share difficulty and content. Counting them as independent observations can make uncertainty appear smaller than it is. For prespecified aggregate features, the analysis therefore estimates 95% percentile bootstrap intervals by resampling whole problems and retaining their attempts together. For problem-level Spearman coefficients, it resamples the observed problem pairs. This latter procedure quantifies variation across observed problems without separately modelling uncertainty from the finite number of attempts within each problem.

An additional descriptive comparison uses only problems that yield both correct and incorrect attempts:

$$\Delta_i = \overline{X}_{i,Y=1} - \overline{X}_{i,Y=0}. \quad (1.18)$$

Averaging these contrasts removes fixed differences between source problems, but still permits differences in response length, token composition, or generation history within a problem. It also describes the subset of mixed-outcome problems. It is not an estimate of what would happen if routing were changed. The project’s GPQA design varies answer-option positions between attempts, so it is excluded from this particular contrast.

Searching many layers, features, experts, and pairs creates a multiplicity problem. Benjamini–Hochberg adjustments provide a declared way to summarise multiple tests within a specified family [13]. They rely on valid input tests and appropriate dependence assumptions. In particular, applying the adjustment to naive attempt-level p -values does not repair their failure to account for repeated problems. Such values remain exploratory diagnostics. A confirmatory intervention should specify its outcomes, targets, and evaluation partitions independently of these scans.

1.6 From observations to routing intervention

A routing intervention modifies the computation in Equations 1.4–1.5 during inference. One possible family of policies adds a context-dependent bias to the router scores:

$$\tilde{z}_t^{(\ell)} = z_t^{(\ell)} + \alpha v^{(\ell)}(q, x_{<t}), \quad (1.19)$$

after which expert selection and mixing weights are recomputed. This equation describes a design space for the thesis, rather than an already validated intervention. The strength α , affected layers, target experts, and activation times are parts of the policy.

Alternative policies reweight experts already selected, mask particular experts, or alter the selected set. These choices test different hypotheses. For example, rescaling all logits by a positive temperature changes their concentration but preserves their ordering and therefore preserves a deterministic top- k set. Increasing the contribution of an already selected expert also differs from forcing an otherwise unselected expert to participate. In all cases, renormalisation and the contribution of shared experts must be specified to interpret the intervention.

A controller intended to act before a reasoning unit can use only information available at that time. Offline labels may help construct or evaluate that controller, but the target unit and its future continuation cannot supply its decision at deployment. Prospective probes are relevant because they test whether useful information exists under this restriction. Their predictions

still require validation on the generation distribution where the controller will operate.

Teacher-forced replay is useful for measuring activations on a fixed saved response, but an intervention during free generation changes subsequent tokens and therefore subsequent states. A replay-based change in a feature or next-token score is not sufficient evidence of improved problem solving. The intervention objective ultimately concerns a contrast such as

$$\Delta_{\pi} = \mathbb{E}_{q,\xi} \left[Y(q, \xi; \pi) - Y(q, \xi; \pi_0) \right], \quad (1.20)$$

where π_0 is the unmodified router, π is the intervention, and ξ represents sampling randomness under a specified evaluation protocol. Episode frequencies, answer correctness, and computational cost should be measured together: increasing the amount of checking language is not itself evidence of better reasoning.

This formulation motivates four connected research questions. Can the episode vocabulary be applied reliably to new traces? Can relevant behaviours be detected from information available before they occur? Which routing measurements remain associated with behaviour and correctness after accounting for major confounders? Finally, does acting on selected routing targets improve outcomes on held-out problems under a controlled generation budget? The annotation, probe, and correlation stages address the measurement and observational questions; the guiding experiments specified in Chapter 3 address the effect of intervening during generation.

Chapter 2

Related Work

Research relevant to routing-guided reasoning spans several levels of model behaviour. Some approaches change the prompts or training signals that produce reasoning traces. Others annotate those traces, decode information from internal representations, or directly modify activations and expert routing. This chapter compares those approaches according to the object they measure or manipulate, and positions the four-stage experimental procedure within that literature.

2.1 Characterising reasoning processes

Chain-of-thought prompting demonstrated that intermediate reasoning demonstrations can improve language-model performance on arithmetic, commonsense, and symbolic tasks [1]. This established generated reasoning text as a practical object of study, but it did not make final accuracy a complete measure of the quality of a reasoning process. Self-consistency samples several reasoning paths and aggregates their answers [14], establishing a distinction between improving an individual attempt and improving answer selection over repeated attempts. Lightman et al. distinguish outcome supervision from feedback on intermediate step validity and release PRM800K for studying process supervision [15]. Their distinction is relevant to this thesis because a correct final answer and a well-formed intermediate process are related but

separate evaluation targets. ProcessBench evaluates detection of the earliest erroneous step in a mathematical solution [16], providing a complementary target to the functional labels used here.

Functional episode annotation addresses a further question: what a step is doing within the solution. Building on Schoenfeld’s framework, Li et al. annotate model-generated mathematical solutions at sentence and paragraph levels and analyse transitions between episode categories [9]. Their sentence-level labels supply the project’s annotation vocabulary and released reference data. A semantic episode differs from a process-reward label: Verify describes an attempt to check something, whereas a validity label evaluates whether a step is correct. The same functional category can therefore occur in both successful and failed traces.

Figure 2.1 shows the transition structure in the original annotated corpus. Its persistence within categories and directed transitions motivate analysing episode order alongside class shares. The later ThinkARM study extends this framework to 15 models and examines how episode transitions vary with correctness and reasoning efficiency [17], making it a direct comparison for our cross-model transition analysis. Gandhi et al. study verification, backtracking, subgoal setting, and backward chaining through controlled behavioural priming and reinforcement learning [18], offering a related vocabulary whose usefulness is tested through training interventions.

The faithfulness experiments of Turpin et al. further limit what can be inferred from generated explanations [2]. They show settings where input cues influence a prediction without being acknowledged in the accompanying rationale. For the present project, the implication is methodological: an episode sequence is an informative description of generated text, but a claim about the mechanism producing it needs independent internal evidence and intervention.

SPIRAL studies reasoning through a different route, using self-play on zero-sum games as a reinforcement-learning training signal [19]. Its evaluation includes mathematical and knowledge-intensive benchmarks also used by the correlation procedure, and it analyses textual reasoning patterns. The project adopts this benchmark suite with its own benchmark-specific repeated-attempt protocol; it does not claim to reproduce every evaluation setting of the SPIRAL



Figure 2.1: Sentence-level episode transitions in the reference corpus. Reproduced from Li et al. [9], Figure 1, p. 18270 (CC BY 4.0); cropped to the original plot. Rows denote the preceding class and columns the following class. These reference-corpus frequencies motivate our directed-transition analysis.

paper. Self-play changes the training process, whereas the thesis’s intended control mechanism acts on routing during inference.

2.1.1 Regulating reasoning effort

Reasoning length is itself a target of intervention. Chen et al. characterise overthinking through outcome and process efficiency and use self-training to reduce redundant reasoning [20]. Conversely, Wang et al. associate premature switching between thoughts with underthinking and introduce a decoding penalty that encourages pursuing an existing path [21]. Muennighoff et al.’s s1 combines a small supervised reasoning dataset with budget forcing, which can stop reasoning or extend it by suppressing its end and appending a continuation cue [22].

These approaches, also discussed by RICE, make clear why more tokens, more verification language, or more frequent transitions cannot be interpreted as unconditional improvements. In the present analysis, episode composition

and routing are therefore considered alongside token use and termination. Behavioural thought switching must also be distinguished from switching expert identities: the former concerns the text’s strategy, whereas the latter concerns successive routing decisions.

2.2 Prompt optimisation for episode annotation

Prompt search provides a way to improve an annotation instrument without training a new classifier. DSPy formalises language-model procedures and optimises their prompts and demonstrations against a chosen metric [23]. OPRO proposes new instructions from a history of candidate prompts and their evaluation scores [24]. ProTeGi instead generates textual critiques of errors and uses them to guide prompt edits and beam search [25]. These methods supply complementary precedents for treating the annotation prompt as an optimisable part of the measurement procedure.

An automatic episode annotator must handle closely related categories, class imbalance, and contextual ambiguity. GEPA addresses the broader problem of optimising language-model systems through reflective prompt evolution [10]. It uses execution and evaluation traces to generate natural-language feedback, proposes prompt revisions, and maintains candidates with complementary strengths through Pareto-based selection. The optimisation operates with fixed model weights.

The automatic episode annotation experiment applies this approach to the seven-label annotation task. It supplies the problem and neighbouring text units, evaluates predictions against the released annotations, and preserves response boundaries in its data splits. A same-model judge can supply detailed feedback during the search, while agreement with the reference labels and per-class recall constraints govern final selection. These are properties of the project’s experimental design, rather than claims about GEPA’s original evaluation.

The distinction between optimisation feedback and final evaluation is central.

A candidate selected after many comparisons on the same validation responses can exploit peculiarities of that partition. Furthermore, a judge that shares the classifier’s model may share its errors. Response-grouped evaluation and an untouched test partition address a different question from whether the search found a better validation prompt. The resulting prompt is intended as a frozen annotation instrument for subsequent experiments; its offline context and reference-label provenance must remain visible when its labels are used to study routing.

2.3 Probing representations of reasoning

Probing asks whether a chosen decoder can extract a property from a model’s representations. Hewitt and Manning demonstrate that a learned transformation of contextual word representations can recover aspects of syntactic tree geometry [26]. Hewitt and Liang subsequently show how control tasks help distinguish information in representations from a probe’s capacity to learn the target task [11]. Together, these works motivate simple, inspectable probes and evaluation designs that make shortcut learning visible. Their linguistic tasks do not establish that reasoning episodes are encoded in the same way; that is an empirical question for this project.

Voita and Titov evaluate probes through minimum description length, accounting for the effort required to extract labels rather than predictive accuracy alone [27]. Elazar et al.’s amnesic probing removes information from representations and measures the resulting behavioural change [28], directly addressing the distinction between decodable information and information used by a model. At the sentence level, Bogdan et al.’s Thought Anchors estimates the impact of reasoning steps by resampling alternatives and continuing generation [29], providing an intervention-based complement to classifying a step’s semantic function.

Sun et al. study reasoning as trajectories through representation space [30]. Their analysis examines step-specific geometry and correctness signals in a Llama family, and they investigate adaptive activation changes toward reference trajectories. Their step categories describe positions in a sequence, rather than

Schoenfeld semantic functions. This distinction makes their layerwise probing protocol relevant without making the two annotation tasks equivalent. They also report more limited transfer for correctness prediction than for step-related structure.

The prospective probing experiment combines a boundary-probe design with the released semantic episode annotations. It extracts representations immediately before the target unit, which creates a prospective prediction task. Its sentence-split replication is useful for comparison with the adopted protocol, but shared response identities and test-based layer selection limit its generalisation claims. The additional grouped study separates responses and layer selection from outer evaluation. Applying the probes to supplied reference solutions tests representation transfer under teacher forcing. Using them to control the model’s own generation requires additional validation.

Chen et al.’s recent preprint, *Beyond the Trace*, connects internal readouts directly to native MoE routing [31]. They construct J64, a 64-axis semantic frame from reasoning states, and learn a routing-based proxy, R64, from expert usage and gate entropy. They evaluate candidate selection, online stopping and resampling, and targeted router edits. This is a close precedent for connecting process interpretation to routing and test-time decisions. The distinction here is the target vocabulary: the present project starts from externally defined Schoenfeld episodes, whereas J64 is derived from model states. Reconstructing a latent readout, predicting an upcoming episode, and improving an answer through intervention remain different evaluation tasks. Their question-grouped evaluation and frozen cross-benchmark transfer also provide useful methodological comparisons.

This line of work supplies possible detection signals and candidate layers. It does not identify a router intervention by itself. A decoder may read information from a representation that is distributed across many experts, or from a component that the model does not use in the way the decoder does. The transition from decodability to controllability therefore remains a separate experimental step.

2.4 Functional perspectives on expert routing

2.4.1 Characterising expert routing behaviour

Early sparse MoE work establishes conditional computation as a way to scale capacity while limiting active computation [3]. Switch Transformer, Mixtral, and DeepSeekMoE explore different selection and expert-sharing designs [4, 5, 6]. These studies support the architectural motivation for treating routing as a control point. They also caution against assuming that its units already correspond to human-readable reasoning functions. In particular, Mixtral’s analysis of syntactic routing and token locality makes lexical controls relevant when comparing experts across reasoning episodes. OLMoE reports domain and vocabulary specialisation together with early stabilisation of routing during pretraining [32], showing that expert organisation depends on the model and training setting. Lo et al. find that routers often select experts with larger output norms and that expert diversity varies with depth [33], motivating our joint inspection of routing statistics, hidden-state norms, and layers.

Olson et al. provide controlled evidence of semantic routing [34]. They compare expert overlap when a fixed word changes sense across contexts and when a fixed context contains semantically similar or dissimilar replacement words. These controls make their study relevant to separating meaning from lexical identity. However, word-sense sensitivity does not establish that experts specialise in multi-sentence reasoning episodes. For the present analysis, it motivates checking whether episode–routing associations persist beyond differences in the words used to express each episode.

Figure 2.2 also highlights a measurement difference: Olson et al. correct overlap for a uniform random-selection baseline, whereas our temporal overlap measures the fraction of selected experts retained between adjacent tokens. These quantities answer different questions and should not be compared as if they shared a common scale.

Ye et al.’s preprint shifts the unit of interpretation from individual experts to paths across layers [35]. Their decomposition separates a routing control component from an orthogonal component invisible to the router. Across the

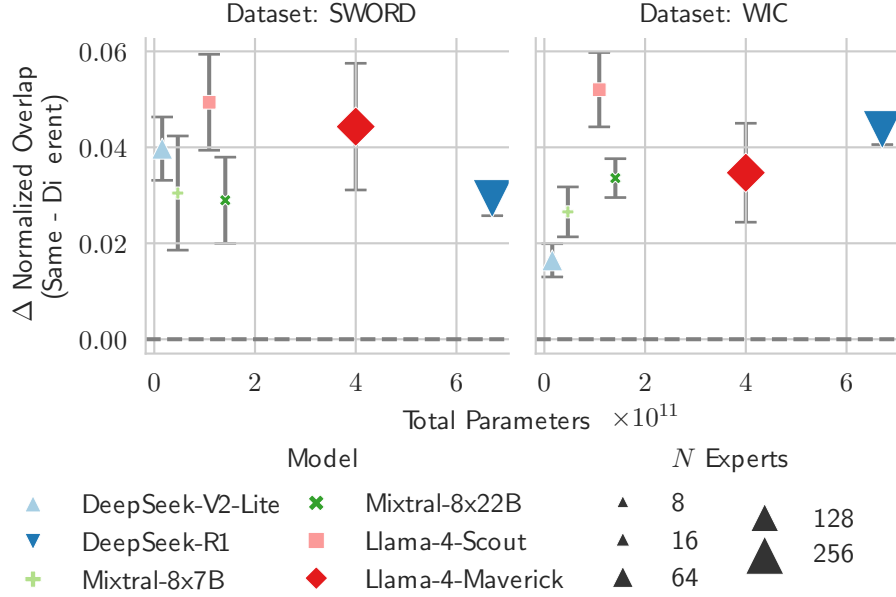


Figure 2.2: Semantic sensitivity of expert overlap across models. Reproduced from Olson et al. [34], Figure 2, p. 18266 (CC BY 4.0); cropped to the original plot. The ordinate is the same-meaning minus different-meaning difference in chance-corrected overlap; SWORDS varies lexical identity and WiC varies word sense. This motivates lexical controls in episode comparisons, without identifying a reasoning function for any individual expert.

architectures studied, they report that individual experts remain polysemantic while expert trajectories group tokens by semantic function across surface forms. This suggests a limitation of per-expert episode correlations: an expert can participate in several functions, with specificity emerging through its combination with other layers. Path-level analysis is consequently a possible extension of the present measurements, rather than a property demonstrated by them.

2.4.2 Inference-time routing control

RICE is a close precedent for reasoning-directed expert intervention [36]. Wang et al. associate experts with reasoning-related marker tokens using nor-

malised pointwise mutual information, then reinforce identified experts during inference. Their reinforcement multiplies an expert’s mixture weight when that expert is already in the selected set; it does not necessarily make an unselected expert participate. The paper’s description of cognitive experts is a working hypothesis. Reported sensitivity to reinforcement strength and normalisation, together with mixed sampled accuracy changes on AIME, makes clear that observed specialisation does not automatically imply a robust intervention benefit.

Table 2.1 reproduces RICE’s sampled evaluation, where pass@1 and pass@8 can move in different directions. This distinction motivates keeping per-attempt correctness, multi-sample coverage, and token use separate in our intervention analysis.

Table 2.1: RICE under stochastic decoding. Adapted from Wang et al. [36], Table 11, p. 24 of the NeurIPS proceedings PDF. Each cell shows baseline $\rightarrow$ RICE; pass@ k values are percentages. The source uses 16 responses per problem, temperature 0.6, and top- p 0.95. Values are published results, not measurements from this thesis.

Model	Benchmark	pass@1	pass@8	Tokens
DeepSeek-R1	AIME24	74.8 $\rightarrow$ 76.0	88.3 $\rightarrow$ 89.2	9,219 $\rightarrow$ 8,317
	AIME25	68.5 $\rightarrow$ 67.7	84.7 $\rightarrow$ 86.3	10,875 $\rightarrow$ 11,441
Qwen3-235B-A22B	AIME24	84.0 $\rightarrow$ 85.0	93.0 $\rightarrow$ 91.6	10,946 $\rightarrow$ 10,706
	AIME25	82.7 $\rightarrow$ 82.1	88.3 $\rightarrow$ 89.7	12,546 $\rightarrow$ 12,373

SteerMoE investigates behaviour-associated expert activation using contrasting inputs, and promotes or suppresses experts by modifying routing scores before top- k selection [37]. Its experiments concern behaviours such as document faithfulness and safety. This provides direct evidence that expert selection can be used as an inference-time control mechanism for the studied settings. Its behavioural targets differ from mathematical episode structure, so the transfer to reasoning-guided routing remains a question. The difference from RICE is also operational: changing scores before selection can change

membership of the active set, whereas reinforcing an already selected expert primarily changes its contribution.

Yoon et al. examine whether the trained router selects favourable expert combinations by comparing standard routes with sampled alternatives at equal expert compute [38]. Their counterfactual analysis uses successful continuations and also studies a final-router update. This is relevant evidence that routing quality can be investigated separately from expert capacity. However, a next-token score on a supplied continuation is a proxy for free-generation success, and layerwise route comparisons do not by themselves identify an episode-level intervention effect.

Su et al. study continuous test-time rerouting under deployment shifts [39]. Their method optimises additive router-logit vectors in selected layers during prefill and at subsequent intervals, using the available context and generated sequence for self-supervision. Generation then continues with the adapted routing until the next update. This offers a comparison with interventions whose expert targets are identified offline: the routing adjustment is optimised online, without external reference data. Its adaptation objective is not an explicit episode vocabulary, so it does not directly test episode-guided control. Comparisons would also need to account for the extra optimisation cost at inference time.

Chen et al.’s Expert-Sample preserves high-confidence expert choices while sampling lower-confidence alternatives to improve multi-sample coverage and verification-based selection [40]. This later work cites RICE but changes the intervention objective: diversity across routes can help pass@ N without necessarily improving the mean correctness of an individual attempt. For our margin experiments, it suggests that ambiguity near the selection boundary may sometimes support useful diversity; increasing the margin is therefore a hypothesis to test, rather than an inherently favourable change.

These studies establish that routing control is an existing research direction. The question developed here is more specific: whether a validated semantic episode vocabulary, prospective internal signals, and outcome-associated routing measurements can jointly support interventions that improve reasoning on held-out problems. Demonstrating that conjunction requires measurements and

evaluations beyond finding experts associated with a few textual markers.

2.5 Activation steering

Inference-Time Intervention shifts activations at selected attention heads toward directions associated with truthful answers, with intervention strength controlling a truthfulness–helpfulness trade-off [41]. Routing edits should also be compared with methods that act directly on representations. Contrastive Activation Addition constructs steering directions from differences between activations on contrasting examples and adds those directions during generation [42]. This changes the residual representation rather than explicitly specifying which experts are selected. It is consequently a useful conceptual baseline for separating the value of a behavioural direction from the value of acting at the router.

Hong et al. study a residual-stream direction associated with the balance between reasoning and memorisation [43]. Their analysis relates scalar projections to graded reasoning scores and includes interventions on the corresponding features. Sun et al.’s trajectory work similarly moves beyond representation analysis to interventions that use reference reasoning trajectories [30]. These papers show why identifying a direction and testing its manipulation belong to distinct stages of a causal investigation.

For an MoE subject, a residual edit may indirectly alter routing as well as other downstream computations. Conversely, a router edit changes the mixture of expert outputs and therefore the residual trajectory that follows. The two intervention families are connected, but their direct targets differ. A routing-focused thesis must specify that target and evaluate its effects under comparable generation conditions. A change in episode labels alone would leave open whether the method improved problem solving or only changed how the response was expressed.

2.6 Statistical evidence for metric selection

The choice of correlation coefficient follows the outcome scale and the scientific question. Liu et al. use point-biserial correlation between continuous quantities, including context length and answer probability, and binary correctness in figurative-language interpretation [12]. This is direct precedent for the attempt-level continuous-feature/0–1-outcome setting. The additional justification is algebraic: point-biserial is Pearson correlation under binary coding, with an interpretable direction given by the group difference.

For graded targets, Hong et al. report Spearman association between residual-stream projections and model-assigned reasoning scores [43]. Hewitt and Manning use rank correlation in evaluating representation-derived syntactic geometry [26]. Xu et al. relate language-model-derived conceptual representations to human behavioural ratings using Spearman correlation and employ bootstrap confidence intervals [44]. These are precedents for rank-based analysis of internal quantities, rather than direct studies of avg@32 correctness. The project’s choice of Spearman for per-problem mean correctness is therefore justified by its monotone association target, not by claiming that these papers uniquely prescribe that coefficient.

Zhang et al. report both Pearson and Spearman correlations when comparing uncertainty-based hallucination scores with graded human judgements [45]. Their use of entropy-related quantities is relevant to this project’s concentration features, although token uncertainty and expert-routing entropy refer to different distributions. Reporting both coefficients can answer distinct linear and monotone questions; it does not turn two analyses of the same data into independent replications.

For the correlation experiment, the sample hierarchy matters as much as the coefficient. Thirty-two attempts from a single problem provide repeated observations of that problem, rather than 32 independent problems. The problem-level analysis and cluster bootstrap described in Section 1.5 address that hierarchy. Xu et al.’s use of 10,000 bootstrap resamples in one analysis provides a precedent for reporting resampling uncertainty, but does not determine an appropriate resample count or sampling unit for a different experiment

[44].

Benjamini and Hochberg’s multiple-testing framework motivates declaring families of feature, layer, and expert comparisons [13]. Multiplicity adjustment cannot remove confounding, repair non-independent tests, or turn a discovered association into a causal effect. The active analysis therefore serves to prioritise hypotheses whose intervention effects must subsequently be tested on separate data.

2.7 Position of our experimental procedure

Table 2.2 relates the active experiments to the evidence required by the thesis objective. Their contributions are complementary: a reliable annotation instrument makes behavioural comparisons possible; a prospective probe tests whether suitable signals are available before a decision; a routing analysis narrows the set of mechanisms worth changing; and guiding experiments evaluate those changes during generation.

Our method connects these stages without requiring an episode-aware controller. The frozen annotator supplies offline behavioural labels, and the probes select measurement locations. Identity and margin policies are then calibrated from routing measurements and correctness, independently of episode labels. This design avoids relying on the transfer accuracy of the probes as online classifiers. Their layer selection remains exploratory under the replicated split and selection protocol.

Generation and measurement provenance impose a further interpretative limit. Quantised generation and replay configurations can differ, so replay measurements need not equal the generator’s internal activations. Guiding policies retain that provenance and are evaluated by changing the actual router used for free generation. Expert-identity calibration also preserves the full (ℓ, e) pair, avoiding the ambiguity of pooled screens that pooled equal numeric expert indices across layers.

The contribution pursued by the thesis is an experimentally grounded connection between semantic reasoning behaviour and controllable sparse computation. Chapter 3 specifies this connection through our four-stage procedure.

Table 2.2: Role of each stage in our routing-intervention procedure. These are research functions, rather than claims of completed causal validation.

Experiment	Evidence sought	Role in intervention research
Episode annotation	Agreement between automatic episode labels and reference annotations on separated responses.	Defines a reproducible behavioural measurement instrument.
Prospective probing	Predictability of upcoming episodes from pre-unit hidden states.	Nominates signals and locations for detecting relevant contexts during generation.
Routing association analysis	Associations of routing and hidden-state features with correctness and reasoning events.	Nominates routing properties and expert-use patterns for controlled tests.
Routing guidance	Paired changes in correctness and generated-token counts on held-out problems.	Tests calibrated expert-identity biases and margin targets during free generation.

Its empirical success is assessed by the effect of frozen routing policies on held-out outcomes and token use.

Chapter 3

Our Method

Our method connects semantic descriptions of reasoning to controlled changes in sparse expert computation. It consists of four stages: optimising an LLM-based episode annotator with GEPA, identifying informative hidden-state locations with probes, measuring associations between routing and reasoning outcomes, and testing routing policies during generation. Model parameters remain frozen throughout. The learned objects are an annotation prompt, linear classifiers, and calibration-derived routing policies.

These stages have distinct roles. The annotator describes the function of text already produced; the probes identify locations where information about an upcoming episode is accessible; the correlation experiment characterises routing on generated solutions; and the guiding experiments test whether changing routing changes performance. Figure 3.1 shows the flow of information. Neither guiding policy requires a predicted episode label at inference time. Episode annotation supplies a behavioural analysis, while the deployed interventions use expert identity or the current router margin.

3.1 Optimising the LLM-as-judge prompt

The first stage operationalises the seven reasoning episodes defined in Section 1.3: Read, Analyze, Plan, Implement, Explore, Verify, and Monitor.

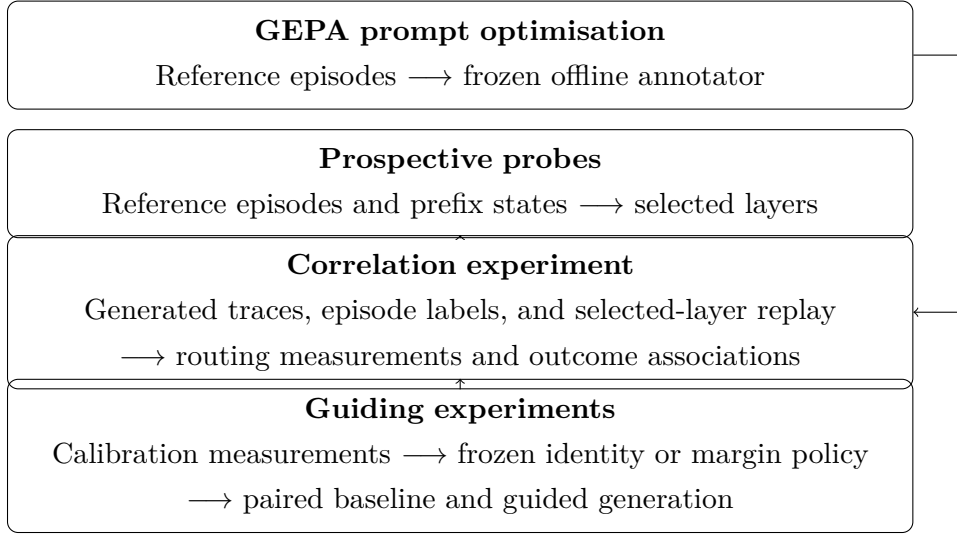


Figure 3.1: Information flow through our method. Annotation and probing use the reference corpus independently. Their outputs meet in the correlation experiment; guiding policies are fitted using calibration problems only.

For problem q and response unit u_j , the annotator receives the context

$$c_j = (q, u_{j-1}, u_j, u_{j+1}), \quad \hat{y}_j = A_\phi(c_j), \quad (3.1)$$

where ϕ is the instruction prompt. The model assigns exactly one label to the current unit. Neighbouring units help distinguish, for example, announcing an operation from performing it, or deriving a result from checking an existing result. The problem statement is visible, but reference answers and rationales are excluded. Since the next unit is available, this is an offline measurement procedure.

GEPA searches over ϕ using performance feedback while leaving the annotator’s weights fixed. The initial context-aware prompt contains the category definitions, boundary rules, and audited synthetic examples covering each category. Complete responses are partitioned before their units are flattened so that neighbouring units cannot cross the annotation training and validation boundaries. Configuration comparison uses nested response-grouped cross-validation; the selected configuration is then fitted to obtain a deployable prompt. The concrete partitions and model settings are specified in Section 4.4.

Two uses of the language model must be distinguished. The episode annotator predicts $\hat{y}_j$ without access to the reference label. During prompt optimisation, a separate call to the same model evaluates that prediction with access to the reference label. This feedback judge supplies an explanation and four scores $s_d \in \{s_{\min}, \dots, s_{\max}\}$: gold agreement, functional fit, contextual coherence, and boundary precision. The reward is

$$R(c_j, \hat{y}_j, y_j) = \sum_{d=1}^4 \omega_d \frac{s_d - s_{\min}}{s_{\max} - s_{\min}}, \quad \omega_d \geq 0, \quad \sum_{d=1}^4 \omega_d = 1. \quad (3.2)$$

Here $s_{\min} < s_{\max}$ delimit the scoring scale and ω_d weights each feedback dimension. Their experimental values are specified in Section 4.4. The resulting score lies in $[0, 1]$. The explanations provide information about category confusions for prompt revision. Exact-match and inverse-frequency weighted rewards provide alternative optimisation conditions.

The feedback score is not the final prompt-selection criterion. Selection compares the candidate and seed prompts against reference annotations, using a prespecified classification measure. A candidate is retained only if it improves that measure and no class recall decreases by more than a prespecified tolerance $\delta \geq 0$. Otherwise, the seed remains the selected prompt. This distinction matters because a same-model judge can reinforce its own preferences; agreement with its feedback is not an independent estimate of annotation accuracy.

The selected instruction is frozen before downstream tagging. Each generated reasoning unit is assigned a label using the same context construction, and its text span is aligned to completion tokens. Unlabelled units remain unlabelled rather than being imputed from probe predictions. Classification quality is assessed through accuracy, balanced accuracy, macro-F1, per-category precision and recall, and valid-output coverage. Changes to the annotation model or reasoning configuration are recorded because a frozen prompt alone does not imply unchanged annotation behaviour.

3.2 Identifying relevant layers with prospective probes

The second stage identifies hidden-state locations informative about the next reasoning episode. It uses the reference annotations directly, without substituting the automatic judge’s labels. The released problem and response are supplied to each frozen subject model through teacher forcing. For a unit whose first token has position b_j , the feature at hidden-state index ℓ is

$$v_j^{(\ell)} = h_{b_j-1}^{(\ell)}. \quad (3.3)$$

This boundary excludes the target unit and all later tokens. Consequently, the probe measures information available in the prefix before the unit is read, rather than classifying its text retrospectively. Synthetic units marking the end of reasoning are removed to avoid learning a structural shortcut.

For each episode c and hidden-state index ℓ , a separate logistic regression predicts the binary target $\mathbb{1}[y_j = c]$:

$$P_{c,\ell}(v_j) = \sigma\left(w_{c,\ell}^\top v_j^{(\ell)} + b_{c,\ell}\right). \quad (3.4)$$

The replication uses raw activations, balanced class weights, and L2 regularisation. The sentence-level split and classifier settings are held constant across models. Accuracy, F1, and AUROC are computed at every index. For each category, the saved protocol selects the index with greatest test accuracy, resolving ties toward the earlier index.

The output used by the correlation experiment is the union of those selected indices that correspond to actual router layers. The resulting measurement locations are reported in Section 4.5. Hidden-state index zero denotes the embedding output; an intermediate index denotes the state entering the decoder block with that index. The final output index has no following router. Indices are therefore filtered for router availability without shifting or renumbering them.

Layer selection narrows the measurement space but does not establish causal importance. Units from the same response occur on both sides of the probe split, and the same test partition is used for selection and evaluation. The

selected scores are therefore exploratory measures of within-corpus decodability. Moreover, a block-input state is not necessarily the vector presented directly to its router after sequence mixing and normalisation. The subsequent experiments preserve this distinction. They use the selected locations for routing measurements, without treating the probes as validated online episode detectors or optimal intervention selectors.

A separate evaluation mode tests generalisation across responses while preserving those original measurement locations. Outer folds hold out complete responses. Within each outer development set, a grouped validation partition selects each target’s layer by AUROC; the selected classifier is refitted on the entire development set before outer testing. This produces fold-specific estimators rather than a new globally selected routing layer. A prefix-position baseline and response-level uncertainty intervals help assess how much signal extends beyond a simple progression-through-trace cue.

3.3 Analysing routing associations

3.3.1 Trace preparation workflow

The correlation stage begins with freely generated benchmark solutions. Each record identifies the dataset, source problem, attempt, generation settings, response, final-answer score, and termination status. Multiple attempts share a source-problem identifier. Correctness is defined by the benchmark answer checker; an unparseable answer to a scored problem is incorrect, while a genuinely unscored task has a missing outcome.

Generation and internal measurement are separate operations. The saved continuation is replayed through the corresponding instrumented model to capture hidden states and router behaviour at the retained layers. The replay procedure preserves the original tokenised prompt and continuation, together with the correspondence between tokens and text positions. This preserves native chat and reasoning markers and permits sentence labels to be associated with the same token positions used for feature extraction.

Generation and replay can use different quantisation configurations of the

same base model. Replay measurements therefore describe the replay model on the saved continuation. Numerical equivalence with the generator is not assumed. The same provenance is carried into policy calibration: transferring an expert identity or a numeric margin range between configurations is an explicit experimental assumption.

The frozen annotator supplies class labels for the tagged analysis. When annotation is restricted to a sampled population, the sampling record specifies which attempts and sentences are covered, as detailed in Section 4.6. Such class statistics describe that selected population. Aggregate routing analyses and analyses of available tagged units retain their own denominators; missing labels do not imply an absence of the corresponding episode. Untagged runs also support separate lexical indicators of backtracking, contradiction, and self-correction, which are not interchangeable with the seven episode labels.

3.3.2 Routing feature aggregation

Let $p_{i,t,e}^{(\ell)}$ be the full router probability for expert e at token t of attempt i , and let $S_{i,t}^{(\ell)}$ be the selected expert set. Our method measures concentration, probability margins, selected mass, selection-boundary gaps, and changes in expert selection across successive tokens. Hidden-state norms, successive-state distances, and hidden/router geometry provide complementary representation measurements. Their definitions are given in Chapter 1. Expert membership is measured anywhere in the native top- k set, not only at rank one. Every intervention target retains the full pair (ℓ, e) , since equal expert numbers at different layers identify different modules.

Features are summarised over the whole continuation, the reasoning span, each available episode class, and absolute-position windows. The position view uses common token boundaries derived from the corpus mean reasoning length, with an overflow window. A short response contributes only to windows it reaches. Class views similarly require observed labelled tokens of the class. Coverage accompanies each estimate so that changes in the analysed population remain visible. Full distributions need not be retained for all analyses: compact scalar summaries and selected expert tensors preserve the measurements

required for correlation and guiding calibration.

3.3.3 Estimating associations with correctness

For a continuous feature f_i and binary correctness Y_i , the attempt-level association is the point-biserial correlation. Positive values mean that the feature is larger in correct attempts. The same convention applies to binary reasoning-event indicators. For repeated-attempt benchmarks, a separate problem-level analysis relates the mean feature to mean correctness using Spearman correlation. With n attempts, this outcome is $\text{avg}@n$, not the probability that at least one attempt succeeds. The primary problem-level analysis requires the complete declared set of unique attempts and complete binary outcomes. Problems containing both correct and incorrect attempts also permit within-problem feature contrasts.

Repeated attempts are not independent observations. For prespecified aggregate associations, uncertainty is estimated by resampling source problems and retaining their attempts together; the attempt-level bootstrap is stratified by dataset. Problem-level Spearman intervals resample complete problem summaries. Benjamini–Hochberg adjustment is applied within the declared testing families. Broad layer, expert, and feature-pair screens remain exploratory, and conventional attempt-level significance tests do not replace uncertainty estimates that account for problem clustering.

The current analysis retains token-limit completions and reports their termination status in a generation-budget audit. Missing or nonfinite measurements are omitted for the relevant feature–outcome pair. This keeps the target population distinct from a completed-response-only analysis. Length, benchmark difficulty, and episode composition can affect both routing and correctness, so an observed association is a candidate for intervention, not evidence that changing the feature will improve an answer.

3.4 Guiding experiments

The final stage tests two policies separately: promoting expert identities associated with correctness and moving router margins toward ranges associated with correctness. Both act on native router outputs during free generation, with expert network parameters unchanged. They preserve the model’s expert count, native top- k dispatch, and shared computation. Policies are fitted separately for each model and cannot exchange expert identities across unrelated models.

3.4.1 Problem-level partitioning

The global workflow partitions source problems within each benchmark into disjoint calibration and evaluation sets. All attempts of a problem remain on the same side. Calibration consumes scored traces with the required routing measurements and records exclusions. Evaluation uses the held-out generation problems irrespective of whether their earlier traces had routing tensors or usable scores. Both correct and incorrect calibration answers are required to estimate an accuracy advantage.

For calibration trace i belonging to problem $q(i)$, let $n_{q(i)}$ be the number of available calibration attempts for that problem and define $w_i = 1/n_{q(i)}$. Each problem then has total weight one. The common baseline for policy fitting is

$$A_0 = \frac{\sum_i w_i Y_i}{\sum_i w_i}. \quad (3.5)$$

This weighting prevents repeated-attempt benchmarks from dominating solely through their number of attempts; datasets with more problems still have more calibration weight. Policies are frozen before evaluation. Any tuning of intervention strength, support thresholds, or binning must use a separate validation partition rather than held-out outcomes.

3.4.2 Expert-identity guiding

Identity guiding uses the frequency with which an expert belongs to the selected pool. For a trace with T_i saved completion positions, define

$$r_{i,\ell,e} = \frac{1}{T_i} \sum_{t=1}^{T_i} \mathbb{1}[e \in S_{i,t}^{(\ell)}], \quad A_{\ell,e} = \frac{\sum_i w_i r_{i,\ell,e} Y_i}{\sum_i w_i r_{i,\ell,e}}. \quad (3.6)$$

The first normalisation prevents long traces from dominating through their number of tokens. $A_{\ell,e}$ is a frequency-weighted correctness estimate, not the success rate of an expert acting in isolation. Its lift is $L_{\ell,e} = A_{\ell,e} - A_0$.

Expert polarity specifies which association is targeted. Positive polarity retains experts with $L_{\ell,e} > 0$; negative polarity retains experts with $L_{\ell,e} < 0$. In either case, eligibility requires occurrence in at least $n_{\min}$ distinct calibration problems. The target budget per layer is bounded by native routing top- k . Eligible experts are retained up to this budget, ordered by decreasing lift magnitude, then problem support, then expert identifier. Let E_ℓ denote this retained set. Scores are normalised magnitudes,

$$v_{\ell,e} = \begin{cases} |L_{\ell,e}| / \max_{j \in E_\ell} |L_{\ell,j}|, & e \in E_\ell, \\ 0, & e \notin E_\ell. \end{cases} \quad (3.7)$$

Layers without eligible experts receive zero scores, and calibration fails if no layer has an eligible target. Thus scores are nonnegative for both polarities: the selected set and the direction of intervention are separate choices.

Fixed additive bias. During generation, the fixed method adds a signed bias to each selected layer’s router logits:

$$\tilde{z}_{t,e}^{(\ell)} = z_{t,e}^{(\ell)} + \alpha v_{\ell,e}, \quad \alpha \in \mathbb{R}. \quad (3.8)$$

Positive strength promotes the selected identities; negative strength suppresses the same identities; zero leaves routing unchanged. In particular, negative strength with positive polarity suppresses success-associated experts, whereas positive strength with negative polarity promotes failure-associated experts. The experimental strengths and polarities are specified in Section 4.7.

Paper-style steering. The second method implements the extreme-score intervention used by SteerMoE, while retaining the calibration estimator above [37]. For each token and selected layer, define the original log-softmax scores $s_{t,e}^{(\ell)} = \log \text{softmax}(z_t^{(\ell)})_e$. For nonzero strength, replace them simultaneously by

$$\tilde{s}_{t,e}^{(\ell)} = \begin{cases} \max_j s_{t,j}^{(\ell)} + \varepsilon, & e \in E_\ell, \alpha = +1, \\ \min_j s_{t,j}^{(\ell)} - \varepsilon, & e \in E_\ell, \alpha = -1, \\ s_{t,e}^{(\ell)}, & e \notin E_\ell. \end{cases} \quad (3.9)$$

The extrema are computed before any replacement, with $\varepsilon > 0$ ensuring strict separation. The adjusted scores determine expert selection at the router’s original numerical precision. If rounding erases the strict separation, the target is moved to the next representable value beyond the original extreme at that precision. All selected experts receive the same target, independently of lift magnitude. This method supports strengths $+1$ and -1 , with zero leaving routing unchanged. Layers with no selected experts are unchanged. When the target set fills all native top- k slots, promotion fixes the selected identities and gives them equal scores after the intervention. A smaller target budget leaves slots available to untargeted experts. This distinction matters when interpreting the strength of the intervention.

Paper-style steering reproduces the intervention rule, not the paper’s contrastive risk-difference expert-selection estimator. Both methods here use our problem-balanced, routing-frequency-weighted accuracy lifts. Native dispatch selects and weights experts from the modified router outputs, so either intervention can change both membership and mixture weights. Neither uses episode labels or waits for a particular margin or reasoning event. Their purpose is to test whether acting on a calibrated association changes held-out performance, without assuming that promotion must help or suppression must hurt.

3.4.3 Margin guiding

Margin guiding instead targets a distributional property without assigning preferred expert identities. Let $p_{(r)}$ denote the probability at descending rank

r before selected-expert renormalisation. One possible metric is the selection-boundary margin

$$m = p_{(k)} - p_{(k+1)}. \quad (3.10)$$

The top-1/top-2 gap is an alternative configuration. For each layer, calibration partitions trace-mean margins into up to B problem-weighted quantile bins, keeping ties together. Among bins supported by at least $n_{\min}$ distinct problems and with accuracy above A_0 , it selects the bin with highest weighted accuracy, breaking ties by support and then lower margin. The observed minimum and maximum define the frozen interval $[a_\ell, b_\ell]$. Layers without an eligible bin are skipped; a policy with no eligible layer is rejected.

At a live token, the desired margin interpolates toward the nearest point of this interval:

$$d = m + \lambda(\min\{b_\ell, \max\{a_\ell, m\}\} - m), \quad 0 \leq \lambda \leq 1. \quad (3.11)$$

Let $r = k$ for the boundary margin, or $r = 1$ for the alternative gap, and let E be the number of routed experts. To decrease the margin, the policy mixes the full probability vector with the uniform distribution u_E :

$$\tilde{p} = (1 - \beta)p + \beta u_E, \quad \beta = \frac{m - d}{m} \quad (d < m). \quad (3.12)$$

To increase it, the policy mixes with a distribution u_r uniform over the current top- r experts and zero elsewhere:

$$\tilde{p} = (1 - \beta)p + \beta u_r, \quad \beta = \frac{d - m}{1/r - m} \quad (d > m). \quad (3.13)$$

If $d = m$, the original logits remain unchanged. Otherwise the adjusted logits are $\log \tilde{p}$, expressed at the original numerical precision; the model then performs its usual expert selection and weighting.

In exact arithmetic these mixtures reach the desired gap and preserve expert ordering away from singular endpoints. Mixture coefficients are bounded strictly below one to avoid fully uniform ties or zero probabilities. Finite precision can still introduce ties and small target errors, so the recorded diagnostics use the achieved margin at the model’s numerical precision. Relative expert weights can change even when top- k membership does not. Learning a range from trace averages and applying it to individual token margins is a specific experimental assumption tested by the generation comparison.

3.4.4 Matched evaluation protocol

Each policy is evaluated against a matched native-routing baseline on identical held-out prompts and attempt identifiers. A baseline is generated for the comparison or reused after compatibility checks on the saved experimental conditions, paired inputs, and recorded responses. The two conditions share the model, tokenised inputs, prompt-formatting conventions, sampling settings, per-attempt seeds, and completion budget. Concrete generation settings are specified in Section 4.7. Repeated benchmarks retain their original attempt counts. Matching seeds pairs the randomness settings but does not force identical sampled trajectories once the distributions change.

Interventions apply during both prompt prefill and decoding at the policy’s selected layers. Native expert dispatch remains responsible for executing the chosen experts. Measurements verify that the intervention acts at every selected layer and record changes in top- k membership and, for margin guiding, achieved margins and distance from the target range. These counts can include padding positions or repeated computations and are not interpreted as independent generated tokens.

The comparison reports final-answer accuracy, generated-token counts, truncations, and paired wrong-to-right and right-to-wrong transitions. All intervention evaluation outcomes Y_i use the common offline answer-equivalence rule in Section 5.4.3. Frozen policies retain their original calibration labels; no policy is refitted using evaluation outcomes. For N matched attempts, the empirical accuracy change is

$$\hat{\Delta}_\pi = \frac{1}{N} \sum_{i=1}^N (Y_i^\pi - Y_i^0) = \frac{N_{\text{wrong} \rightarrow \text{right}} - N_{\text{right} \rightarrow \text{wrong}}}{N}. \quad (3.14)$$

Token counts include reasoning and final-answer text. Benchmark-specific results accompany pooled comparisons, since pooling attempts gives more weight to repeated-attempt benchmarks. Experimental conditions are checked before comparison so that changes in prompts, scoring, numerical execution, or sampling are not silently attributed to the routing policy.

Together, these experiments distinguish a calibrated association from the measured effect of acting on it. Our method specifies the complete path

from reference labels to held-out intervention evaluation; whether either policy improves accuracy or reduces token use is an empirical result, not an assumption of our procedure.

3.5 Evaluating expert reuse with ordered cache traces

This experiment tests whether fixed-identity guiding makes expert requests more reusable under a limited cache. It separates two effects: changing the routing trajectory through guiding, and reserving cache slots for selected identities. Baseline versus guided generation with ordinary least-recently-used (LRU) replacement measures the first contrast; ordinary LRU versus target pinning on the same guided trace measures the second. A frequency-pinned baseline controls for reserving equally many slots using calibration usage.

Ordered router-hook top- k sets cover all 24 OSS, 40 Qwen and 30 Gemma layers. All prefill chunks are excluded, so N returned tokens correspond to $N - 1$ recorded decode steps at every layer. Source hashes, paired inputs, layer coverage and trace lengths are validated. Hook top- k is not an independent observation of native dispatch; tied scores may differ.

Each answer starts with a cold per-layer cache. Initial pinned-expert loads count, even if a pin is never subsequently used. The four conditions are baseline LRU, baseline with calibration-frequency pins, guided LRU, and guided routing with target pins; remaining slots use LRU. Pins are restricted to each model’s policy layers, with equal pin counts for frequency and target controls. Other layers use LRU. Simultaneous top- k requests cannot evict one another; sorted expert IDs break simultaneous recency ties.

OSS uses four of 32 experts and budgets of 8, 16 and 32 slots per layer. Qwen uses eight of 256 and Gemma eight of 128, with budgets of 16, 32 and 64 slots. Capacities are equal across conditions within a model. They do not represent equal bytes or equal fractions of expert capacity across models. Shared experts, dense operations and KV-cache memory are outside the expert-load metric. No SSD reads, transfer bandwidth or latency are measured.

The primary loading outcome is the number of expert loads per complete answer. Loads per recorded decode token provide a complementary locality measure, but a reduction in this rate can be offset by a longer answer. Both are reported alongside paired answer accuracy, output length, and termination. The simulator counts expert loads rather than elapsed time; reduced loading demand is a hypothesis for an offloaded implementation to test, not a direct measurement of speed.

Chapter 4

Experimental Setup

This chapter describes the computing environment, model configurations, datasets, and evaluation protocols used in the experimental work. The three principal studies address complementary questions: whether an automatic annotator can identify reasoning episodes, whether upcoming episodes can be predicted from hidden representations, and whether routing measurements are associated with answer correctness and reasoning events. Model weights remain fixed throughout these studies. The guiding stage additionally fits routing policies on calibration problems and evaluates them on held-out problems, as specified in Section 3.4. The ordered-cache experiment adds a matched evaluation of expert-loading demand under the protocol in Section 3.5.

4.1 Computing infrastructure

Most experiments were conducted on the University of Bologna NLP group’s GPU cluster, principally using one NVIDIA GeForce RTX 5090 with 32 GB of device memory. Annotation experiments also used one NVIDIA GeForce RTX 3090 with 24 GB of device memory [46].

Table 4.1 describes the capacity of the principal experimental node. Annotation and probe experiments requested one GPU and 64 GiB of host memory. Generation, annotation, and activation extraction were performed sequentially to limit simultaneous device-memory use.

Table 4.1: Resources of the principal experimental node. GPU specifications follow the cluster guide; CPU and host-memory capacities are the scheduler values displayed by the dashboard [46, 47].

Resource	Configuration
GPU	One NVIDIA GeForce RTX 5090
GPU memory	32 GB
Scheduler CPU capacity	32 CPU resources
Scheduler host memory	121 GB

The 32 GB device-memory budget influenced the setup directly. Quantised weights made the chosen models practical to load, while long contexts still required space for inference caches and intermediate tensors. The extraction stages therefore retain boundary vectors or compact statistics instead of retaining complete activation tensors.

4.2 Model configuration

The experimental configuration distinguishes the models used for annotation, representation analysis, and generation. Quantisation reduces memory requirements, but numerical differences between model configurations must be considered when interpreting internal measurements.

4.2.1 Model roles

The representation and routing studies use three MoE models: Qwen3.5-35B-A3B, GPT-OSS-20B, and Gemma-4-26B-A4B. All three contribute hidden-state probes, routing measurements, and generation experiments, including the fixed-expert guiding and cache evaluations.

Qwen3.5-35B-A3B has approximately 35 billion total parameters and 3 billion active parameters, 40 layers, and a hidden dimension of 2,048. Each MoE block has 256 routed experts, of which eight are selected per token, together with a shared expert. The backbone alternates Gated DeltaNet and

gated-attention blocks [7].

GPT-OSS-20B has 24 layers and a hidden dimension of 2,880. Each MoE block selects four of 32 routed experts per token. Gemma-4-26B-A4B has 30 layers and a hidden dimension of 2,816, with eight of 128 routed experts selected per token. These differences in expert counts and selection widths enter the interpretation of routing concentration and cache coverage. Across all three models, the experiments use textual problems and continuations; they do not evaluate visual inputs.

The annotator is a separate model. The Qwen3.6 GEPA experiments use Qwen3.6-27B with 4-bit quantisation. The model-alignment study uses Qwen3.8-27B with a different 4-bit quantisation configuration. These annotators supply episode predictions and optimisation feedback; the representation and routing measurements come from the three MoE models listed above.

4.2.2 Model execution configurations

For Qwen, the probe experiment uses the official Qwen3.5-35B-A3B model with GPTQ 4-bit weights. The correlation study uses the same quantised model for generation, while replay applies 4-bit quantisation to the corresponding base model at loading time. The generator and replay model therefore share a base model but differ in quantisation and numerical execution.

GPT-OSS generation uses the original MXFP4 checkpoint. Its probe and replay configurations dequantise those checkpoint values to BF16; this preserves the quantised weight values rather than recovering unquantised weights. Gemma generation uses the NVIDIA NVFP4 checkpoint, while its probes and replay use the corresponding Google instruction-tuned checkpoint loaded with bitsandbytes NF4 quantisation. Thus each model’s internal measurements must be interpreted in the context of its own extraction configuration.

Replay supplies the recorded problem and continuation to a frozen model through teacher forcing, without generating a replacement answer. The resulting routing and hidden-state measurements describe the replay model’s processing of those tokens. They are not direct recordings of the generator’s internal activations, and numerical equivalence between the two configurations is not

assumed.

4.3 Experimental data

4.3.1 Annotated reasoning traces

The annotation and probe studies use the released Schoenfeld reasoning corpus of Li et al. [9]. The corpus used here contains 38 DeepSeek-R1 responses to SAT problems and 3,125 annotated units. Each unit has one of the seven labels introduced in Section 1.3. The SAT problem, answer choices, and available textual descriptions of tables or figures provide context. Reference answers and rationales are excluded from the annotator’s model-visible input.

The annotation study retains all 3,125 units. The probe study removes the 38 synthetic end-of-reasoning units, one per response, all labelled Monitor. Its retained corpus therefore contains 3,087 units. This removes a structural marker that could otherwise provide a shortcut for Monitor classification and matches the sentence count in the source study. Table 4.2 shows the resulting class distribution.

Table 4.2: Episode support in the local corpus. Only the synthetic closing units are removed for the probe experiment.

Episode	Annotation study	Probe study
Read	377	377
Analyze	791	791
Plan	225	225
Implement	675	675
Explore	208	208
Verify	629	629
Monitor	220	182
Total	3,125	3,087

The two studies deliberately use different split protocols. GEPA partitions complete responses before flattening their units. With seed 42, the final-fit

partition contains 26 training, six validation, and six test responses, corresponding to 2,382, 407, and 336 units. The probe replication instead uses a target-stratified 80/20 sentence split with seed 42: 2,469 training units and 618 test units for each binary target. All 38 response identifiers occur in both probe partitions. Consequently, the probe split measures within-corpus predictability and does not provide a response-disjoint generalisation estimate. The additional grouped probe study reuses these 3,087 units but holds entire responses out, with the same five outer folds for all three encoders and all seven targets (Section 4.5.1).

4.3.2 Mathematical reasoning benchmarks

The completed correlation run covers the six mathematical benchmarks in Table 4.3. The dataset selection and repeated-attempt convention follow the project’s adaptation of the SPIRAL evaluation suite [19]. There are 1,547 source problems and 4,647 generated attempts. These are corpus sizes before any analysis-specific filtering.

Table 4.3: Benchmark coverage of the correlation corpus. Counts refer to source problems and generated attempts, not to the number of independent observations in every statistical analysis.

Dataset	Problems	Attempts per problem	Total attempts
MATH-500 [48]	500	1	500
AIME 2024 [49]	30	32	960
AIME 2025 [50]	30	32	960
OlympiadBench [51]	675	1	675
AMC 2023 [52]	40	32	1,280
Minerva Math [53]	272	1	272
Total	1,547	—	4,647

MATH-500, AIME, AMC, and Minerva use their published test partitions. OlympiadBench uses the English subset of open-ended, text-only mathematics competition problems. Although the source release designates this subset as

training data, the present study uses it exclusively for evaluation, without updating the subject model’s weights.

A separate qualitative probe-transfer stage uses 20 examples each from GSM8K, MATH, PRM800K, and ProcessBench. It teacher-forces their supplied solutions or reconstructed rated paths, rather than generating new ones. These 80 examples serve to inspect predicted episode labels and are not included in the correlation benchmark totals.

4.4 Episode-annotation protocol

Each annotation request contains the problem and the previous, current, and next response units. The model returns one episode label for the current unit. The initial context-aware instruction contains 21 audited synthetic demonstrations, three per class, chosen to illustrate ambiguous category boundaries without copying held-out annotations. Since the next unit is visible, this protocol is an offline annotation procedure.

GEPA searches over instructions while keeping the model fixed [10]. The experiments include exact-match and inverse-frequency weighted rewards, followed by an informative same-model judge reward. In the latter configuration, the judge scores gold agreement, functional fit, contextual coherence, and boundary precision from one to five, so $s_{\min} = 1$ and $s_{\max} = 5$ in Equation 3.2. The weights are $(\omega_1, \omega_2, \omega_3, \omega_4) = (0.40, 0.25, 0.15, 0.20)$, respectively, and the weighted score is normalised for optimisation. Gold labels are available to this feedback stage, but are not included in the classifier request.

Prompt selection remains based on reference-label performance. The default selection measure is balanced accuracy; accuracy and macro-F1 are also evaluated. A candidate is retained only if it improves the selected measure and does not lower any class recall by more than $\delta = 0.10$ on validation data. This gate prevents a gain in frequent classes from concealing a substantial loss in a less frequent episode.

Configuration comparisons use five-fold nested response-grouped cross-validation over the 32 development responses, with the six test responses excluded. Each outer fold has its own inner training and validation partition.

The final-fit studies then use the 26/6/6 partition described above. The Qwen3.8 alignment starts from the 40-example prompt selected by the Qwen3.6 fit, uses the heavy GEPA budget and medium reasoning effort, and retains the imported seed after the validation gate rejects the new candidate. The selected instruction is subsequently frozen for downstream annotation.

Reported classification measures include exact accuracy, balanced accuracy, macro-F1, per-class precision and recall, Cohen’s κ , and valid-output coverage. Kendall’s τ_b is recorded among the supplementary statistics but is not a primary measure for the nominal episode categories. The final-fit runs do not evaluate the 336-unit test partition. Moreover, those responses had already been inspected in preliminary experiments, so their designation as a locked test does not make them a newly untouched evaluation set.

4.5 Prospective hidden-state probes

Probe extraction supplies each SAT problem as a user message and its released DeepSeek-R1 response as an assistant continuation. Annotated units are matched verbatim and monotonically to the response before tokenisation. For a unit beginning at token b_j , the stored feature is the hidden state at $b_j - 1$. The target unit and later text are therefore unavailable at the measurement boundary.

For Qwen, all 41 hidden-state indices are retained: index 0 is the embedding output and indices 1–40 are successive block outputs. Each feature has 2,048 dimensions. GPT-OSS supplies 25 indices with 2,880 dimensions, and Gemma supplies 31 indices with 2,816 dimensions. Adapting the layerwise binary-probe protocol of Sun et al. [30] from step-position labels to Schoenfeld episodes, a separate one-versus-rest logistic regression is fitted for each label at each index, giving $7 \times 41 = 287$ classifiers for Qwen, 175 for GPT-OSS, and 217 for Gemma in the sentence-split replication. Training uses raw activations, balanced class weights, L2 regularisation with inverse regularisation strength $C = 1$, and a limited-memory Broyden–Fletcher–Goldfarb–Shanno optimisation method with at most 2,000 iterations. No standardisation or principal-component analysis is applied.

Accuracy, binary F1, and AUROC are computed on the 618-unit test partition. For each target, the selected index maximises accuracy on that same partition. This selection and the shared response identities make the best-index scores exploratory. In the transfer inspection, the seven selected binary classifiers are applied independently. Their scores do not constitute calibrated multiclass probabilities; choosing their maximum produces an inspection label rather than a reference annotation.

Table 4.4 records the probe-selected indices retained for routing extraction. Qwen’s Explore selection at hidden-state index 40 is omitted because it has no following router.

Table 4.4: Probe-derived router locations used in the experiments. Indices are zero-based decoder indices and are specific to each model’s probe run.

Subject model	Retained router indices
Qwen3.5-35B-A3B	27, 34, 35, 36, 38, 39
GPT-OSS-20B	15, 19, 21, 23
Gemma-4-26B-A4B	16, 19, 21, 22, 25, 27, 29

Only boundary activations are retained. For Qwen, at 32-bit floating-point precision, the $3,087 \times 41 \times 2,048$ values require approximately 0.97 GiB. This choice retains the information required for refitting probes without preserving every token’s full hidden-state trajectory.

4.5.1 Response-grouped evaluation

The additional evaluation uses the same recorded activations and partitions complete responses into five outer folds, approximately balancing the seven episode classes (seed 42). Unit texts, labels, and response assignments are identical across encoders. Outer test folds contain 8, 8, 7, 8, and 7 responses; each response is tested once. Within each development fold, a random partition at the response level reserves 20% of responses for validation, using seed $42 + f$ for fold $f \in \{1, \dots, 5\}$. This leaves 24 fit responses in every fold and six or seven validation responses. Every partition contains both classes for every

binary target.

For each target, all hidden-state indices are fitted on the inner fit set. The index with highest validation AUROC is selected, breaking ties toward the lowest index. That classifier is then refitted on all 30 or 31 outer development responses. The untouched test responses determine the final AUROC, balanced accuracy, binary F1, and accuracy. Regularisation and the fixed 2,000-iteration budget match the replication; there is no search over thresholds or hyperparameters. The position baseline uses the same logistic regression with two features: $\log(1 + j)$ for the number j of preceding annotated units, and $\log(1 + t)$ for the number t of preceding response tokens. It excludes total length and future content. Token counts depend on each encoder’s tokenizer, so the three baselines need not be identical.

The primary summary is the unweighted mean of five outer-fold metrics, followed by an equal average across the seven targets. Units have equal weight within each fold metric; response grouping governs separation and uncertainty rather than assigning equal sentence-metric weight to responses. The 95% percentile intervals use 5,000 draws (seed 42), resampling whole responses with replacement independently within each test fold. The same draws pair probe and position predictions and all targets. Draws without both binary classes are excluded for AUROC and balanced accuracy, with valid counts retained. These intervals condition on the saved folds, selected layers, and fitted models; they omit refitting and split-seed variation and are not adjusted for multiple comparisons. Differences from the legacy scores change both splitting and selection criterion and do not isolate a leakage effect. External-domain transfer and online generation remain separate evaluation questions.

4.6 Benchmark measurement procedure

4.6.1 Generation settings

Benchmark solutions are generated zero-shot at temperature 0.6 and top- $p = 0.95$, with a completion budget of 32,768 tokens. The generation configuration provides a 49,152-token context, leaving room for the input prompt in addition

to the completion. The default sampling seed is zero, with the per-request seed formed as $s_{i,a} = s_0 + 1000i + a$, where i is the problem index within a dataset and a is the attempt index. The saved trace records the actual seed and sampling settings.

The Qwen generation configuration permits up to eight simultaneous sequences and processes at most 8,192 tokens per batch. Previously computed prompt representations may be reused. Its multi-token prediction proposes three speculative tokens at a time; the saved GPT-OSS and Gemma generation manifests specify no draft model. These settings govern computational scheduling rather than the number of attempts per problem; long contexts can reduce the number of sequences processed simultaneously. The principal configuration uses one GPU with a memory utilisation limit of 90%.

4.6.2 Replay extraction at selected layers

After generation, recorded continuations are replayed through the corresponding frozen model. Extraction retains the model-specific decoder indices in Table 4.4: six locations for Qwen, four for GPT-OSS, and seven for Gemma. Each set is the compatible union of indices selected by that model’s probe experiment. Qwen’s final probe index 40 is excluded because it has no following decoder router.

The indexing convention is significant. A probe hidden-state index denotes a block output, while replay pairs the block-input hidden state at the retained decoder index with that block’s router distribution. The hidden vector is therefore not the exact normalised router input after the intervening sequence-mixing computation. Probe-based layer selection is a way to restrict measurement locations, not evidence that these locations are optimal intervention targets.

At each retained layer, extraction computes the routing and representation summaries defined in Chapter 1: entropy-derived concentration, top-1/top-2 margin, selected top- k probability mass, selection-boundary margin, selected-versus-unselected mean probability gap, top-1 switching, top- k set overlap, hidden-state norm, successive-state distance, and hidden/router geometry. Here $k = 8$ for Qwen and Gemma and $k = 4$ for GPT-OSS. Geometry calculations

use at most 128 uniformly sampled tokens per trace by default. Full hidden and router tensors are reduced during extraction; compact scalar features and selected expert identities are retained. Original layer indices remain attached to the measurements.

The correlation analysis includes whole-continuation, full-reasoning, class-specific, and absolute-position views. Position analysis uses ten disjoint windows defined relative to the corpus mean reasoning length, followed by an overflow window. These are common absolute token windows, rather than ten equal fractions of each response. A shorter trace has no observation in a window it never reaches. Coverage is consequently reported alongside each view. Class-specific views use available sentence annotations. Full-reasoning and position views use all reasoning tokens independently of annotation coverage.

4.6.3 Sentence annotation

The frozen GEPA annotator labels generated reasoning using Qwen3.8-27B with NVFP4 quantisation, temperature zero, low reasoning effort, a 4,096-token completion budget, and a 32,768-token context. Attention states are retained at 8-bit floating-point precision, and multi-token prediction is disabled. This configuration differs from the quantised model and medium-effort setting of the alignment study, so agreement measured there is not automatically an estimate of downstream annotation accuracy.

Before tagging, the Qwen sampling protocol selects attempt zero for repeated problems and at most 100,000 sentences across benchmarks. Benchmark quotas are proportional to their observed error rates; sentences are sampled uniformly within each benchmark with seed 42. The sampling record identifies the selected units. This outcome-dependent sample has a different population from the full repeated-attempt corpus. Class-specific analyses produced from it therefore describe the labelled subset rather than the full corpus.

4.7 Guiding configuration

The guiding experiments instantiate the policies defined in Section 3.4. Source problems are split within each benchmark into 70% calibration and 30% evaluation using seed 42, with all attempts of a problem assigned to the same partition.

Expert-identity calibration requires support from at least $n_{\min} = 4$ distinct problems. The default target budget per layer equals native top- k : eight of 256 routed experts for Qwen, four of 32 for GPT-OSS, and eight of 128 for Gemma. Smaller target budgets are supported. The completed Qwen fixed-bias evaluations use strengths $\alpha \in \{+1, +2, -1\}$ with positive polarity. The policy also permits negative polarity. Paper-style steering uses $\varepsilon = 0.01$; its strengths are restricted to $+1$ and -1 , with zero leaving routing unchanged.

Margin calibration uses at most $B = 5$ problem-weighted quantile bins and requires at least $n_{\min} = 4$ distinct problems per eligible bin. The default metric is the selection-boundary margin; the top-1/top-2 gap is an alternative configuration.

The default stochastic evaluation uses temperature 0.6, top- $p = 0.95$, and a 32,768-token completion budget. Repeated benchmarks retain their original attempt counts. The Qwen and GPT-OSS margin comparisons at concurrency 25 reuse compatible saved baselines after the checks described in Section 3.4. Interventions apply during both prefill and decoding. Previously computed prompt states are not reused, and speculative decoding is disabled, ensuring that the intervention is applied throughout each evaluated sequence.

4.8 Ordered cache evaluation configuration

The completed cache study compares GPT-OSS, Qwen and Gemma on the same 100 held-out source problems, with one baseline and one fixed positive bias- $+1$ generation per model and problem. Source-problem identities match across all three cohorts. The deterministic selection contains 9 AIME24, 9 AIME25, 12 AMC23, 24 MATH-500, 23 Minerva and 23 OlympiadBench problems; smaller held-out pools are exhausted before the larger pools. Each

model retains its own frozen calibration policy. Per-model paired prompts, seeds and generation settings match. Generation uses temperature 0.6, top- p 0.95, a 32,768-token output limit, and synchronous single-sequence execution. These are separate generations from the 1,395-attempt detailed runs and do not constitute independent seed replications of those runs.

4.9 Outcome analysis protocol

Mathematical answers are assessed for symbolic equivalence. When symbolic comparison is unavailable, the evaluation uses exact comparison of normalised text or numerical values and records the criterion applied. For an attempt with a known reference answer, an unparseable response is counted as incorrect in the all-attempt evaluation. This denominator is distinct from an accuracy calculated only over successfully parsed answers.

For benchmarks with one attempt per problem, the reported score is empirical pass@1. For the repeated AIME and AMC evaluations, it is avg@32: the mean correctness over 32 attempts for each problem, averaged over problems. It does not measure whether at least one of the 32 attempts succeeds. A pooled attempt-level score gives more weight to benchmarks with more attempts, so benchmark-specific scores remain necessary for interpretation.

Continuous features are related to binary correctness or event indicators using point-biserial correlation. At the problem level, mean features are related to avg@32 using Spearman’s rank correlation. The primary repeated- problem analysis requires the full set of unique attempt identifiers, complete binary outcomes, and verification that each generated attempt is included only once. Problems with both correct and incorrect attempts also support a paired correct-minus-incorrect feature contrast.

The saved analysis uses 500 bootstrap resamples and seed 42 for 95% percentile intervals. Resampling occurs at the source-problem level, keeping repeated attempts together. Benjamini–Hochberg adjustment is applied within the declared testing families [13]; layerwise, feature-pair, and expert-use screens remain exploratory. The initial expert-use screen pools equal numeric expert indices across retained layers. It therefore cannot identify a particular layer-

specific expert module without a subsequent analysis preserving the full (ℓ, e) identity.

Token-budget handling. The analysis retains supplied token-limit completions and records their termination status in a generation-budget audit. Missing or nonfinite features are omitted only for the relevant analysis pair. Problem-level avg@32 requires a complete set of attempts and binary outcomes. Estimates remain conditional on the configured generation budget.

4.10 Reproducibility of the experimental protocol

Reproducibility requires an explicit connection between each result and its experimental conditions. The study records data partitions, random seeds, selected annotation instructions, optimisation feedback, classifier parameters, model configurations, measurement locations, and the number of observations included in each analysis. Generation, annotation, and replay are documented separately because they may use different model configurations.

The response-grouped probe evaluation retains fold assignments, validation measurements, fitted classifiers, and one held-out prediction per unit and target. These records distinguish model selection from final evaluation and allow the reported aggregate statistics to be reconstructed.

The available documentation does not uniquely determine every numerical component of the experimental environment. Exact numerical reproduction therefore requires matching the original model versions, quantisation settings, and inference conditions. Random seeds alone do not guarantee identical activations or generations across different configurations.

For guiding, calibration partitions, selected policies, paired generations, and routing measurements establish the provenance of each comparison. The split, calibration, intervention, and evaluation rules are specified in Section 3.4. Verification that an intervention changes routing establishes that the treatment was applied; evidence of improved accuracy requires the corresponding held-out

comparison.

Chapter 5

Results

This chapter follows the four stages of our experimental procedure. The annotation and probing experiments assess the available measurements of reasoning episodes. The observational analysis then examines routing and correctness on generated mathematical solutions. Finally, matched generation experiments test whether acting on selected routing measurements changes performance. These stages provide different kinds of evidence: agreement with an annotation corpus, accessibility of information in representations, association with an outcome, and the effect of an implemented intervention. They are interpreted separately before drawing conclusions about guidance.

Unless stated otherwise, intervention results use stochastic decoding. Complete matched comparisons and the interrupted Qwen paper-style experiment are reported separately; missing guided attempts never enter the partial-result denominator. The analysis includes all completed margin comparisons available and uses existing generations without additional model inference. Intervention tables use the consistent offline scoring rule in Section 5.4.3, while calibration and observational analyses retain their original labels. The qualitative inspection keeps its original sampled attempts and historical outcome frame.

5.1 Automatic annotation of reasoning episodes

5.1.1 Prompt optimisation results

The judge experiments show that prompt optimisation can improve agreement on a development corpus, but the improvement depends on the starting prompt and selection protocol. The preliminary response-grouped experiments gave 66.07% test accuracy for an optimised base instruction and 69.94% for the matched few-shot condition. These runs preceded the context-aware protocol and used per-unit exact match. They therefore motivate the use of examples, without establishing the performance of the final annotator on generated benchmark traces.

The later nested cross-validation study evaluated 2,789 units from 32 responses. Macro-F1 selection with a light optimisation budget achieved accuracy 0.6734, balanced accuracy 0.6796, and macro-F1 0.6505; the heavy budget achieved 0.6669, 0.6737, and 0.6437. Increasing search effort did not improve this comparison. The class-recall gate retained the seed in two of five light-budget folds and three of five heavy-budget folds. Since the outer folds evaluated the selected prompt rather than both seed and candidate, these results describe the selection procedure and do not directly estimate the paired outer-fold benefit of optimisation.

The completed Qwen3.6 context-aware final fit used 407 development units from six responses. Its same-model feedback combined gold agreement with functional fit, contextual coherence, and boundary precision. As shown in Table 5.1, exact accuracy increased from 0.6683 to 0.7543, with corresponding improvements in balanced accuracy and macro-F1. The candidate corrected 49 seed errors and introduced 14 new errors, a net gain of 35 units. This is a substantial development-set change, but selection repeatedly used the same validation responses; it is not a fresh test estimate.

The final Qwen3.8 alignment started from the previously optimised, 40-example instruction. Its new candidate reduced accuracy from 0.7174 to 0.7125 and balanced accuracy from 0.7400 to 0.7288. Plan recall fell from 0.7500

Table 5.1: Context-aware final fits on 407 reused development units. The Qwen3.8 seed is the earlier optimised Qwen3.6 prompt; these are successive development stages, not independent model comparisons.

Judge	Prompt	Accuracy	Balanced acc.	Macro-F1
Qwen3.6	Seed	0.6683	0.6765	0.6132
Qwen3.6	GEPA candidate	0.7543	0.7769	0.7270
Qwen3.8	Seed	0.7174	0.7400	0.6938
Qwen3.8	GEPA candidate	0.7125	0.7288	0.6743

to 0.5625, exceeding the allowed drop of 0.10. The selection gate therefore retained the imported seed. The internal optimisation reward increased even though exact-label performance declined, illustrating why that reward cannot substitute for independently specified reporting metrics. The stronger Qwen3.6 final-fit numbers and the Qwen3.8 numbers belong to successive deployments and prompt-development stages; they are not a controlled comparison of model capabilities.

The final-fit evaluation leaves the 336-unit locked partition unevaluated. Its responses had nevertheless been inspected during earlier experiments. It cannot serve as a genuinely unseen final assessment merely because it is excluded from subsequent evaluation. Moreover, success on the released episode corpus does not establish label quality on long solutions sampled from the three benchmark generators.

5.1.2 Annotation behaviour at deployment

The frozen Qwen3.8 NVFP4 deployment was checked on the same 407 development units under low and medium reasoning effort. Both settings achieved accuracy 0.7101, while balanced accuracy and macro-F1 differed slightly (Table 5.2). Approximately 10.1% of labels changed between settings. Equal accuracy therefore masks disagreement about individual sentences, which matters when those labels define the tokens used in a class-conditioned routing analysis.

Table 5.2: Frozen Qwen3.8 NVFP4 judge on the same 407 development units. Equal aggregate accuracy does not imply identical labels.

Effort	Accuracy	Balanced acc.	Macro-F1
Low	0.7101	0.7415	0.6863
Medium	0.7101	0.7444	0.6832

For the generated benchmark corpus, the refreshed Gemma and GPT-OSS annotations use a common stratified procedure: 100,000 selected sentences per model, with seed 42 and quotas based on dataset, saved correctness, and reasoning-length quartile. One completion per source problem is eligible for annotation. GPT-OSS has 100,000 valid labels on 1,510 problems; Gemma has 99,999 on 1,503 problems, with one missing judge response. The missing sentence is excluded from class features while its trace remains partially labelled. The four exported parts are pieces of one sample, not independent replicates.

Coverage extends to all six benchmarks, but equal sentence budgets do not produce equal cohorts. For example, OlympiadBench contributes 54,556 valid Gemma labels and 70,215 GPT-OSS labels. Qwen retains a separate annotation sampling protocol and deterministic judge deployment. Class proportions and class-conditioned correlations must therefore be read with their actual model-specific denominators. None of these annotations has an independent human accuracy estimate on the generated mathematical solutions. The annotation stage supplies an explicit, inspectable measurement instrument, with unresolved transfer and boundary errors.

5.2 Prospective episode information in hidden states

5.2.1 Within-corpus decodability

All three encoders were evaluated on the same 3,087 labelled units, with 2,469 training and 618 test units. The probes use the hidden state of the prefix

immediately before the labelled unit, rather than a representation computed after reading it. Table 5.3 reports each target’s accuracy-selected hidden-state index and the corresponding F1 and AUROC. The indices denote hidden-state outputs and should not be equated directly with decoder-block indices without the extraction convention described in Chapter 3.

Table 5.3: Prospective boundary probes on the same sentence-level split. Each model reports the accuracy-selected hidden-state index, F1, and AUROC. Selection uses the evaluation split; these are within-corpus estimates.

Target	Index	Qwen		Index	GPT-OSS		Index	Gemma	
		F1	AUROC		F1	AUROC		F1	AUROC
Read	34	0.704	0.934	19	0.686	0.906	21	0.675	0.898
Analyze	36	0.647	0.835	21	0.540	0.767	27	0.611	0.816
Plan	39	0.485	0.855	15	0.434	0.852	25	0.500	0.907
Implement	38	0.771	0.930	21	0.745	0.905	29	0.735	0.922
Explore	40	0.489	0.888	23	0.405	0.842	16	0.447	0.835
Verify	27	0.726	0.912	21	0.667	0.880	19	0.603	0.872
Monitor	35	0.482	0.903	15	0.344	0.774	22	0.382	0.831

There is ranking information for every target in every model: selected-layer AUROC ranges from 0.835 to 0.934 for Qwen, 0.767 to 0.906 for GPT-OSS, and 0.816 to 0.922 for Gemma. Implement has the largest F1 in all three models, at 0.771, 0.745, and 0.735, respectively. Read and Verify are also more accessible at the evaluated decision threshold than several of the rarer classes. Monitor illustrates the difference between ranking and thresholded classification: Qwen reaches AUROC 0.903 but F1 only 0.482, while GPT-OSS has F1 0.344. A high majority-class accuracy is particularly uninformative for these imbalanced one-versus-rest tasks.

The estimates have two sources of optimism. All 38 responses appear in both sentence partitions, so test sentences share response context with training sentences. In addition, the reported index is selected by accuracy on the same test partition used for evaluation. The table consequently supports prospective within-corpus decodability, not response-held-out prediction. It also does not establish that the selected hidden-state indices are the best sites for a causal

intervention. The probe supplies a measurement-location heuristic rather than a validated online controller.

5.2.2 Probe validation on held-out responses

The additional grouped study tests the same representations on complete responses that are absent from both fitting and layer selection. Table 5.4 reports macro-AUROC 0.852 (95% interval [0.838,0.868]) for Qwen, 0.780 ([0.761,0.797]) for GPT-OSS, and 0.788 ([0.772,0.805]) for Gemma. The corresponding legacy means are 0.894, 0.847, and 0.869. These differences change both the split and the layer-selection criterion, so they cannot be attributed solely to response overlap. The grouped results nevertheless establish a stronger claim than the original table: upcoming-label information remains accessible on held-out responses within the released SAT corpus.

Table 5.4: Response-grouped prospective probes on 38 responses and 3,087 units. AUROC is averaged equally across five outer folds and seven targets. Brackets are 95% response-bootstrap intervals (5,000 draws), conditional on saved folds, selected layers, and fits. The paired gain subtracts the prefix-position baseline. Legacy scores select layers by sentence-test accuracy, so the two protocols differ in both splitting and selection.

Model	Legacy AUROC	Grouped AUROC	Position AUROC	Paired gain
Qwen	0.894	0.852 [0.838, 0.868]	0.566 [0.552, 0.594]	+0.286 [0.259, 0.302]
GPT-OSS	0.847	0.780 [0.761, 0.797]	0.571 [0.557, 0.600]	+0.209 [0.179, 0.227]
Gemma	0.869	0.788 [0.772, 0.805]	0.566 [0.553, 0.594]	+0.221 [0.195, 0.237]

The prefix-position baseline reaches macro-AUROC 0.566, 0.571, and 0.566, respectively. Each encoder exceeds its baseline, and all 21 per-target paired AUROC intervals exclude zero; these intervals are not adjusted for multiple comparisons (Table 5.5). Figure 5.1 shows both predictions with response-bootstrap uncertainty. Thus the signal exceeds this particular two-feature positional rule. The control does not exclude more flexible position effects, problem-type or formatting shortcuts, or correlations between episodes and lexical prefix content.

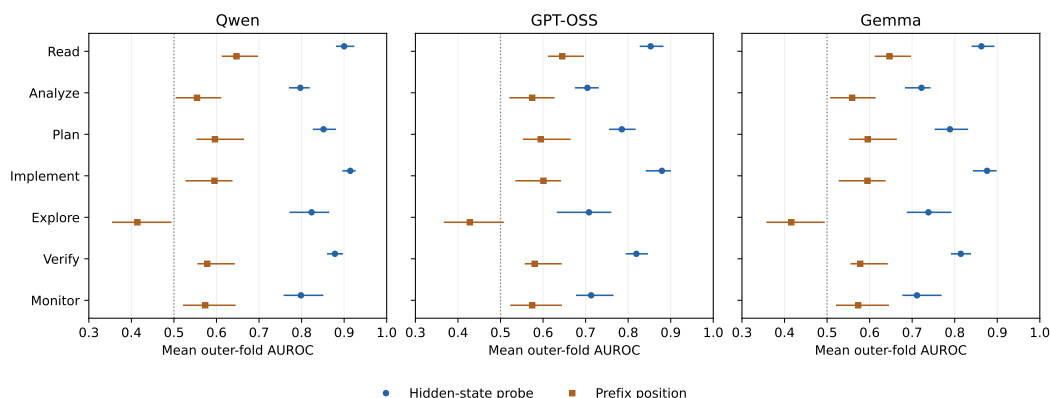


Figure 5.1: Prospective AUROC on response-disjoint outer test folds. Points are means of five fold scores and horizontal bars are 95% conditional response-bootstrap intervals. Both the hidden-state probe and the simple prefix-position baseline are fitted without the test responses.

Ranking performance still does not imply reliable seven-class annotation. Macro binary F1 is 0.521 for Qwen, 0.400 for GPT-OSS, and 0.426 for Gemma at the unchanged decision rule. In Qwen, Explore reaches AUROC 0.823 but F1 only 0.341; Monitor reaches 0.799 and 0.355. The selected layers also vary across folds: Qwen Read uses indices 23–36 and Implement 18–32. This variability argues against interpreting one selected index as a unique reasoning location. The study intentionally produces fold-specific probes; the existing routing experiments retain their original measurement layers.

Across the five folds there are 1,505 Qwen, 945 GPT-OSS, and 1,155 Gemma fits, including 35 selected-layer refits and 35 position baselines per model. The numbers of inner candidate fits reaching the fixed iteration limit are 0, 2, and 2, respectively; the selected-layer refits have 0, 0, and 0 convergence warnings. All position fits converge, and all 5,000 bootstrap draws retain binary support for every reported target. Diagnostics and indices are retained rather than used to change the solver budget after evaluation.

The remaining inferential limits are the small number of responses and the scope of the test. The intervals condition on this single split plan, selected layers, and fitted classifiers; they omit refitting and split-seed variation. Same-corpus response generalisation does not establish transfer to new benchmarks,

Table 5.5: Response-grouped results by target. AUROC and the paired AUROC gain over the position baseline include 95% conditional response-bootstrap intervals. Balanced accuracy (BA) and binary F1 are mean-fold point estimates at the unchanged decision threshold; their intervals are recorded separately. Indices list the validation-selected hidden states in outer-fold order, not a globally selected layer.

Model	Target	AUROC	Gain over position	BA	F1	Indices (folds 1–5)
Qwen	Read	0.900 [0.881, 0.924]	+0.253 [0.209, 0.288]	0.805	0.599	34, 29, 36, 32, 23
Qwen	Analyze	0.797 [0.770, 0.819]	+0.243 [0.178, 0.296]	0.725	0.586	22, 26, 35, 22, 22
Qwen	Plan	0.852 [0.826, 0.881]	+0.255 [0.196, 0.292]	0.754	0.440	22, 29, 29, 29, 33
Qwen	Implement	0.914 [0.896, 0.927]	+0.319 [0.277, 0.385]	0.825	0.696	32, 31, 30, 19, 18
Qwen	Explore	0.823 [0.772, 0.865]	+0.409 [0.323, 0.478]	0.674	0.341	35, 31, 29, 34, 24
Qwen	Verify	0.878 [0.860, 0.897]	+0.300 [0.232, 0.325]	0.798	0.634	24, 24, 24, 20, 15
Qwen	Monitor	0.799 [0.758, 0.851]	+0.225 [0.156, 0.292]	0.671	0.355	21, 36, 22, 34, 31
GPT-OSS	Read	0.853 [0.827, 0.883]	+0.208 [0.164, 0.241]	0.724	0.528	21, 18, 17, 16, 13
GPT-OSS	Analyze	0.704 [0.675, 0.731]	+0.130 [0.072, 0.185]	0.609	0.416	16, 20, 17, 20, 20
GPT-OSS	Plan	0.785 [0.755, 0.818]	+0.191 [0.118, 0.232]	0.619	0.286	20, 18, 20, 23, 7
GPT-OSS	Implement	0.879 [0.841, 0.901]	+0.278 [0.233, 0.335]	0.768	0.640	16, 20, 18, 21, 18
GPT-OSS	Explore	0.708 [0.632, 0.760]	+0.280 [0.155, 0.369]	0.561	0.178	23, 22, 19, 11, 22
GPT-OSS	Verify	0.819 [0.794, 0.847]	+0.239 [0.174, 0.264]	0.699	0.517	16, 18, 18, 17, 16
GPT-OSS	Monitor	0.713 [0.678, 0.766]	+0.139 [0.078, 0.211]	0.589	0.233	16, 9, 21, 13, 22
Gemma	Read	0.863 [0.840, 0.893]	+0.216 [0.174, 0.253]	0.747	0.547	16, 20, 16, 19, 21
Gemma	Analyze	0.722 [0.683, 0.743]	+0.163 [0.088, 0.215]	0.653	0.488	6, 20, 20, 23, 22
Gemma	Plan	0.789 [0.753, 0.832]	+0.193 [0.129, 0.241]	0.655	0.337	19, 29, 16, 23, 16
Gemma	Implement	0.876 [0.843, 0.898]	+0.281 [0.244, 0.337]	0.779	0.644	24, 29, 20, 23, 25
Gemma	Explore	0.738 [0.687, 0.792]	+0.322 [0.229, 0.402]	0.572	0.185	17, 19, 17, 29, 19
Gemma	Verify	0.814 [0.791, 0.838]	+0.236 [0.167, 0.262]	0.725	0.543	25, 15, 28, 16, 15
Gemma	Monitor	0.711 [0.677, 0.769]	+0.138 [0.070, 0.217]	0.593	0.240	28, 0, 29, 25, 25

annotation validity on the encoders’ own generations, or a causal role for the probed directions. The legacy external Qwen inspection below addresses a different population and decision rule.

5.2.3 External inspection of the Qwen probes

The external inspection covers 80 benchmark examples and 581 sentence boundaries. The seven-way argmax display assigns 286 units to Implement,

126 to Read, and 89 to Monitor, while Explore is never selected. Independent binary probes do not automatically yield calibrated multiclass probabilities: 74 units have no score above 0.5, whereas 180 exceed that threshold for more than one class. Taking an argmax still produces one label in both cases.

A subsequent rule-assisted pragmatic review provides a provisional reference, not independent human consensus or benchmark gold. The probes agree with its primary label on 292/581 units (50.3%) and with any accepted label on 353/581 (60.8%). The difference reflects 187 units for which the review accepts overlapping functions. Implement has the strongest accepted-membership F1, 0.729, while Explore has zero recall. These results identify useful qualitative structure and substantial transfer limitations. They do not validate the probes as a replacement for the contextual sentence judge, and the external review provides no direct evaluation of the Gemma or GPT-OSS probes.

5.3 Routing associations in generated solutions

5.3.1 Observational corpus characteristics

Each model’s observational corpus contains 4,647 attempts from 1,547 source problems. MATH-500, OlympiadBench, and Minerva contribute 500, 675, and 272 single-attempt problems. AIME24 and AIME25 each contribute 30 problems with 32 attempts, and AMC23 contributes 40 with 32 attempts. Consequently, the pooled attempt denominator weights each repeated-attempt problem 32 times as heavily as a single-attempt problem. Benchmark-specific results are necessary even when the aggregate population is complete.

The accuracy denominator includes missing or invalid extracted answers when gold is known. For Qwen, 3,568 of 4,647 attempts are correct. Of the 412 token-limit completions, only two are correct; accuracy among the remaining 4,235 attempts is 0.842. Excluding limit hits would therefore substantially change the population and apparent accuracy. Likewise, accuracy restricted to parseable answers would answer a different question. The retained failures make length and termination central to the interpretation of routing–correctness associations.

Table 5.6: All-attempt accuracy in the observational generation corpora. Invalid answers with known gold are counted as incorrect, and token-limit completions remain included. These are generation populations, not the held-out guiding baselines.

Benchmark	Attempts	Qwen	Gemma	GPT-OSS
MATH-500	500	0.690	0.706	0.664
AIME24	960	0.851	0.887	0.709
AIME25	960	0.805	0.824	0.696
OlympiadBench	675	0.477	0.517	0.422
AMC23	1,280	0.967	0.976	0.900
Minerva	272	0.268	0.301	0.114

5.3.2 Routing metric associations with correctness

Table 5.7 shows that there is no common sign for the association between router confidence and correctness. Qwen’s coefficients are positive on all six benchmarks. Gemma is negative on AMC23 and Minerva; GPT-OSS is negative on MATH-500, AIME25, AMC23, and Minerva. Confidence summarises the concentration of the router distribution, not the probability that the mathematical answer is correct. Different model architectures, quantisation settings, and distributions of generated solutions can change its association with success.

The Qwen pointwise 95% source-problem bootstrap intervals exclude zero for confidence on five benchmarks. AIME24 gives $r = 0.533$ with interval $[0.382, 0.678]$, while AIME25 gives $r = 0.493$ with $[0.302, 0.608]$. AMC23 has $r = 0.175$ with the much wider interval $[-0.116, 0.678]$. These intervals use 500 resamples and account for repeated attempts by retaining whole problems. They do not control for difficulty, length, or stopping behaviour, and they are pointwise intervals across an exploratory family of features and scopes.

Other measurements also resist a single interpretation. Qwen’s leading-expert margin is positively associated with correctness across benchmarks, but the selection-boundary margin is approximately zero or negative on AIME24 and AMC23. The former separates the first two experts; the latter concerns

Table 5.7: Whole-continuation router confidence versus correctness: point-biserial coefficients, with all attempts retained. Each model has the same benchmark counts, but these marginal correlations do not estimate controlled model differences.

Benchmark	Qwen	Gemma	GPT-OSS
MATH-500	+0.199	+0.080	-0.061
AIME24	+0.533	+0.392	+0.210
AIME25	+0.493	+0.313	-0.075
Olympiad	+0.243	+0.027	+0.051
AMC23	+0.175	-0.256	-0.077
Minerva	+0.112	-0.037	-0.160

the inclusion boundary of the dispatched set. They should not be treated as interchangeable confidence measures. Likewise, selected probability mass and the mean selected-versus-rest probability gap are algebraically related for a fixed expert count and dispatch size; agreement between their correlations is not independent corroboration.

Token count is negatively associated with correctness in all reported model-benchmark combinations except Gemma on AMC23. For Qwen AIME24 and AIME25, its coefficients are -0.726 and -0.727 . This does not imply that shortening a solution would improve it: harder problems, unsuccessful search, and completion limits can jointly increase length and reduce accuracy. Hidden-state measurements also vary across models. On AIME25, Qwen’s step-distance coefficient is $+0.404$, whereas GPT-OSS gives -0.404 . These descriptive differences argue against transferring a scalar routing or representation rule without model-specific validation.

Figures 5.2 and 5.3 extend the confidence comparison to the other routing and representation measurements, with their uncertainty. Separating the complete continuation from the reasoning span makes the measurement scope explicit. The panels compare descriptive estimates; interval overlap is not a paired test of differences between models.

Figure 5.4 examines dependence among the candidate measurements them-

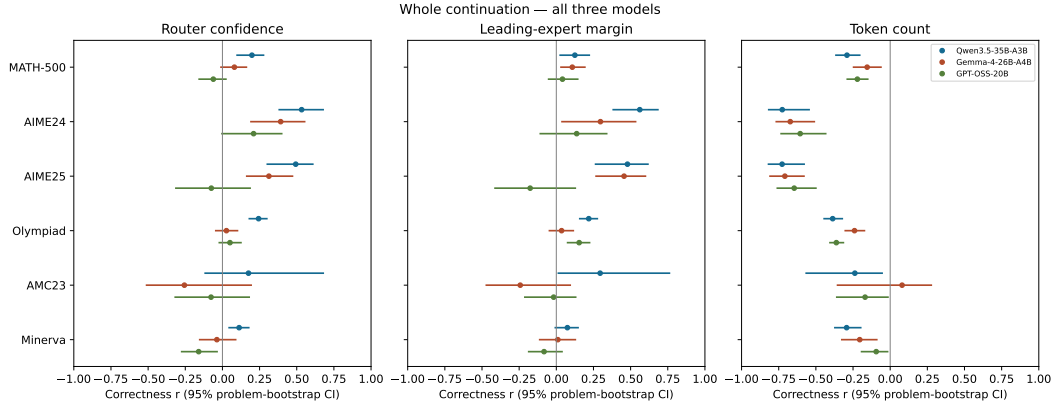


Figure 5.2: Whole continuation: saved correctness correlations and pointwise source-problem bootstrap intervals. Overlap or separation of intervals is not a paired test of model differences.

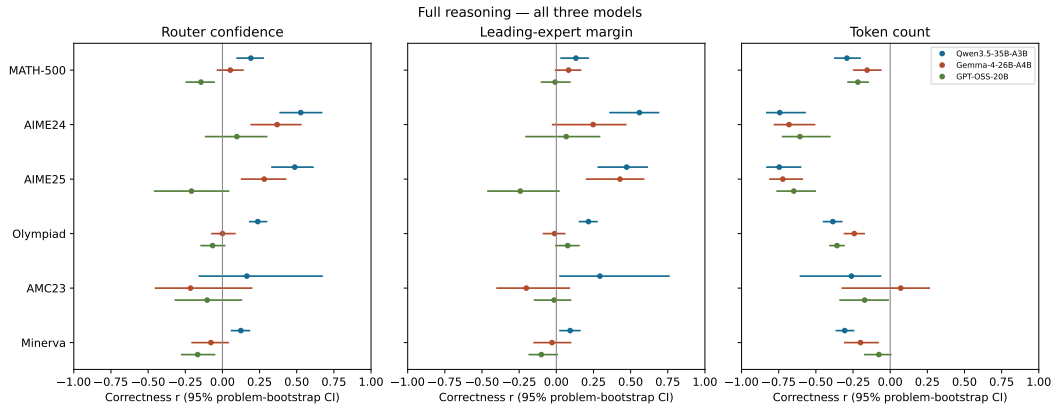


Figure 5.3: Full reasoning: saved correctness correlations and pointwise source-problem bootstrap intervals. Overlap or separation of intervals is not a paired test of model differences.

selves. This matters because several apparent outcome signals can reflect the same variation in length or router concentration. For notation, T denotes token count, C confidence, M the leading-expert margin, S selected probability mass, B the selection-boundary margin, G the selected/rest mean gap, H hidden-state norm, D hidden-state step distance, R hidden/router geometry, W top-one switch rate, and O adjacent selected-set overlap. Definitions are given in the methods chapter.

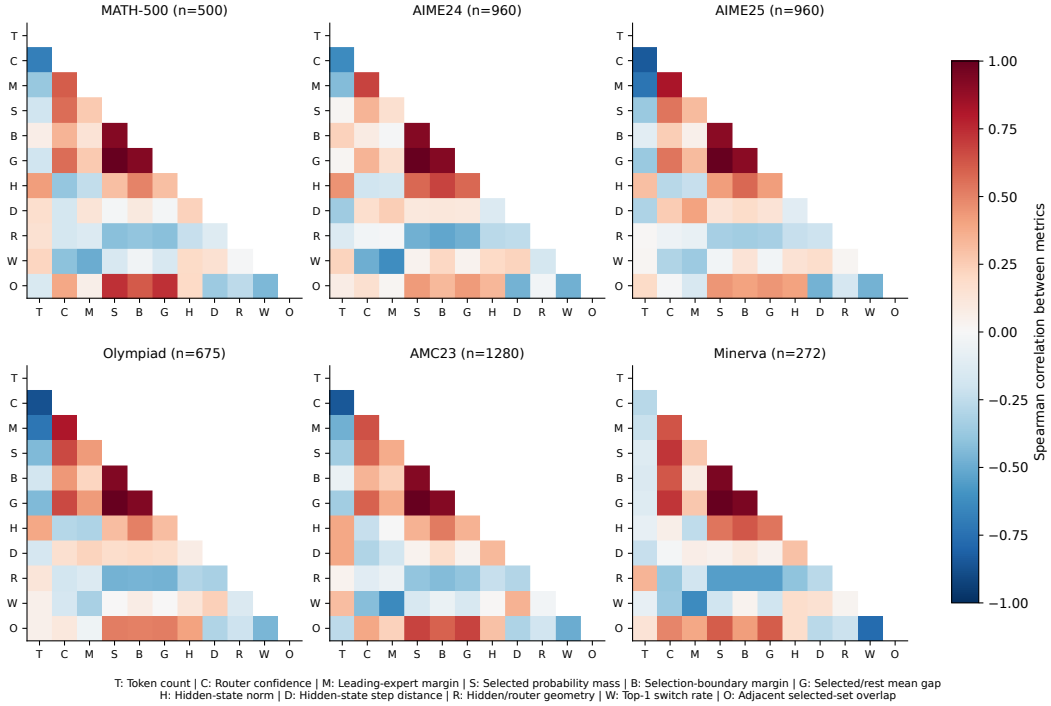


Figure 5.4: Whole-continuation metric–metric Spearman correlations across all six benchmarks. Lower triangles display each association once; diagonals and upper triangles are omitted. Repeated attempts and limit hits remain included. These are descriptive metric relationships, not correctness effects.

A correlation coefficient alone does not identify useful operating values. Figure 5.5 therefore shows accuracy against observed confidence and margin ranges for Qwen. These bins describe the saved population and do not define validated intervention thresholds.

5.3.3 Routing associations across analysis scales

Aggregation changes the estimand. For Qwen, mean problem-level confidence correlates with avg@32 at Spearman $\rho = 0.632$ on AIME24 and 0.705 on AIME25, but at -0.200 on AMC23, whose interval includes zero. The positive attempt-level AMC23 coefficient and negative problem-level rank coefficient are therefore not interchangeable evidence. They compare different units and different outcome variation.

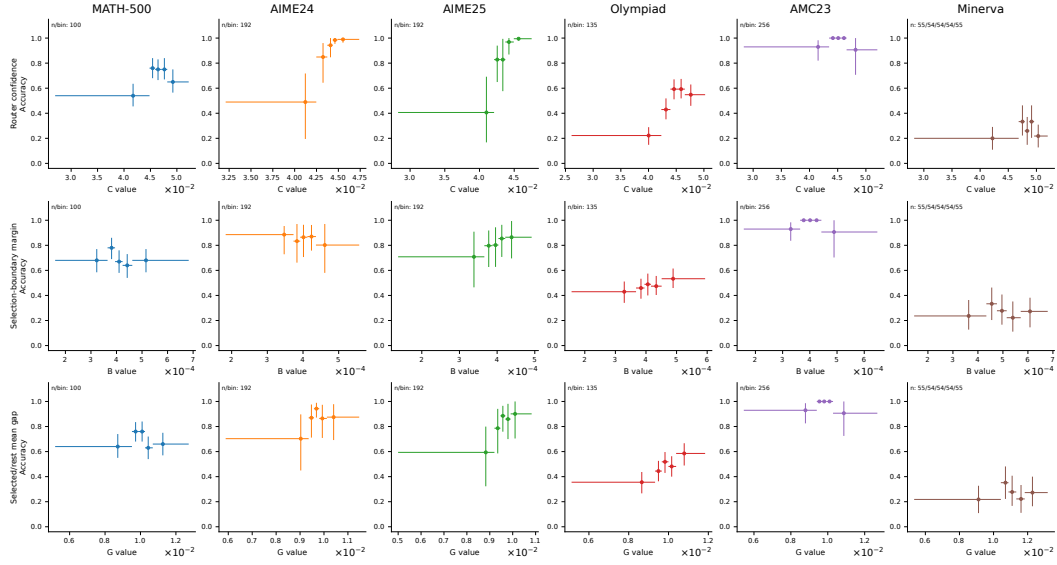


Figure 5.5: Accuracy at observed router-metric values. Each benchmark uses up to five quantile bins; horizontal bars show actual value ranges, points show mean values and accuracy, and panel labels give attempt counts per bin (left to right, or a shared count when equal). Vertical bars are pointwise 95% percentile intervals from 500 whole-source-problem bootstrap resamples within each fixed bin (seed 42). Degenerate intervals can occur in all-correct/all-incorrect bins and do not imply certainty. All attempts and limit hits are retained. Bins can share source problems, so their estimates are not independent; neither bin selection nor length and termination confounding is addressed by these intervals.

Figure 5.6 compares these problem-level associations across the three models. Its units are the 30, 30, and 40 source problems in AIME24, AIME25, and AMC23, respectively; all 32 attempts contribute to each problem’s feature mean and accuracy.

Within-problem contrasts provide a complementary view. Restricting Qwen to problems with both correct and incorrect attempts, mean confidence is higher in the correct attempts by 0.00268 on AIME24, 0.00123 on AIME25, and 0.01102 on AMC23. Their saved whole-problem intervals are $[0.00152, 0.00412]$, $[0.00065, 0.00176]$, and $[0.00455, 0.01709]$, respectively. These estimates use only

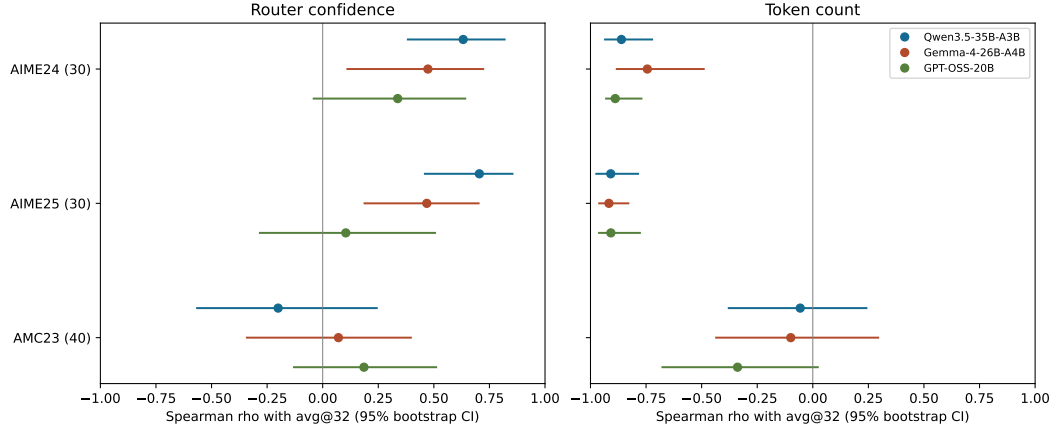


Figure 5.6: All models: whole-continuation problem means versus avg@32, with saved pointwise problem-bootstrap intervals. All 32 attempts enter each problem mean.

14, 15, and 6 mixed-outcome problems. They remove differences in problem identity but leave attempt-level length, termination, and trajectory differences unresolved.

Class-conditioned features retain only tokens assigned to a given episode class, while correctness still describes the whole generated answer. Figure 5.7 illustrates the resulting heterogeneity. For Gemma Implement tokens, confidence correlates with correctness at $+0.498$ on AIME24 (30 attempts) and -0.367 on AMC23 (40 attempts). The corresponding GPT-OSS coefficients are $+0.164$ and -0.098 . Gemma’s AMC23 Plan, Explore, and Monitor cohorts contain only correct attempts, so their correctness correlations are undefined. Nonzero token coverage is not sufficient to estimate an association when the outcome has no variation. Neither a missing coefficient nor a sparse estimate should be interpreted as evidence that the class has no role.

The complementary Qwen class-conditioned screen appears in Figure 5.8. In addition to confidence, it includes hidden-state step distance, showing that class-conditioned associations depend on the measurement as well as the benchmark. The counts printed in each cell identify the available cohort rather than a common matched class sample.

Absolute position windows introduce a further selection effect. In Qwen,

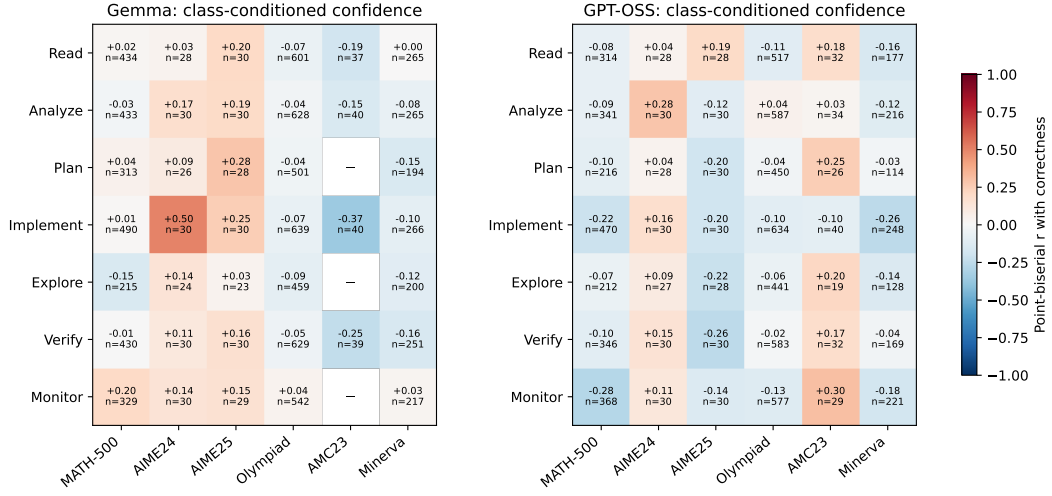


Figure 5.7: Class-conditioned router confidence versus whole-attempt correctness for the stratified Gemma and GPT-OSS cohorts. Cells show point-biserial coefficients and finite-attempt counts. Missing coefficients are undefined, not zero. The models share an annotation procedure but not matched labelled cohorts; this is a descriptive comparison.

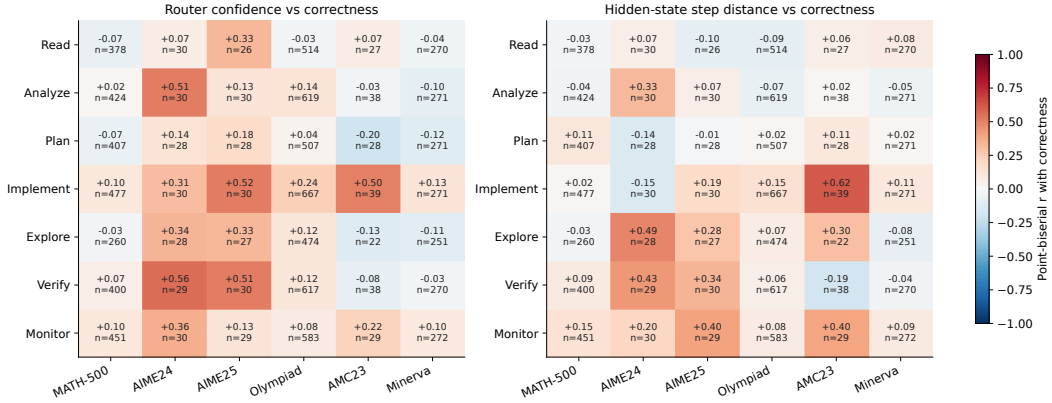


Figure 5.8: Class-conditioned confidence and hidden-state step distance versus whole-attempt correctness. Cells show point-biserial r and finite trace count. Classes use sampled Qwen3.8 labels on Qwen3.5 generations, with different token-bearing cohorts. No significance markings are implied.

the mean reasoning length of 12,498.1 tokens defines ten windows of about 1,250 tokens and an overflow region. Coverage falls from 4,647 traces in the

first window to 1,789 in overflow. Later-window correlations concern solutions that continued long enough to reach those windows; they are not within-trace changes on a fixed cohort. All available reasoning tokens enter position views irrespective of sentence annotation, so this change in coverage is distinct from missing class labels.

Table 5.8 gives the Qwen reasoning-length distribution used to interpret these windows. Figure 5.9 shows the corresponding position-dependent confidence associations and pointwise intervals. The changing token-bearing population is essential to interpreting the later windows.

Table 5.8: Reasoning length in tokens, all attempts including limit hits. This replaces the position-coverage count plot; lengths are observed under the generation budget.

Dataset	Attempts	Mean	Median	90th percentile
MATH-500	500	6,226.5	4,250	12,772
AIME24	960	15,908.4	14,011	32,766
AIME25	960	17,874.9	16,620	32,768
Olympiad	675	13,402.5	9,150	32,768
AMC23	1280	8,452.8	7,822	14,929
Minerva	272	9,805.0	7,465	19,533

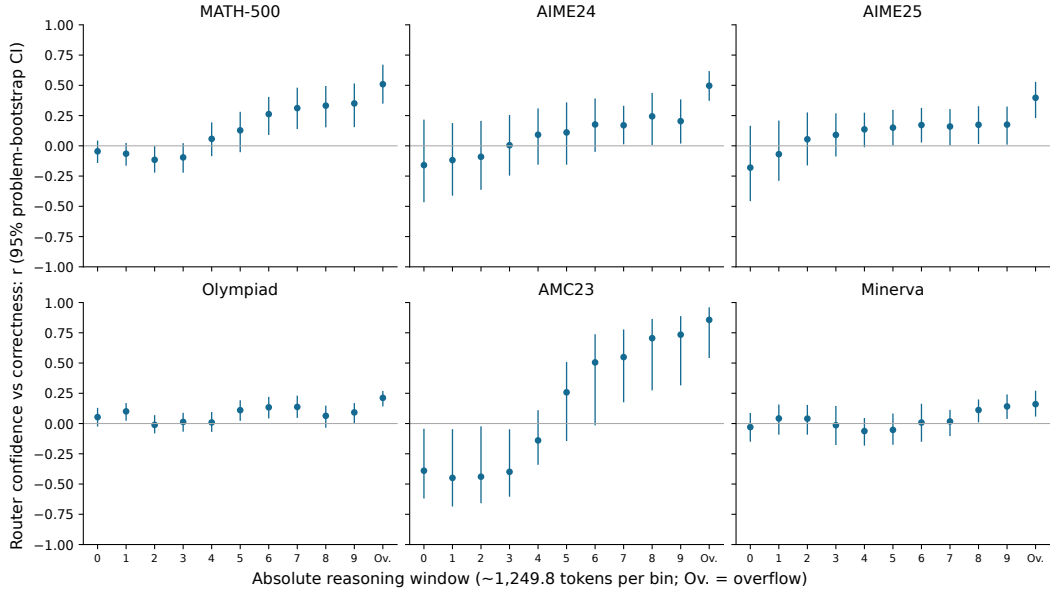


Figure 5.9: Router confidence versus whole-attempt correctness in absolute reasoning windows, with saved pointwise 95% source-problem bootstrap intervals. Each benchmark has its own panel. Missing estimates remain blank. Later windows use changing token-bearing cohorts; these are not within-trace causal effects.

Finally, the observational expert-identity screen is exploratory. It tests many top-one, selected-pool, and pair-use features, and a maximum selected after inspecting them is subject to selection bias. Some pooled screens combine numeric identities across measured layers, whereas an intervention requires a

specific (ℓ, e) pair. A visually strong association cannot be promoted directly to a portable expert target. The calibration procedure therefore retains layer-qualified identities and evaluates its policies on separate source problems.

5.3.4 Outcome associations of sampled reasoning composition

The frequency of a predicted reasoning activity is distinct from the router features measured within that activity. Figure 5.10 shows the sampled label and benchmark composition for Gemma and GPT-OSS. These models share a stratified annotation procedure but retain different selected problems and traces; equal label budgets do not imply matched populations.

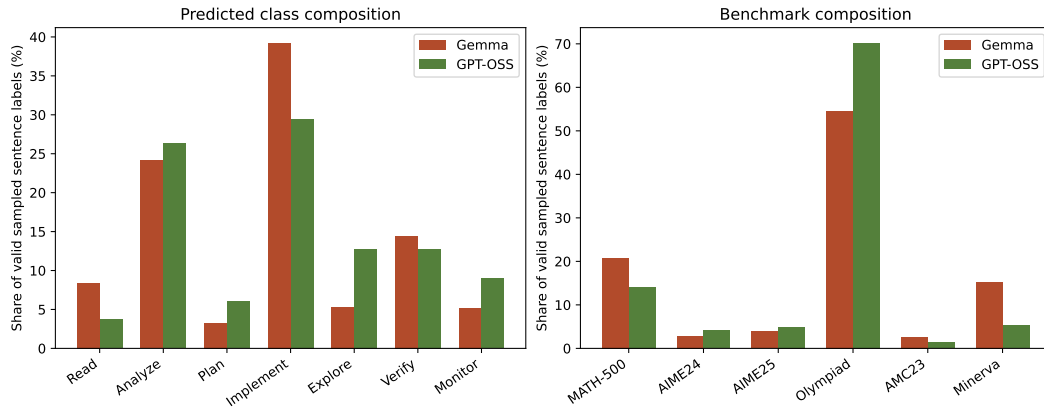


Figure 5.10: Gemma and GPT-OSS stratified label composition. Denominators are 99,999 and 100,000 valid sampled sentences. Equal sample budgets do not imply equal class or benchmark proportions. These sentence shares are neither complete-corpus token shares nor adjusted model effects.

For Qwen, Table 5.9 reports mean sentence shares, with each labelled sample-zero attempt receiving equal weight. An observed sample without a class contributes zero for that class; attempts without labels are excluded. These quantities describe the annotation sample, not complete-trace class prevalence. Figure 5.11 locates the sampled sentences along relative reasoning progress, while Figure 5.12 relates their shares to correctness and output length.

Because class shares sum to one, their correlations cannot be interpreted as independent class effects.

Table 5.9: Mean sampled sentence share (percent), with equal weight per labelled attempt. Empty-label attempts are excluded; absent classes in an observed sample have zero share. These are not full-corpus class proportions.

Dataset	Attempts	Read	Analyze	Plan	Implement	Explore	Verify	Monitor
MATH-500	499	8.9	19.5	9.3	29.3	4.1	11.8	17.1
AIME24	30	6.7	33.9	2.9	28.3	7.0	14.4	6.8
AIME25	30	5.7	33.2	4.8	27.9	6.7	13.4	8.3
Olympiad	675	6.9	26.8	6.0	29.7	6.4	13.2	11.0
AMC23	40	6.6	25.9	5.8	32.0	3.8	15.2	10.6
Minerva	272	9.4	19.8	8.4	21.8	7.7	10.3	22.6

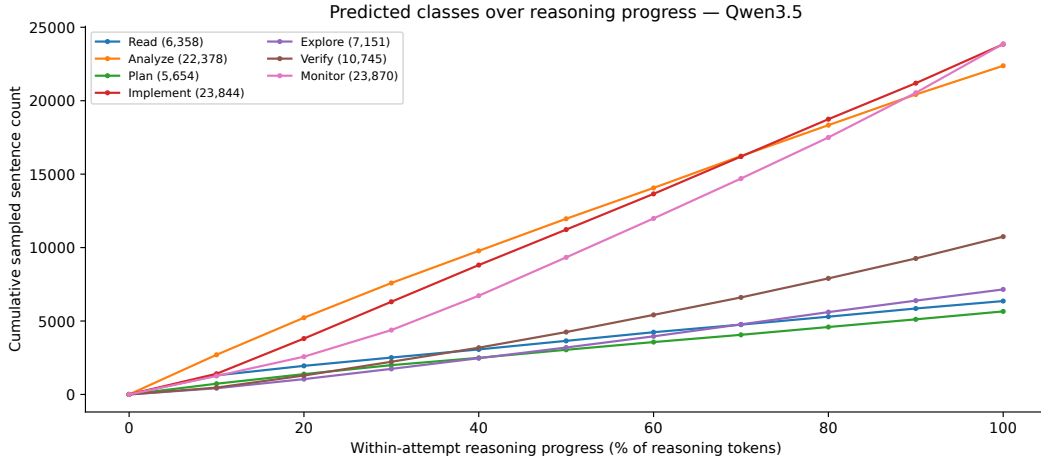


Figure 5.11: Cumulative counts of sampled sentences by predicted class, pooled across benchmarks. Counts start at zero; a sentence is counted when its last reasoning token is reached. Progress is normalized by each attempt’s full reasoning-token length, unlike the absolute windows in Figure 5.9. Legend totals equal the rightmost endpoints. These are raw sampled counts, not cumulative percentages or estimates of full-corpus counts; longer traces and benchmark sampling quotas affect them.

Figures 5.13 and 5.14 show observed accuracy and generation cost across sampled class-share bins. The attempt counts make sparse bins visible. Lines

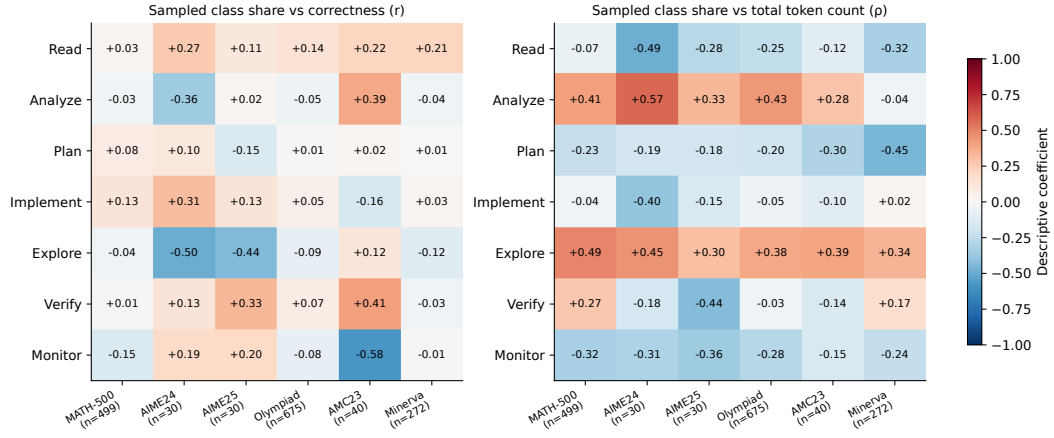


Figure 5.12: Sampled class proportions versus whole-attempt correctness (point-biserial r) and total continuation-token count (Spearman ρ). All labelled attempts enter each class, including observed zero proportions. These differ from correlations of router features within class-selected tokens. No significance claim is made; class shares sum to one and cannot be interpreted as independent effects.

connect occupied bins and do not fit an optimal class composition or establish a causal response.

5.3.5 Directed class transitions by answer correctness

For each model, we separate labelled attempts by whole-answer correctness and count directed transitions between originally adjacent labelled sentences. Rows are source classes and columns are destination classes. Self-transitions are retained; missing labels and sampling gaps are never bridged. Each cell is the total observed count divided by the number of labelled attempts in that outcome group: equivalently, the mean per-attempt count, including zero-pair attempts. Empty-label attempts are excluded. These are counts per attempt, not row-normalized probabilities or full-trace transition estimates.

The two panels per model pool the six benchmarks and all six panels share one colour scale. Counts and denominators are recorded separately for each benchmark. Outcome groups differ in benchmark mix, reasoning length, and

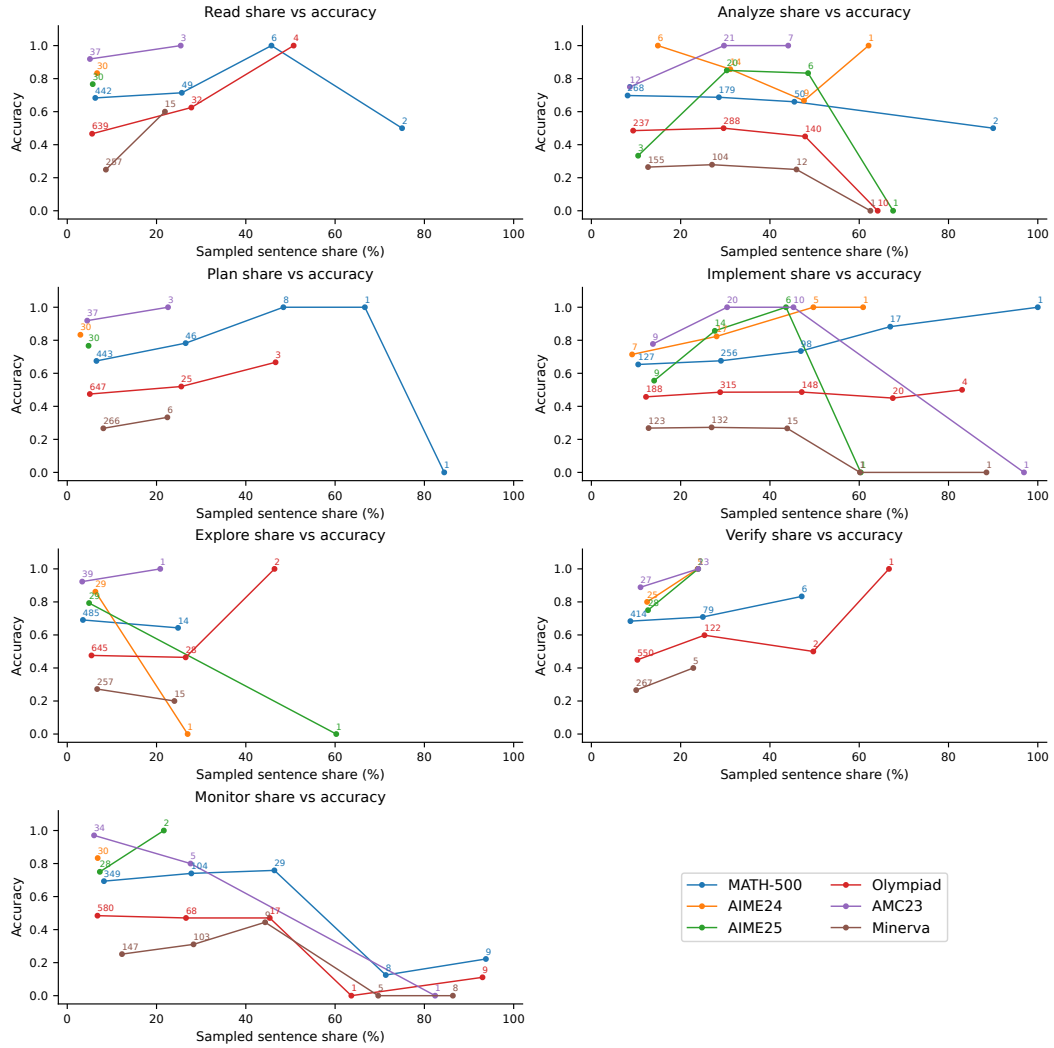


Figure 5.13: Actual outcome levels versus sampled class proportions. Fixed share bins are $[0, 20)$, $[20, 40)$, $[40, 60)$, $[60, 80)$, and $[80, 100]$ percent. Each point is at the mean observed share and shows accuracy; small labels give attempt counts. Empty bins are omitted. Colours distinguish benchmarks; lines only connect occupied bins. Sparse bins, sentence sampling, length, and difficulty can explain apparent trends; these descriptive plots do not identify optimal class proportions or a validated accuracy-length Pareto frontier.

label coverage; Qwen also retains a different annotation protocol from Gemma and GPT-OSS. Differences are descriptive associations, not evidence that a

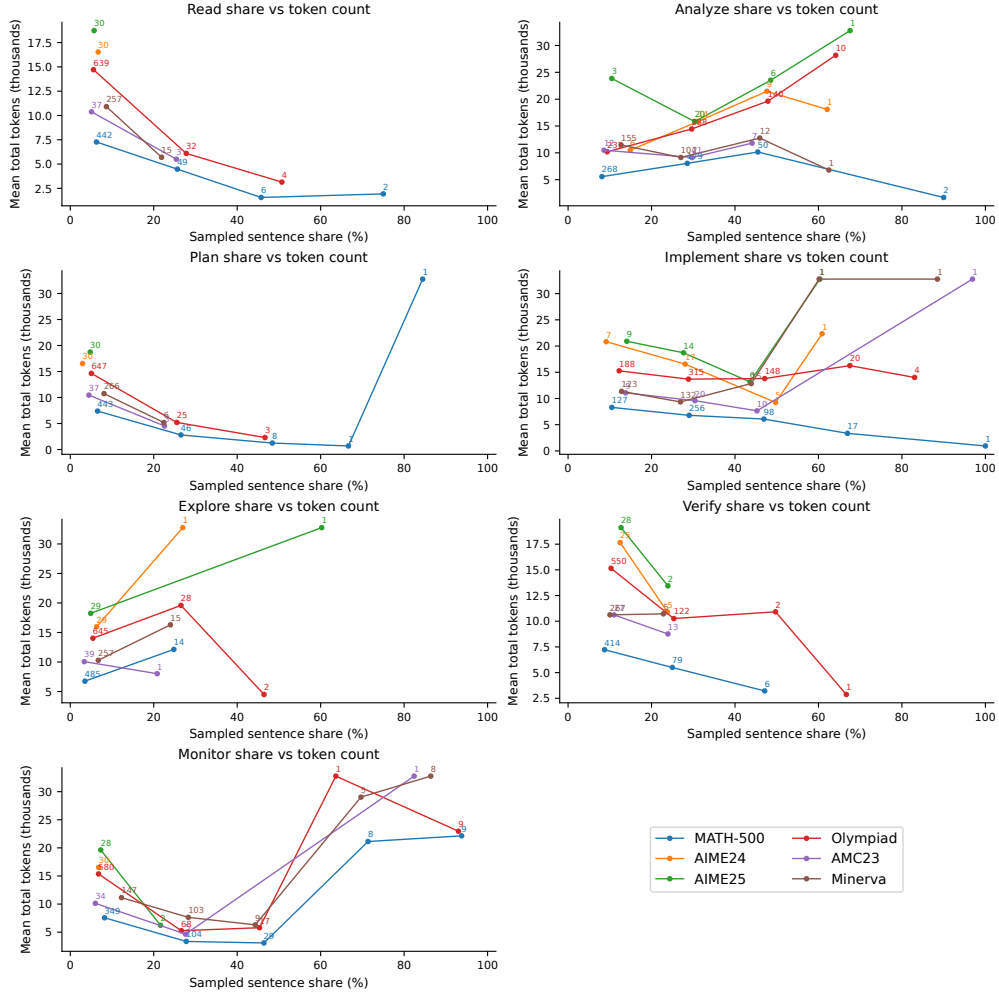


Figure 5.14: Mean total continuation tokens versus sampled class share. Bin boundaries, colours, sample-count labels, and sampling limitations are as in Figure 5.13. Token count includes the final answer and token-limit completions; lines are descriptive, not intervention effects.

transition causes correctness or a controlled comparison between models.

Figure 5.18 complements the correctness-stratified mean-count matrices above. It pools outcomes within each benchmark and normalises each source-class row to percentages, describing the observed destination distribution conditional on an adjacent labelled pair. These percentages have a different denominator from mean counts per attempt; a missing row is undefined, not a

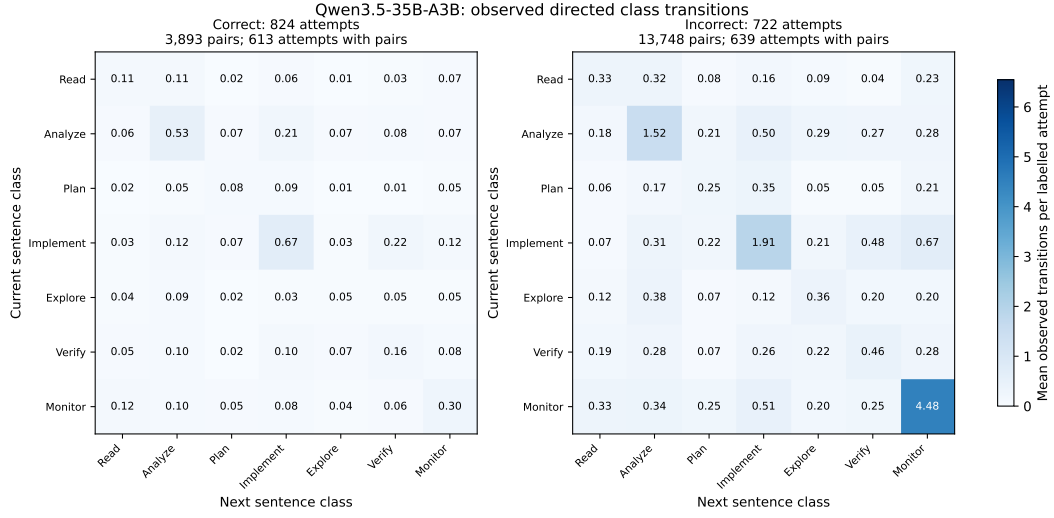


Figure 5.15: Qwen3.5-35B-A3B: directed class transitions for correct and incorrect answers. Cells show mean observed counts per labelled attempt, including self-transitions on the diagonal. Panel headings give denominators and adjacent-pair coverage; zero-pair attempts remain in the mean. The shared colour scale is identical across all three models. Sampling and population differences limit interpretation.

zero transition probability.

5.4 Held-out routing interventions

5.4.1 Statistical basis of intervention comparisons

The global intervention split assigns 1,082 source problems to calibration and 465 to evaluation. The evaluation set contains 150 MATH-500, 9 AIME24, 9 AIME25, 203 OlympiadBench, 12 AMC23, and 82 Minerva problems. Keeping 32 attempts for each AIME and AMC problem produces 1,395 matched attempts. All attempts from one problem remain on the same side of the split. For the original Qwen policies, calibration uses 3,014 scored traces from 1,001 problems after excluding 238 traces with missing saved correctness. This exclusion differs from counting invalid known-gold answers as incorrect in the observational

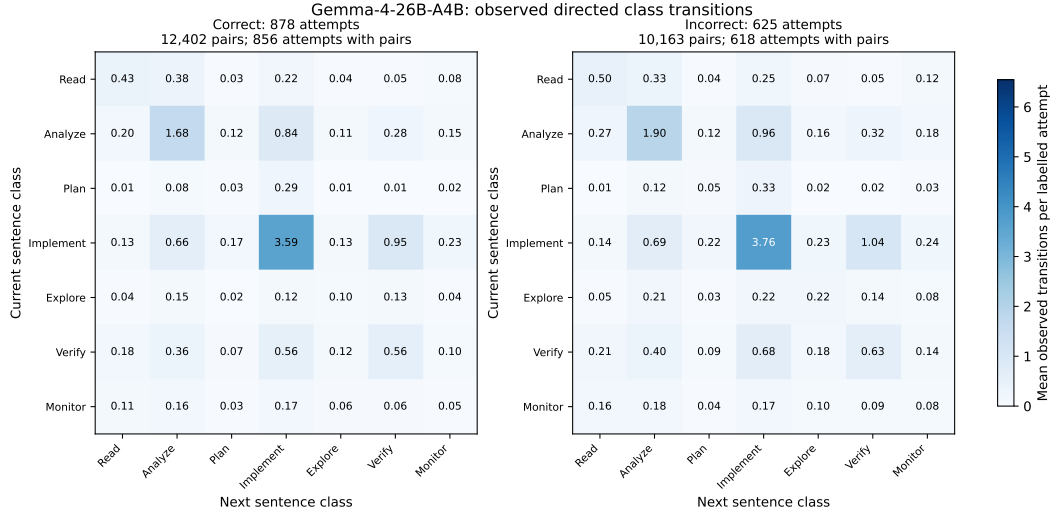


Figure 5.16: Gemma-4-26B-A4B: directed class transitions for correct and incorrect answers. Cells show mean observed counts per labelled attempt, including self-transitions on the diagonal. Panel headings give denominators and adjacent-pair coverage; zero-pair attempts remain in the mean. The shared colour scale is identical across all three models. Sampling and population differences limit interpretation.

analysis, and can affect the selected expert targets.

Each row in the complete-comparison tables compares baseline and guided runs with matching source problems, tokenised prompts, per-attempt sampling settings, and numerical inference conditions. Baselines are not interchanged between rows. Stochastic generation uses temperature 0.6, top- $p = 0.95$, and a 32,768-token output limit. The primary accuracy summaries are attempt-weighted. Section 5.4.8 adds benchmark-level results and a secondary analysis giving equal weight to each source problem; neither aggregation is a benchmark macro-average.

Accuracy-change intervals use 5,000 paired, dataset-stratified source-problem bootstrap resamples with seed 42. A sampled problem carries all of its attempts and both conditions together. This preserves the paired contrast and dependence between repeated attempts. The intervals are conditional on the fitted policy and saved decoding runs. They do not incorporate policy

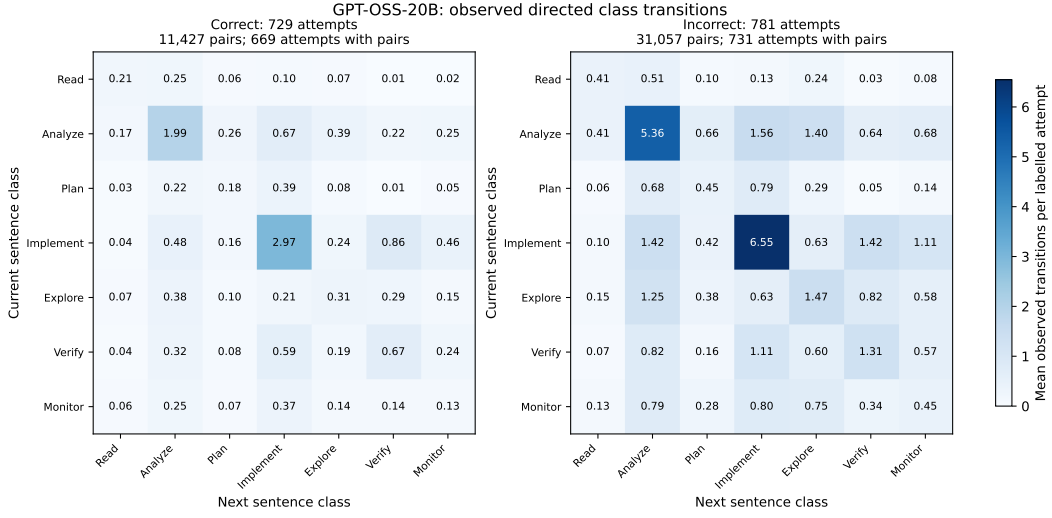


Figure 5.17: GPT-OSS-20B: directed class transitions for correct and incorrect answers. Cells show mean observed counts per labelled attempt, including self-transitions on the diagonal. Panel headings give denominators and adjacent-pair coverage; zero-pair attempts remain in the mean. The shared colour scale is identical across all three models. Sampling and population differences limit interpretation.

refitting, independent decoding seeds, or adjustment for the multiple tested strengths and methods. In particular, each AIME evaluation has only nine source problems. An interval containing zero does not establish equivalence, and an interval excluding zero is interpreted within this exploratory design.

5.4.2 Expert-selection comparison with SteerMoE

The comparison separates expert detection from the intervention applied to those experts. SteerMoE uses paired contrastive inputs and counts top- k expert inclusions separately in the two populations. Its score is the token-pooled activation-rate difference

$$\Delta_{\ell_e}^{\text{paper}} = \frac{A_{\ell_e}^+}{N^+} - \frac{A_{\ell_e}^-}{N^-}, \quad (5.1)$$

where A counts tokens selecting the layer-expert identity and N counts all tokens in the corresponding population. Detection therefore uses routing

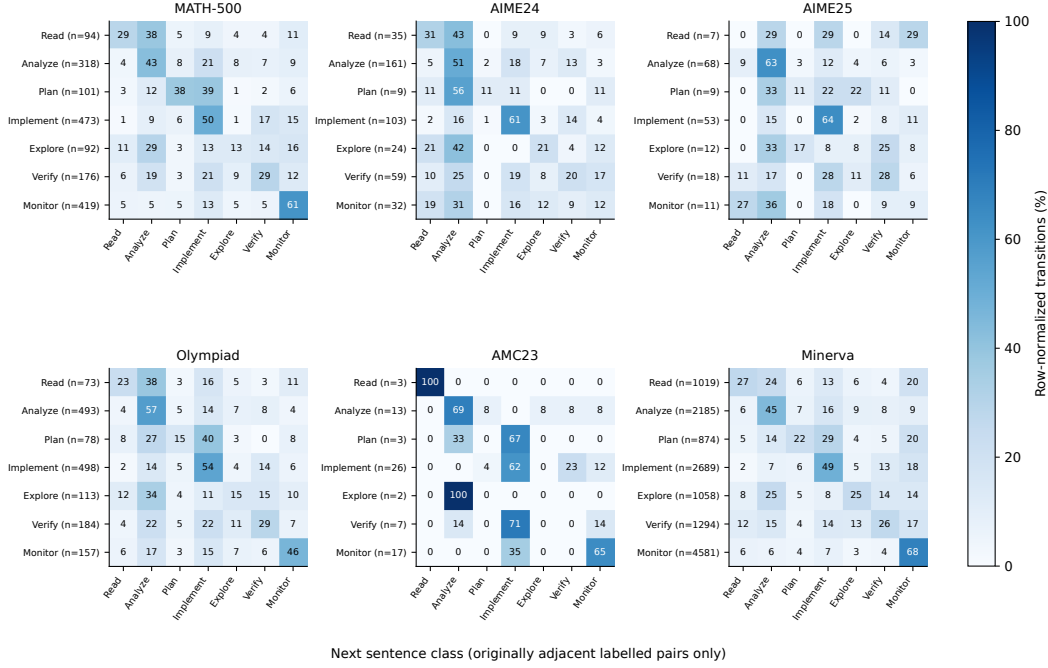


Figure 5.18: Observed class-state transitions by benchmark. Rows are the current class, columns the next class; cells are row percentages and row labels give pair counts. Only originally consecutive sentence indices with both labels are counted. Gaps in sampling are never bridged. Self-transitions are retained; empty rows are undefined. This describes the observed adjacent-pair sample, not the complete transition process or an effect of changing state.

frequencies, not hidden-state similarity. Positive and negative differences identify opposite behavioral associations; steering changes the selected experts’ log-softmax scores to the original maximum plus ϵ or minimum minus ϵ [37], Sections 3.1–3.2. The authors’ selection procedure ranks layer–expert identities jointly by absolute risk difference. Our policy instead assigns a separate budget to each of a preselected set of layers.

Calibration estimator. For calibration attempt r , let $y_r \in \{0, 1\}$ be correctness, $f_{r\ell e}$ the fraction of saved routing tokens selecting expert e at layer ℓ , and $w_r = 1/m_{p(r)}$, where $m_{p(r)}$ is the number of usable attempts for its source

problem. With $W = \sum_r w_r$, the calibration score is

$$\pi = \frac{\sum_r w_r y_r}{W}, \quad L_{\ell e} = \frac{\sum_r w_r f_{r\ell e} y_r}{\sum_r w_r f_{r\ell e}} - \pi. \quad (5.2)$$

Thus calibration gives every problem total weight one and first normalizes routing counts within each trace. It contrasts observed correct and incorrect completions; it does not construct paired counterfactual behavioral inputs. Experts need at least four supporting problems in the audited policies. Within each selected layer, eligible experts are ordered by signed lift, then support, then expert ID, and capped at native top- k (eight for Qwen and Gemma, and four for GPT-OSS). Fixed bias scales each retained lift by the largest retained magnitude; paper-style steering uses only membership in the selected set, not those relative magnitudes.

Why activation-based scores need not select the same experts. Define $\mu_1 = E_w[f_{r\ell e} \mid y_r = 1]$, $\mu_0 = E_w[f_{r\ell e} \mid y_r = 0]$, and $\bar{f} = E_w[f_{r\ell e}]$. Algebra applied to our estimator gives

$$L_{\ell e} = \frac{\pi(1 - \pi)}{\bar{f}}(\mu_1 - \mu_0), \quad \bar{f} > 0. \quad (5.3)$$

Under these same weights, lift and activation-rate difference have the same sign but can have different rankings: lift additionally divides by the expert’s mean activation frequency. This identity does not equate either score to the paper’s token-pooled estimate on different, paired inputs.

The saved frequency masses, activation-weighted accuracies, and calibration baseline suffice to reconstruct $\mu_1 - \mu_0$ without replaying a model. Table 5.10 changes only the ranking score and retains our weights, sign filter, support threshold, layers, and target budget. The audit also reconstructs every saved selected set and normalized score. Only 14 of 48 Qwen positive layer–expert identities remain selected; the four-target GPT-OSS policies retain 6 of 16 positive identities and 7 of 16 negative identities. This demonstrates selection sensitivity even before changing the data construction or token weighting. It does not measure intervention performance for these alternative sets.

Table 5.10: Selection sensitivity using saved calibration sufficient statistics. RD here uses the same problem-balanced trace weights, support filter, polarity, layer set, and per-layer budget as lift; it is not a reproduction of the paper's token-pooled contrastive estimator. Overlap counts layer-expert identities.

Model	Polarity	Layers	Overlap	Overlap per layer
GPT-OSS	negative	4	7/16	15: 1/4, 19: 2/4, 21: 1/4, 23: 3/4
GPT-OSS	positive	4	6/16	15: 2/4, 19: 1/4, 21: 1/4, 23: 2/4
Qwen	positive	6	14/48	27: 4/8, 34: 3/8, 35: 2/8, 36: 1/8, 38: 3/8, 39: 1/8

Interpretation of the intervention comparison. The completed Qwen fixed runs share selection statistics and target sets. The completed GPT-OSS fixed and paper-style conditions use four targets per layer, with the same selected identities within each polarity. These comparisons retain our selector and do not evaluate the full SteerMoE detection and steering procedure.

The completed OSS comparison already holds the selected identities fixed within each polarity while changing the intervention rule. A selector comparison should instead fix the split, generation protocol, layers, budget, and intervention, changing only the ranking estimator. The present OSS budget is four targets per layer, matching native dispatch; smaller budgets of one or two targets would test less extensive promotion. Matched random sets would help separate selector information from generic perturbation. A faithful detection replication additionally requires paired contrastive inputs and token-pooled activation counts. The analysis uses saved calibration statistics and does not evaluate new intervention runs for the alternative selected sets.

5.4.3 Consistent offline answer-equivalence audit

The qualitative inspection identified equivalent final answers rejected by the historical normalized-string/numeric fallback. A uniform offline pass therefore rescores both conditions in every documented identity and margin evaluation: 13 complete full stochastic comparisons, two complete greedy comparisons, the 39-problem Minerva comparison, and the interrupted Qwen subset. It reads 40,097 stored condition records, including reused baseline copies; this is not a

count of independent generated solutions. The original responses, experimental conditions, policies, and judgments are preserved, and revised decisions are documented separately.

The rule first extracts the last nonempty balanced boxed candidate, then uses the existing answer-marker and last-number fallbacks when no such box is present. Only that candidate is compared with the gold answer. It normalizes display notation, plain-text labels, common units, currency, and explicit list separators. Mathematical equivalence is assessed through symbolic comparison and numerical evaluation. Variables and structured objects use strict comparison; exact-number answers require equality, while decimal/scientific scalar answers allow relative tolerance 10^{-6} with no absolute floor. Thus zero does not match a small nonzero answer merely because its magnitude is below 10^{-6} . Scientific notation is parsed numerically, rather than as a symbolic expression involving e . Text denoting multiple answers is not silently reduced to its first number. Standard unit annotations follow the parser’s normalization conventions; this is not a dimensional-analysis check.

The pass changes 4,054 stored judgments from incorrect to correct and one from correct to incorrect. It also changes the extracted candidate in 2,065 records. The single downward change concerns a capped Qwen baseline whose extracted box contains the literal placeholder “answer” rather than its earlier saved numeric fallback. These counts include repeated baseline copies. Changes occur on every benchmark, mainly OlympiadBench, MATH-500, and Minerva. The analysis reruns the same 5,000 paired, dataset-stratified source-problem bootstrap resamples (seed 42), including the equal-problem sensitivity statistic. The 494-ID interrupted comparison remains separate and its accuracy counts are unchanged.

No revised positive accuracy interval excludes zero in the complete comparisons under either reported weighting. In particular, the earlier positive secondary OSS findings and greedy Qwen margin interval do not survive the common scoring rule. The Qwen concurrency-25 margin lower endpoint is exactly zero. Strong negative effects under paper-style promotion remain. Generation costs and termination conditions are unchanged.

The checker is tested on equivalent and distinct fractions, text labels,

Table 5.11: Selected changes after applying the same revised scoring rule to both conditions. Changes and 95% intervals are percentage points. Attempt weighting is primary; problem weighting is secondary. Historical values are retained here only to document the audit and are not mixed into current result tables.

Condition	Weighting	Historical change	Revised change	Revised 95% CI
Qwen margin (c=25)	Attempt	+1.08	+0.93	[0.00, +1.86]
GPT-OSS margin	Attempt	-3.15	-0.86	[-2.87, +1.15]
GPT-OSS fixed positive +2	Attempt	-1.43	-3.30	[-6.16, -0.93]
GPT-OSS fixed positive +2	Problem	+4.87	-0.10	[-3.12, +3.08]
GPT-OSS fixed negative +1	Problem	+4.94	+2.35	[-0.58, +5.44]
Qwen greedy margin	Attempt	+2.15	+0.86	[-1.29, +3.01]

scientific notation, ordered answers, units, currency, multiple-answer text, and very small numerical values, as well as final-candidate extraction. It does not provide independent human adjudication. There are 699 records with an unparsed candidate or reference, 332 with unsupported embedded text, and 30 parser/comparison errors; these receive an incorrect decision unless normalized display equality already succeeds. Error categories and changed judgments remain available for further review. A two-second parser/comparison timeout and a 2,048-character limit for nonnumeric expressions bound degenerate inputs. Calibration and observational correlation labels retain their original values: the revised evaluations test the existing frozen policies, not policies refitted with revised labels.

5.4.4 Sensitivity to expert-identity policy configuration

Table 5.12 distinguishes polarity, which defines the selected expert set, from signed strength, which defines whether that set is promoted or suppressed. The fixed method uses the lift-weighted additive bias. The study also considers the paper-style extreme-score replacement described in Section 3.4.2; the completed comparisons in this table evaluate both fixed bias and paper-style steering.

For Qwen fixed positive-polarity guiding, strength +1 changes accuracy by

Table 5.12: Complete stochastic identity comparisons. B/G denotes baseline/guided accuracy in percent; changes and intervals are percentage points. Each row uses its own matched baseline and 1,395 attempts on 465 problems. Intervals use 5,000 paired, dataset-stratified problem resamples, seed 42.

Model	Method	Polarity	Strength	B accuracy	G accuracy	Change	95% CI
GPT-OSS	Fixed	Negative	+1	77.85	78.42	+0.57	[-1.58, +2.65]
GPT-OSS	Paper	Negative	+1	77.85	18.28	-59.57	[-67.24, -51.40]
GPT-OSS	Paper	Positive	-1	77.85	75.63	-2.22	[-4.16, -0.43]
GPT-OSS	Paper	Positive	+1	77.85	16.20	-61.65	[-69.32, -53.41]
GPT-OSS	Fixed	Positive	-1	77.85	76.85	-1.00	[-3.44, +1.22]
GPT-OSS	Fixed	Positive	+1	77.85	77.85	+0.00	[-2.15, +2.01]
GPT-OSS	Fixed	Positive	+2	77.85	74.55	-3.30	[-6.16, -0.93]
Qwen	Fixed	Positive	+1	84.95	84.66	-0.29	[-1.15, +0.65]
Qwen	Fixed	Positive	-1	84.09	85.16	+1.08	[-0.14, +2.37]
Qwen	Fixed	Positive	+2	84.09	81.29	-2.80	[-5.16, -0.72]
Gemma	Fixed	Positive	+1	87.67	87.60	-0.07	[-1.15, +1.00]

-0.29 points with interval $[-1.15, +0.65]$, and strength -1 gives $+1.08$ with $[-0.14, +2.37]$. Increasing promotion to strength $+2$ gives -2.80 points with $[-5.16, -0.72]$. Thus stronger promotion reduces primary attempt-weighted accuracy under the revised scorer.

Gemma-4-26B-A4B NVFP4 uses fixed positive identity bias $+1$ on 1,395 held-out attempts from 465 problems, with concurrency 16 and the frozen model-specific policy. Eight experts are targeted at each of layers 16, 19, 21, 22, 25, 27 and 29. Calibration uses replay routing from the corresponding Gemma checkpoint under a different quantization; expert-identity transfer is therefore an assumption. No held-out result is used to refit targets or choose the strength.

The original scorer gives 1,117 baseline and 1,113 guided correct answers. Under the same offline extracted-answer equivalence contract used for the other intervention evaluations, the counts are 1,223 and 1,222: 87.67% versus 87.60%. The change is -0.07 percentage points, with a paired, dataset-stratified problem-bootstrap 95% interval $[-1.15, +1.00]$ (5,000 resamples, seed 42). There are 33 wrong-to-right and 34 right-to-wrong changes. Mean generated tokens decrease

from 11,695.48 to 11,562.61 (-1.14%), and token-limit hits decrease from 67 to 66. These results show a small observed accuracy difference, not established non-inferiority or an accuracy improvement.

All seven GPT-OSS identity conditions use four targets at each of layers 15, 19, 21, and 23, matching native routing top- $k = 4$. Each evaluates 1,395 matched attempts from 465 held-out problems at concurrency 25, with a frozen chat-template date of 22 September 2026. Rescored baseline accuracy is 77.85% (1,086/1,395) in every condition. Fixed positive strengths +1, +2, and -1 give changes of 0.00, -3.30 , and -1.00 points. Only the strength-+2 interval excludes zero: $[-6.16, -0.93]$. Fixed negative-polarity promotion gives +0.57 points with $[-1.58, +2.65]$ and does not establish a benefit.

Paper-style promotion at strength +1 reduces accuracy to 16.20% (226/1,395) for positive targets and 18.28% (255/1,395) for negative targets. Changes are -61.65 points with $[-69.32, -53.41]$ and -59.57 with $[-67.24, -51.40]$, respectively. Mean generated length rises from 6,497 to 22,489 and 23,639 tokens; token-limit hits increase from 74 to 670 and 815. Suppressing the positive set with paper-style strength -1 gives 75.63% accuracy and a smaller but below-zero primary change of -2.22 points ($[-4.16, -0.43]$); mean length is 6,698 tokens with 106 limit hits.

Paper-style steering adapts the extreme-score intervention with $\epsilon = 0.01$ while retaining our accuracy-lift selector [37]; it is not a full reproduction of SteerMoE. The four-target budget does not prevent severe degradation from promotion. Promoting a full top- k set fixes membership and equalises its routing scores; this is a possible mechanism, not a verified explanation of termination.

5.4.5 Partial results of the interrupted Qwen paper-style intervention

The Qwen positive-expert paper-style intervention at strength +1 was interrupted after 494 of 1,395 planned attempts (35.4%). 901 attempts have no saved guided result. This policy targets eight experts at each of six layers, matching the model routing top- $k = 8$. The intervention uses epsilon 0.01,

concurrency 25, and a 32,768-token completion budget. Only saved completed attempts are analysed.

On the 494 matched baseline and guided attempts (16 source problems), baseline accuracy is 70.45% (348/494), and guided accuracy is 0.81% (4/494), a change of -69.64 percentage points. There are 4 wrong-to-right and 348 right-to-wrong flips. Mean generated length changes from 19,352 to 32,768 tokens (medians 17,252 and 32,768); token-limit hits change from 110/494 to 494/494.

Table 5.13: Partial Qwen paper-style intervention. Accuracy and token counts compare only the same completed attempts in each condition. B/G denotes baseline/guided. A dash indicates no completed guided attempts, not zero accuracy.

Benchmark	Completed/planned	Problems	Accuracy B/G (%)	Change (pp)	Mean tokens B/G	Limit hits B/G
AIME24	288/288	9	61.81/0.00	-61.81	20,947/32,768	74/288
AIME25	206/288	7	82.52/1.94	-80.58	17,121/32,768	36/206
AMC23	0/384	0	—	—	—	—
MATH-500	0/150	0	—	—	—	—
Minerva	0/82	0	—	—	—	—
Olympiad	0/203	0	—	—	—	—

These are descriptive partial results, excluded from the complete-run comparison and weighting tables. The completed subset is not a random sample: benchmark submission order, scheduling, and generation length can affect which requests have finished. Multiple attempts from one problem are also dependent. Consequently the pooled difference is not an estimate for the full 1,395-attempt evaluation, and no population confidence interval or significance claim is reported. Missing attempts are not counted as wrong or filled from the baseline. Increased token-limit incidence does not by itself establish repetitive looping.

Each completed guided attempt has a unique matching baseline, a binary correctness outcome, identical tokenised inputs, and the same sampling settings. The comparison also preserves model, inference, answer-evaluation, and prompt-formatting conditions. Routing measurements confirm that the intervention was applied at all selected layers.

5.4.6 Qualitative inspection of paper-style failures

A reproducible, assistant-assisted excerpt inspection examines 22 saved historically baseline-correct/guided-incorrect pairs from strength-+1 paper-style promotion. For each complete GPT-OSS polarity, four cases are sampled without replacement from outputs ending at the token limit and four from outputs ending normally. Six additional capped cases are sampled from the interrupted Qwen positive run. Sampling uses sorted attempt IDs and seed 20260924, with separate streams per condition and termination stratum. The OSS samples cover eight and seven distinct problems; the Qwen sample covers four. Selection conditions on recorded correctness loss and termination, and repeated attempts can share a problem. The resulting counts describe this sample, not the prevalence of failure modes.

The review examines fixed beginning, middle, and ending excerpts, together with the most frequent exact 16-word span, against the corresponding baseline answer. Repetition is described from the text, not inferred from output length alone. The whole-text diagnostic counts case-sensitive, whitespace-delimited spans, allowing overlap; it does not capture every form of repetition, particularly unspaced symbol chains. This is neither independent human annotation nor exhaustive verification of the derivations.

Table 5.14: Descriptions of 22 cases sampled as correctness losses under the original scorer. Capped/stopped gives the sampled termination strata. Repetition can coexist with mathematical errors; the arithmetic-error column describes the remaining case without an observed long cycle. Formatting disagreement denotes an equivalent final answer rejected by the saved scorer. Counts are conditional on the sampling design.

Condition	Capped/stopped	Repetition	Arithmetic error	Formatting	No box
GPT-OSS positive	4/4	6	0	2	4
GPT-OSS negative	4/4	7	1	0	4
Qwen positive (partial)	6/0	6	0	0	6

Generation pathologies. All 14 sampled capped outputs contain repeated arithmetic or verbal cycles and lack a boxed-answer marker. Five of the eight normally ended OSS outputs also contain cycles before returning an incorrect boxed answer. Across these 19 repetitive outputs, the most frequent exact 16-word span occurs 62–9,906 times; the matched baseline maxima range from one to four. For example, positive OSS case P03 repeats “390?” while its baseline returns 49. Qwen case Q06 repeatedly emits “Wait, wait.” while its baseline returns 82. Negative OSS case N06 repeats an unresolved sum and eventually returns 112 instead of 144. Thus repetition can accompany both failure to finish and a normally terminated wrong answer. The separate non-cyclic arithmetic example N05 correctly enumerates three, two, and two choices but evaluates $3 \times 2 \times 2 = 6$, returning 6 instead of 12.

Scoring disagreement. Two normally ended positive OSS cases expose formatting sensitivity in the original answer-evaluation procedure. Case P05 expresses the correct fraction $\frac{14}{15}$ using an inline division symbol; case P08 gives “East” instead of the reference label “east”. Both were classified as incorrect, whereas their baselines matched the reference notation and were classified as correct. The original procedure compared normalised text or numerical values but failed to recognise equivalence in these two cases. The classification concerns final-answer equivalence, not the validity of every intermediate step. The subsequent offline audit in Section 5.4.3 corrects both examples and applies the same rule to all saved baseline and guided evaluations. Current accuracy and bootstrap tables use those revised decisions. This qualitative sample retains its original case identifiers and selection based on historical judgments; P05 and P08 are consequently no longer correctness losses under the revised rule. Resampling after correction would answer a different qualitative question.

Scope of interpretation. The review supplies direct text examples of repetition, incorrect arithmetic, and scoring artefacts. It does not isolate why the routing intervention causes a particular trajectory, quantify population-level looping, or extend the interrupted Qwen result to its missing attempts. Full-population termination counts remain those in the generation-diagnostic

tables. The review records the selected attempts, examined excerpts, repetition measurements, and individual case assessments.

5.4.7 Effects of margin guiding

All three complete stochastic margin comparisons have intervals containing zero after consistent rescoring. Qwen at concurrency 16 gives +0.07 points ($[-0.86, +1.01]$); Qwen at concurrency 25 gains 13 net correct attempts, or +0.93 points ($[0.00, +1.86]$). GPT-OSS loses 12 net correct attempts, or -0.86 points ($[-2.87, +1.15]$). The Qwen concurrency-25 lower endpoint is zero, so its positive point estimate does not establish a direction under this interval criterion.

Table 5.15: Complete stochastic margin-guiding comparisons at strength 1. B/G denotes baseline/guided accuracy in percent; changes and intervals are percentage points. Concurrency is the maximum number of active sequences. Each row has its own matched baseline; differences between rows are not a controlled concurrency experiment.

Model	Concurrency	B	G	Change	95% CI
Qwen	16	84.30	84.37	+0.07	$[-0.86, +1.01]$
Qwen	25	84.09	85.02	+0.93	$[+0.00, +1.86]$
GPT-OSS	25	77.85	76.99	-0.86	$[-2.87, +1.15]$

The small margin effects do not establish benefit or harm after rescoring. The two Qwen configurations have different baseline conditions, so their point-estimate difference cannot be attributed specifically to concurrency. No independent configuration-selection set was used.

Table 5.16 reports generation cost and limit hits. In the positive Qwen margin comparison, mean generated tokens change from 13,830 to 13,810 and limit hits from 154 to 146. GPT-OSS margin guiding changes mean tokens from 6,497 to 6,409 with an accuracy interval containing zero. A reduction in average tokens is therefore not sufficient to claim useful efficiency. These are descriptive token statistics; no paired token-cost confidence interval or

equal-accuracy compute comparison is available [here](#).

Table 5.16: Generation diagnostics for the stochastic comparisons. Means count all generated tokens, including final answers; limit hits are counts among 1,395 attempts. B/G denotes baseline/guided, and c denotes concurrency. Token differences are descriptive and have no uncertainty interval here.

Condition	Mean tokens B	Mean tokens G	Limit hits B	Limit hits G
Qwen fixed positive +1	13863	13706	140	146
Qwen fixed positive +2	13830	14092	154	169
Qwen fixed positive -1	13830	13919	154	138
GPT-OSS fixed positive +1	6497	6580	74	72
GPT-OSS fixed positive +2	6497	8629	74	164
GPT-OSS fixed positive -1	6497	6371	74	60
GPT-OSS fixed negative +1	6497	6900	74	87
GPT-OSS paper positive +1	6497	22489	74	670
GPT-OSS paper positive -1	6497	6698	74	106
GPT-OSS paper negative +1	6497	23639	74	815
Qwen margin (c=16)	13848	13840	144	143
Qwen margin (c=25)	13830	13810	154	146
GPT-OSS margin (c=25)	6497	6409	74	83
Gemma fixed positive +1	11695	11563	67	66

The original Qwen margin run also contains detailed routing measurements. Across selected layers, the top-eight identity set changes on 32.7–41.8% of 20,657,039 token-level routing evaluations per layer. Mean distance to the target interval decreases, from $2.19\text{--}4.71 \times 10^{-4}$ to $0.285\text{--}0.902 \times 10^{-4}$. The intervention modifies the requested margin, but finite-precision arithmetic does not preserve expert membership in this run. These measurements concern the modified router scores rather than an independent observation of the experts subsequently used for computation. Their denominator can include prompt processing, padding positions, and repeated computations. The outcome cannot be attributed solely to a membership-preserving gap adjustment.

The separate greedy comparisons use 465 attempts on 465 problems. Qwen identity guiding gives -0.22 points ($[-2.58, +2.15]$), and margin guiding gives $+0.86$ ($[-1.29, +3.01]$). The 39-problem Minerva-only identity comparison

changes from 24 to 25 correct answers, or +2.56 points ($[-5.13, +12.82]$). All three intervals contain zero. These distinct decoding protocols and populations remain separate from the full stochastic comparisons.

5.4.8 Sensitivity of aggregate intervention effects

Table 5.17 separates the benchmark contributions to the thirteen completed stochastic comparisons reported here. The primary pooled statistic weights attempts equally, so a problem with 32 attempts contributes 32 times as much as a problem with one. As a secondary sensitivity analysis, let n_q be the number of attempts for evaluation problem q and define

$$\hat{\Delta}_{\text{problem}} = \frac{1}{P} \sum_{q=1}^P \frac{1}{n_q} \sum_{i:q(i)=q} (Y_i^\pi - Y_i^0), \quad P = 465. \quad (5.4)$$

This estimates the change in mean per-problem correctness over the saved attempts. It is not pass@32 or a benchmark macro-average: a benchmark’s weight remains proportional to its number of evaluation problems. The attempt-weighted statistic remains the primary analysis; the secondary statistic tests how the interpretation depends on the evaluation unit.

Both statistics use the same 5,000 paired bootstrap samples of whole problems within each benchmark. Only the aggregation of a sampled problem’s outcomes changes. Recomputed attempt-level and benchmark intervals agree with the original comparisons. The sensitivity analysis also records benchmark-specific counts and intervals for both weighting schemes. All intervals remain exploratory and unadjusted for multiple comparisons.

For Gemma fixed positive strength +1, equal-problem weighting gives -0.01 points ($[-2.16, +1.96]$), compared with the primary -0.07 points ($[-1.15, +1.00]$). Neither weighting establishes a beneficial or harmful accuracy direction.

The Qwen margin benchmark breakdown in Table 5.19 uses corrected judgments for both conditions. Its positive point estimate comes from AIME24 and AIME25, offset by one loss on MATH-500. The primary interval includes zero.

Table 5.17: Benchmark-level accuracy changes in percentage points for the fourteen reported stochastic interventions. Positive/negative denotes expert polarity; c denotes concurrency. Entries are point estimates, not significance decisions. Benchmark counts (problems/attempts) are shown in the header.

Condition	MATH-500 150/150	AIME24 9/288	AIME25 9/288	Olympiad 203/203	AMC23 12/384	Minerva 82/82
Qwen fixed positive +1	+0.00	-2.78	+0.00	-0.49	+0.52	+3.66
Qwen fixed positive +2	+0.00	-5.90	-1.74	-0.99	-3.91	+0.00
Qwen fixed positive -1	-1.33	+1.39	+2.08	+1.97	+0.00	+3.66
GPT-OSS fixed positive +1	+5.33	+1.39	-0.35	-2.46	-1.30	-1.22
GPT-OSS fixed positive +2	+2.00	-8.33	-7.64	-2.46	-0.26	+3.66
GPT-OSS fixed positive -1	+3.33	-2.08	-3.12	-2.96	+0.78	-1.22
GPT-OSS fixed negative +1	+4.67	-1.04	-1.39	+0.00	+1.04	+4.88
GPT-OSS paper positive +1	-47.33	-56.60	-76.39	-60.10	-68.75	-24.39
GPT-OSS paper positive -1	+2.67	-3.47	-5.21	-3.45	-0.78	+0.00
GPT-OSS paper negative +1	-47.33	-56.60	-74.31	-56.16	-63.80	-29.27
Qwen margin (c=16)	+0.67	+1.39	+1.39	-3.45	+0.26	-2.44
Qwen margin (c=25)	-0.67	+2.43	+2.43	+0.00	+0.00	+0.00
GPT-OSS margin (c=25)	+2.67	-1.04	-0.69	-3.45	-1.30	+1.22
Gemma fixed positive +1	-0.67	+1.04	-1.74	+1.48	+0.26	-2.44

Qwen margin guiding at concurrency 25 gains seven correct attempts on AIME24 and seven on AIME25, with zero net change on AMC23. MATH-500 loses one correct attempt; Minerva and OlympiadBench each have zero net change. The AIME improvements involve nine improved problems and one worsened problem. Equal-problem weighting gives -0.12 points with $[-2.07, +1.80]$, compared with the primary $+0.93$ and $[0.00, +1.86]$. Neither weighting establishes a beneficial direction.

For GPT-OSS fixed positive strength +2, the revised equal-problem estimate is -0.10 points ($[-3.12, +3.08]$), while fixed negative-polarity promotion gives $+2.35$ ($[-0.58, +5.44]$). The previous secondary positive intervals do not survive rescoring. For positive strength +2, MATH-500 and Minerva gain three correct attempts each, OlympiadBench loses five, and AIME24, AIME25, and AMC23 lose 24, 22, and one. Qwen fixed strength +2 gives -0.68 equal-problem points ($[-2.76, +1.33]$), and GPT-OSS paper-style suppression

Table 5.18: Sensitivity to the evaluation unit: accuracy changes and 95% intervals in percentage points. Attempt weighting is the primary statistic; equal-problem weighting first averages correctness over each problem’s attempts and then averages the 465 problems. Both use the same 5,000 paired, dataset-stratified problem resamples (seed 42). Equal problem weights are not equal benchmark weights. This is a secondary exploratory analysis without multiplicity adjustment.

Condition	Attempt change	95% CI	Problem change	95% CI
Qwen fixed positive +1	-0.29	[-1.15, +0.65]	+0.39	[-1.51, +2.31]
Qwen fixed positive +2	-2.80	[-5.16, -0.72]	-0.68	[-2.76, +1.33]
Qwen fixed positive -1	+1.08	[-0.14, +2.37]	+1.14	[-0.75, +2.94]
GPT-OSS fixed positive +1	+0.00	[-2.15, +2.01]	+0.42	[-2.37, +3.12]
GPT-OSS fixed positive +2	-3.30	[-6.16, -0.93]	-0.10	[-3.12, +3.08]
GPT-OSS fixed positive -1	-1.00	[-3.44, +1.22]	-0.51	[-3.41, +2.49]
GPT-OSS fixed negative +1	+0.57	[-1.58, +2.65]	+2.35	[-0.58, +5.44]
GPT-OSS paper positive +1	-61.65	[-69.32, -53.41]	-50.15	[-54.88, -45.45]
GPT-OSS paper positive -1	-2.22	[-4.16, -0.43]	-0.83	[-3.76, +2.23]
GPT-OSS paper negative +1	-59.57	[-67.24, -51.40]	-49.13	[-53.80, -44.56]
Qwen margin (c=16)	+0.07	[-0.86, +1.01]	-1.66	[-3.67, +0.45]
Qwen margin (c=25)	+0.93	[+0.00, +1.86]	-0.12	[-2.07, +1.80]
GPT-OSS margin (c=25)	-0.86	[-2.87, +1.15]	-0.50	[-3.49, +2.46]
Gemma fixed positive +1	-0.07	[-1.15, +1.00]	-0.01	[-2.16, +1.96]

Table 5.19: Benchmark contributions to Qwen margin guiding at concurrency 25. B/G denotes baseline/guided correct counts; net is guided minus baseline correct attempts. Changes and pointwise 95% intervals are percentage points, with 5,000 paired problem resamples (seed 42).

Benchmark	Problems	Attempts	Correct B/G	Net	Change	95% CI
MATH-500	150	150	148/147	-1	-0.67	[-2.00, +0.00]
AIME24	9	288	178/185	+7	+2.43	[+0.00, +5.56]
AIME25	9	288	242/249	+7	+2.43	[+1.04, +4.17]
Olympiad	203	203	171/171	+0	+0.00	[-3.45, +3.45]
AMC23	12	384	384/384	+0	+0.00	[+0.00, +0.00]
Minerva	82	82	50/50	+0	+0.00	[-6.10, +6.10]

gives -0.83 ($[-3.76, +2.23]$); their below-zero primary intervals do not persist under this secondary weighting. GPT-OSS paper-style positive and negative promotion remain strongly negative: -50.15 ($[-54.88, -45.45]$) and -49.13 ($[-53.80, -44.56]$) points. No complete stochastic comparison has a positive interval excluding zero under either weighting.

5.5 Expert reuse under a fixed-cache proxy

5.5.1 GPT-OSS

A separate full-diagnostics rerun evaluates GPT-OSS fixed positive identity steering at strength $+1$, using the frozen policy and the same 1,395 held-out attempts on 465 problems. The compatible baseline is reused with output-hash provenance; the guided generations are new. This diagnostic rerun is kept separate from the historical minimal-diagnostics comparison and is not an independent problem-set replication. No new target or strength is selected from these results.

Under the same offline answer-equivalence scoring contract as the main intervention analysis, accuracy is $1086/1395 = 77.85\%$ for baseline and $1088/1395 = 77.99\%$ for guidance. The change is $+0.14$ percentage points, with a paired, dataset-stratified source-problem bootstrap 95% interval $[-2.37, +2.37]$ (5,000 resamples, seed 42). There are 94 wrong-to-right and 92 right-to-wrong changes. Mean generated tokens increase from 6,497.2 to 6,813.9 ($+4.88\%$), and token-limit hits increase from 74 to 77. Thus the observed aggregate accuracy difference is small, but accuracy preservation and an efficiency improvement are not established.

The routing counters permit a fixed resident-cache proxy at the four steered layers, 15, 19, 21 and 23. Four target experts per layer would occupy 16 layer-expert slots in total. Define H as the fraction of top- k expert selections covered by that fixed resident set; the proxy for nonresident selections is $1 - H$. Both pre-bias and post-bias counters are evaluated on the guided trajectory’s hidden states. The pre-bias counts are therefore not routing measurements from the separately generated baseline.

Table 5.20: GPT-OSS diagnostic rerun: static-cache coverage as a percentage of expert selections. Target and frequency caches each have four slots per layer. Frequency pins are selected from calibration statistics only. All columns use states visited by guided generation.

Layer	Target, pre-bias	Target, post-bias	Frequency, pre-bias
15	5.19	31.40	30.40
19	8.70	45.23	37.42
21	2.20	22.58	41.36
23	4.23	41.48	38.32
All four	5.08	35.17	36.87

Target-cache coverage rises from 5.08% to 35.17%. Relative to pinning those same targets without the bias on these states, this corresponds to 31.70% fewer nonresident selections: $(H_{\text{after}} - H_{\text{before}})/(1 - H_{\text{before}})$. However, a same-budget cache of calibration-frequent experts already covers 36.87% of pre-bias selections. Guided target pinning therefore has 2.70% more nonresident selections than this control. A hindsight-optimal static cache covers 37.39% before the bias; against it, guided target pinning has 3.55% more nonresident selections. The hindsight cache is an analytical control, not a deployable policy or a bound on adaptive caching.

The supported result is greater reuse of the boosted expert identities, alongside a small observed aggregate accuracy difference. For this OSS policy, that concentration does not outperform the calibration-frequency static-cache control in the measured proxy. This distinction prevents increased target frequency alone from being interpreted as a systems gain.

These counters include prefill, decode and engine padding, cover only the four selected layers, and compute top- k from hook logits rather than separately observing native dispatch. They retain neither temporal order nor per-attempt routing, so no cache-effect confidence interval is available. The proxy assumes an already resident pinned set and excludes cold-start loads, dynamic caching, shared loads across batched tokens and prefetch. It measures neither SSD bytes nor latency and must not be extrapolated to whole-model speed. A small

accuracy loss could be acceptable if it buys sufficient speed, but that trade-off requires ordered decode traces and ultimately an equal-memory offloaded-inference benchmark. Longer guided outputs make total loads and latency per answer especially important.

5.5.2 Gemma

For the Gemma fixed-bias condition in Table 5.12, the routing histograms measure executed router evaluations, including prefill, padding and repeated computations. Their population therefore differs from one decode-only routing trace per scored answer. Pre-bias and post-bias counters refer to the same guided hidden states, not separate baseline generations.

Table 5.21: Gemma full-diagnostics fixed guiding: expert-selection coverage in percent, aggregated across recorded router evaluations. Each static cache contains eight experts per measured layer. Frequency pins use calibration frequencies.

Layer	Target, pre-bias	Target, post-bias	Frequency, pre-bias
16	0.05	3.26	35.42
19	0.03	1.06	39.41
21	0.04	1.26	43.19
22	0.11	2.02	53.53
25	0.16	5.31	43.37
27	1.38	6.42	41.37
29	2.67	15.90	36.38
All seven	0.64	5.03	41.81

The target set’s share of expert selections increases from 0.64% to 5.03%. For an already resident target cache this gives 4.43% fewer nonresident selections relative to the same targets before bias. However, a same-budget frequency-selected cache covers 41.81% before bias (and 40.64% after it). Guided target pinning consequently has 63.20% more nonresident selections than the pre-bias frequency cache. The bias increases use of the selected identities but does not

make them a competitive static cache set for this model and strength.

These aggregate counters describe only seven policy layers and do not measure ordered cache loads, cold-start cost, SSD traffic or latency. They establish neither a whole-model speed effect nor a confidence interval for cache coverage. The ordered experiments below test adaptive-cache controls and all-layer loads separately.

5.6 Ordered cache experiments across three models

The completed study applies the protocol in Section 3.5 to the matched 100-problem cohort described in Section 4.8. Baseline and fixed positive bias-+1 generations are compared under equal per-model cache capacities. The results distinguish changes in accuracy and response length from changes in simulated expert-loading demand.

Accuracy and output length. Offline extracted-answer equivalence rescoring gives baseline/guided correct counts of 76/75 for OSS, 83/82 for Qwen, and 84/84 for Gemma. Their accuracy changes have 95% intervals of $[-7, +5]$, $[-5, +3]$ and $[-4, +4]$ percentage points, respectively. These use 5,000 paired, dataset-stratified source-problem resamples with seed 42. The original saved scorer instead gives 53/48, 62/61 and 59/59; those labels remain unchanged in the source artifacts. No model establishes accuracy non-inferiority from these intervals alone.

OSS outputs are 22.04% longer on average; Qwen outputs are 1.06% shorter and Gemma outputs 1.61% shorter. Token-limit counts are 2/6, 7/7 and 3/5, respectively. The distinct length responses make total loads per answer necessary alongside loads per token.

Qwen: a promising loading reduction. Guided target pinning reduces mean simulated loads per answer by 6.20%, 6.98% and 7.81% at 16, 32 and 64 slots per layer. It also uses fewer loads than the same-budget frequency-pinned

Table 5.22: Matched 100-problem cache experiments. B/G denotes baseline/guided. Accuracy uses offline equivalence rescoring; changes and intervals are percentage points. Token changes are relative percentages.

Model	Correct B/G	Change [95% CI]	Mean tokens B/G	Token change (%)	Capped B/G
GPT-OSS	76/75	-1.0 [-7.0, +5.0]	4,418.9/5,392.9	+22.04	2/6
Qwen	83/82	-1.0 [-5.0, +3.0]	12,116.1/11,988.0	-1.06	7/7
Gemma	84/84	+0.0 [-4.0, +4.0]	9,645.4/9,490.5	-1.61	3/5

Table 5.23: Simulated mean expert loads per complete answer, including cold-start pin loads, across all routed layers. Capacity is slots per layer; frequency and target pins use the same slot budget within each model.

Model	Slots	Baseline LRU	Baseline frequency	Guided LRU	Guided targets
GPT-OSS	8	134,007	134,747	158,903	162,649
GPT-OSS	16	40,080	40,068	45,581	46,092
GPT-OSS	32	723	723	717	717
Qwen	16	2,400,863	2,410,567	2,266,939	2,252,038
Qwen	32	1,900,161	1,901,191	1,772,502	1,767,504
Qwen	64	1,184,307	1,184,273	1,092,516	1,091,858
Gemma	16	827,247	821,327	825,811	912,591
Gemma	32	354,596	352,565	353,664	378,813
Gemma	64	60,310	60,284	59,939	61,556

baseline. Ordinary LRU on guided traces reduces observed loads per answer by 5.58%, 6.72% and 7.75% relative to baseline LRU. Reserving slots for the targets then reduces loads by a further 0.66%, 0.28% and 0.06% relative to guided LRU. These descriptive contrasts show that ordinary adaptive replacement captures most of the observed benefit; explicit target pinning contributes a smaller increment. They do not establish separate statistical effects for those increments. The accuracy point estimate is one fewer correct answer out of 100, and output length is slightly lower. Per-token reductions are 5.20%, 5.99% and 6.82%, with all three 95% intervals below zero. All per-answer intervals still include zero: $[-16.35, +5.50]\%$, $[-17.12, +4.72]\%$ and $[-14.90, +0.08]\%$. Thus per-token locality improves under the reported resampling criterion, while a

Table 5.24: Relative change (percent) for guided target pinning versus baseline LRU, with paired dataset-stratified problem-bootstrap 95% intervals. Negative values mean fewer simulated loads. Intervals use 5,000 resamples, seed 42, and no multiplicity correction.

Model	Slots	Loads/answer: change [CI]	Loads/token: change [CI]
GPT-OSS	8	+21.37 [-8.92, +62.53]	-0.55 [-2.74, +1.66]
GPT-OSS	16	+15.00 [-13.33, +53.22]	-5.77 [-9.06, -2.56]
GPT-OSS	32	-0.78 [-1.05, -0.50]	-18.71 [-39.53, +8.83]
Qwen	16	-6.20 [-16.35, +5.50]	-5.20 [-6.34, -4.00]
Qwen	32	-6.98 [-17.12, +4.72]	-5.99 [-7.19, -4.65]
Qwen	64	-7.81 [-14.90, +0.08]	-6.82 [-10.26, -2.14]
Gemma	16	+10.32 [+2.69, +18.68]	+12.12 [+10.23, +14.44]
Gemma	32	+6.83 [-0.72, +15.07]	+8.57 [+5.84, +11.73]
Gemma	64	+2.07 [-4.74, +9.97]	+3.73 [+0.08, +7.76]

population-level reduction in total loads per answer remains unresolved. These are not end-to-end speed measurements.

Gemma: pinning rare targets is costly. Guided target pinning increases loads per answer by 10.32%, 6.83% and 2.07% at 16, 32 and 64 slots. At 16 slots the 95% interval is [+2.69, +18.68]%; at 32 and 64 slots the intervals include zero. Loads per token increase at all three budgets. Ordinary LRU on the guided traces instead gives 825,810.8, 353,664.2 and 59,938.7 loads per answer, slightly below the respective baseline-LRU means. Explicit target pinning therefore consumes capacity that adaptive LRU uses more effectively. This agrees with the full-diagnostics finding that the accuracy-associated Gemma targets remain infrequent even after bias. There is no aggregate accuracy difference in the 100-problem sample.

OSS: local savings are outweighed by longer answers. At 16 slots, guided target pinning reduces loads per token by 5.77% ([-9.06, -2.56]%), but increases observed loads per answer by 15.00% ([-13.33, +53.22]%). At eight

slots, the per-answer increase is 21.37% ($[-8.92, +62.53]\%$). Ordinary LRU uses fewer loads than target pinning on the same guided traces. At 32 slots all OSS experts fit, so the small reduction in loads concerns initial loading of distinct experts, not reduced eviction under memory pressure.

Interpretation. The three models do not support a universal cache benefit from promoting accuracy-associated identities. Qwen is the strongest candidate for a subsequent offloaded-inference benchmark; Gemma exposes the cost of pinning rare targets, and OSS shows how longer outputs can offset per-token locality. A small accuracy loss could be acceptable in exchange for sufficient speed, but establishing that trade-off requires actual answer latency under equal memory and transfer budgets. Predictive prefetching is not evaluated here.

Cache intervals resample paired problems within datasets (5,000 resamples, Python random seed 42), recomputing ratios of total loads and token counts. They condition on the frozen policies and saved generations and are not adjusted for multiple budgets or models. They do not include independent seed or policy-fitting variation. Prefill warming, cross-answer reuse, batching, prefetch and transfer costs are excluded. Weights remain resident during generation; instrumentation time cannot be interpreted as SSD-backed inference time. The matched cohort strengthens descriptive comparison but does not isolate architectural causes of differences between models.

5.7 What the combined evidence establishes

The four stages do not support a single unqualified claim that reasoning can be improved by increasing the use of apparently successful experts. The judge supplies usable but imperfect labels on a development corpus; the probes reveal prospective information on held-out responses within their SAT corpus, beyond the simple position baseline; and the routing analysis identifies associations that depend on model, benchmark, scope, and aggregation. The intervention stage provides the strongest test of actionability. After consistent offline rescoring, no complete intervention establishes a beneficial accuracy direction using the

reported 95% interval criterion. Qwen fixed positive strength +2 loses 2.80 attempt-weighted points ($[-5.16, -0.72]$); GPT-OSS fixed positive strength +2 loses 3.30 ($[-6.16, -0.93]$). GPT-OSS paper-style suppression loses 2.22 points ($[-4.16, -0.43]$), and positive/negative paper-style promotion loses 61.65 and 59.57 points with intervals well below zero. All stochastic margin intervals contain zero: Qwen concurrency 25 gives +0.93 points ($[0.00, +1.86]$), while GPT-OSS gives -0.86 ($[-2.87, +1.15]$). Equal-problem weighting removes the below-zero intervals for fixed strength +2 and paper-style suppression, but paper-style promotion remains strongly negative. The earlier positive OSS secondary intervals and greedy Qwen margin interval do not survive rescoring. These exploratory intervals remain conditional on saved attempts and frozen policies and are not adjusted for multiple comparisons.

This pattern separates the existence of measurable routing structure from the usefulness of a particular control rule. The evidence supports continued testing of carefully specified policies, with attention to numerical behaviour and evaluation protocols. It does not establish a transferable routing heuristic, nor does it rule out improvements from policies, selection procedures, and controls that were not evaluated.

Chapter 6

Conclusions

This thesis investigated whether reasoning annotations and internal routing measurements can support interventions in sparse mixture-of-experts models without changing model weights. Our procedure connects automatic sentence annotation, prospective representation probing, observational routing analysis, and matched free-generation experiments. Its main empirical finding is a gap between observable structure and reliable control: reasoning-related information and routing–correctness associations are measurable, but promoting associated experts does not generally improve held-out mathematical accuracy under the tested policies.

6.1 Answers to the research questions

The first question concerned automatic episode annotation and the accessibility of upcoming episode information in prefix representations. The annotation experiments demonstrate that context-aware prompts can achieve useful agreement on the available labelled corpus and that prompt selection benefits from explicit class-level safeguards. The final alignment retained its seed when an optimised candidate damaged Plan recall, even though its internal optimisation reward increased. The NVFP4 deployment check further showed that unchanged aggregate accuracy can conceal changes in individual labels. Automatic annotations are therefore informative measurements with uncer-

tainty, not independently established ground truth for the generated benchmark solutions.

All three frozen encoders contain linearly accessible information about upcoming episode labels on held-out responses from the SAT corpus. With response-disjoint outer folds and inner-validation layer selection, macro-AUROC is 0.852 for Qwen, 0.780 for GPT-OSS, and 0.788 for Gemma; all exceed the simple prefix-position baseline. This strengthens the answer to the decodability question beyond the original sentence split. It does not establish transfer to other domains or to the models’ own generations, and thresholded classification remains uneven across labels. The external Qwen inspection reinforces this distinction: Implement transfers more coherently than several other classes, while the provisional review does not support a robust seven-class online detector.

The second question concerned relationships between routing, reasoning activity, and correctness. The three-model corpus shows substantial but heterogeneous structure. Confidence is positively associated with correctness across Qwen benchmarks, but its sign varies in Gemma and GPT-OSS. Class conditioning does not remove that variation, and problem-level aggregation can yield a different association from pooling attempts. The analysis therefore identifies hypotheses for model-specific interventions rather than a common scalar measure of successful reasoning. Length, termination, problem difficulty, and annotation coverage remain plausible contributors to the observed patterns.

Directed class-transition matrices additionally separate correct and incorrect attempts for all three models while retaining original adjacency and sampling gaps. The expert-selection audit shows that activation-based lift and risk difference need not choose the same targets, even with identical calibration weights and budgets: overlap is 14/48 layer–expert identities for Qwen and 6/16 for GPT-OSS positive and 7/16 for GPT-OSS negative policies.

The third question concerned accuracy, token use, and expert-loading demand under calibrated routing policies. After consistent offline rescoring, no completed intervention establishes an accuracy improvement under the reported interval criterion. Stronger fixed promotion can harm accuracy, and paper-style promotion produces the clearest degradation. Margin effects remain unresolved.

Some conclusions depend on whether attempts or source problems receive equal weight, so the evaluation objective must be specified before replication. These findings concern the tested policies and do not establish that the selected experts are intrinsically beneficial or harmful.

Gemma’s fixed-bias +1 evaluation is a useful complementary result: substantially greater use of the targeted identities accompanies only a small observed accuracy difference. This shows that changing expert reuse need not produce a large measured accuracy loss in every model. It does not establish accuracy equivalence, and the targeted identities remain less frequent than those selected by the calibration-frequency cache.

The interrupted Qwen paper-style run shows severe degradation on its completed subset: 4/494 correct answers and 494/494 token-limit hits. Its coverage is restricted to AIME24 and part of AIME25, so it remains separate from the complete-run evidence. Promoting a full top- k target set fixes the selected identities and equalises their hook scores; this is a possible mechanism, not a verified causal explanation of the observed termination behaviour.

The qualitative review supplies direct examples of repetition in both capped and normally ended outputs, while also identifying two formatting disagreements with the saved scorer. It does not estimate failure-mode prevalence or establish the routing mechanism.

Token counts and routing coverage supply complementary evidence, but neither alone measures useful efficiency. A small accuracy loss may be acceptable for a sufficiently large reduction in answer cost; evaluating that trade-off requires a stated quality tolerance and direct cost measurements under comparable conditions.

The matched 100-problem ordered-cache experiments (Section 5.6) reveal model-dependent effects. Qwen provides the most promising evidence of better locality: ordinary LRU already captures most of the observed loading reduction on guided traces, and target pinning adds little. Per-token reductions have intervals below zero for the reported guided-target comparison, while total-load benefits remain uncertain. Gemma demonstrates the capacity cost of pinning rare targets, and OSS demonstrates how longer answers can offset local savings. The sample does not establish accuracy non-inferiority for any model. These

findings motivate a direct efficiency test for Qwen, while providing no universal caching benefit or measured offloaded-inference speedup.

6.2 Research contributions

The methodological contribution is an integrated and auditable experimental path from functional descriptions of reasoning to routing perturbations. The completed control experiments test the actionability of calibrated routing associations. They do not evaluate a controller driven by online episode predictions: annotation and probing support measurement and target exploration, and their usefulness as control inputs remains to be tested. The workflow preserves problem identities, generation settings, sentence spans, replay provenance, and frozen calibration policies. It separates sentence-level annotation quality from prefix-state decodability, and both from the effect of intervening during generation. This makes it possible to identify where a promising signal fails to support the next inference.

The observational contribution is a comparison of routing and representation measurements across three sparse models, six benchmarks, multiple reasoning classes, and response positions. Its most consequential lesson is heterogeneity: an association that appears strong in one population may be weak, undefined, or oppositely signed in another. The distinction between attempt-level, problem-level, and within-problem evidence is necessary for interpreting repeated mathematical sampling.

The intervention contribution is a set of paired held-out evaluations that expose both weak effects and substantial degradation. The large losses under GPT-OSS paper-style promotion do not support a monotonic reading of the calibration association under that intervention rule. They do not show that the selected experts are intrinsically harmful: forcing a fixed target set can disrupt coordinated, input-dependent computation. The changes after rescoring also show why conclusions require validated answer equivalence as well as an explicit evaluation objective. Acting on a routing statistic changes an entire generation trajectory, and the measured effect belongs to that complete policy.

6.3 Limitations

The annotation reference contains few responses and ambiguous functional boundaries. Repeated prompt development on its validation responses limits generalisation claims, and the generated benchmark labels lack an independent human assessment. Class-conditioned results inherit these errors and the sampling scheme. Although Gemma and GPT-OSS share a stratified procedure, their labelled problem cohorts differ; Qwen uses a different deployment and sampling protocol.

The original probe scores and routing-layer choices remain exploratory because responses cross the sentence split and layer selection uses test outcomes. The separate grouped study removes those two evaluation overlaps, but has only 38 responses from one corpus and one split seed. Its bootstrap conditions on the fitted classifiers and selected layers; neither external transfer nor online detection is established by that result. Exceeding a two-feature position baseline also does not eliminate richer shortcuts. The correlation study retains all attempts and respects problem clusters in the main bootstrap intervals, but clustering does not remove confounding. Late-position views condition on continued generation, class views condition on available labels and class tokens, and missing-outcome exclusions in policy calibration differ from the observational scoring denominator. Generation-to-replay transfer across quantisations adds uncertainty about the measured router state.

The failure inspection identified answer-formatting sensitivity, addressed by consistent offline rescoring of the saved intervention evaluations. The revised scorer still has a defined extraction hierarchy, numerical tolerance, unit conventions, and unsupported formats; it is not independent human adjudication. The observational analysis and frozen policy calibration retain their original correctness labels. The current results therefore evaluate those existing policies under a revised outcome rule; they do not estimate what a policy refitted using revised calibration labels would do.

Intervention uncertainty is conditional on frozen policies and saved attempts. The confidence intervals do not include variation from recalibration, new decoding seeds, or configuration search, and no multiple-comparison adjustment

is applied. Nine held-out problems in each AIME benchmark limit benchmark-specific precision. The Qwen comparisons with different baseline conditions and concurrency settings do not isolate one experimental factor across rows. The seven OSS identity conditions share those settings and retain the same targets within polarity, allowing a more direct comparison of intervention rules. Finally, margin diagnostics reveal membership changes in finite precision, while identity interventions change routing weights as well as selected sets. These experiments do not isolate all of the mechanisms through which the policies affect generation.

6.4 Directions for further work

The next measurement step is an independent, response-grouped annotation assessment on generated mathematical solutions. It should include examples from every model and benchmark, explicit treatment of ambiguous units, and adjudication of a representative sample. The judge and its decoding settings should be frozen before that assessment. The completed response-grouped probe study should be extended to new domains and the encoders' own generations, with repeated response splits and thresholds and abstention rules selected on development data. More flexible position controls would further test shortcut explanations. These extensions would assess whether the available episode information supports a reliable online measurement.

For observational modelling, matched problem cohorts and common annotation protocols would make cross-model comparisons more interpretable. Analyses should separate within-problem variation from between-problem difficulty and account for length and termination without simply discarding failed completions. Candidate features and hypotheses should be chosen before a fresh evaluation, reducing the temptation to treat the strongest result in a broad screen as a confirmed mechanism.

The offline answer-equivalence audit is complete for the saved intervention evaluations. A further validation study should adjudicate residual parser failures and ambiguous or structured answers, assess robustness to tolerance and extraction choices, and examine calibration-label sensitivity before refitting

policies on development data.

The next generation study should replicate the Qwen margin configuration and test GPT-OSS identity guiding with independent decoding seeds and an untouched evaluation set. The attempt-weighted or problem-weighted objective must be specified in advance, with benchmark-level reporting retained. Strength, target-selection rules, and other settings should be selected on a separate validation partition. The OSS negative-lift controls and target budgets matching native dispatch are already complete. Matched random-expert controls and smaller budgets of one or two targets per layer remain useful for distinguishing selector information from generic perturbation and from forcing a full top- k target set. Applying interventions separately during prefill and decoding would localise when their effects arise. Recording actual dispatched identities and mixture weights would also clarify the numerical behaviour of the margin operation.

The next efficiency experiment should evaluate actual offloaded Qwen inference. Baseline and guided routing should first use ordinary LRU with identical expert-memory budgets, weight formats, hardware, and transfer settings; target pinning can then be added as a separate contrast. The study should measure complete-answer latency, decode throughput, SSD bytes read, host-to-device bytes transferred, and cache misses alongside accuracy, output length, and token-limit hits. Cold-start and warmed-cache conditions should be reported separately. Repeated timing measurements and independent decoding seeds would separate runtime variation from generation variation. A quality-loss tolerance should be fixed in advance so that any speed gain can be assessed against an explicit acceptable accuracy trade-off. Predictive prefetching would require its own comparison after this baseline is established.

A later extension could condition guidance on a validated prediction of the current reasoning activity. The completed interventions do not yet test such a controller: they use fixed identities or current router margins, without an online episode prediction. Any class-conditioned policy would need to account for detector errors and forbid access to the future continuation used to define its reference labels. Its evaluation should jointly report accuracy, token use, and failure modes under a matched budget.

The evidence thus supports routing as a measurable and consequential part of mathematical generation, while leaving reliable guidance an open empirical problem. Progress depends on validating the measurements, specifying what an intervention changes, and testing its benefit on data that did not determine the policy.

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